

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

**An Online Ordering System for Brand-New and Preloved
Clothes with Product Listing, Checkout, and Buyer-Seller
Management.**

A Capstone Project Proposal

In partial fulfillment for the requirements in Capstone 1 and
Research Subject

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TABLE OF CONTENTS

CHAPTER 1 – INTRODUCTION

1.1 Background of the Study	8
1.2 Statement of the Problem	11
1.3 Objectives of the Study	13
1.3.1 General Objective	13
1.3.2 Specific Objectives	13
1.4 Scope and Delimitation	14
1.5 Significance of the Study	16
1.6 Definition of Terms	18

CHAPTER 2 – REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Literature	20
2.2 Foreign Studies	47
2.3 Local Literature	66
2.4 Local Studies	102
2.5 Synthesis of the Literature and Studies	142
2.6 Research Gap	145
2.7 References	147

CHAPTER 3 – METHODOLOGY

3.1 Research Design	166
3.2 System Development Methodology	167
3.2.1 Requirements Analysis	168
3.2.2 System Design	169
3.2.3 Development	169
3.2.4 Testing	169
3.2.5 Deployment (Prototype)	170

3.3 System Architecture	170
3.3.1 Presentation Layer	171
3.3.2 Application Layer	172
3.3.3 Data Layer	173
3.4 System Features and Modules	174
3.4.1 Authentication Module	174
3.4.2 Product Management Module	174
3.4.3 Shopping Cart Module	175
3.4.4 Order Processing Module	176
3.4.5 Preloved-Specific Management Module	176
3.4.6 User Management Module	177
3.4.7 Analytics and Sales Reporting Module	177
3.4.8 Summary of Modules	178
3.5 Database Design	178
3.5.1 Main Database Tables	178
3.5.2 Data Integrity and Optimization	181
3.6 Development Tools and Technologies	181
3.7 Data Gathering Techniques	182
3.8 Testing Procedures	184
3.8.1 Unit Testing	184
3.8.2 Integration Testing	185
3.8.3 Security Testing	185
3.8.4 User Acceptance Testing (UAT)	186
3.9 Evaluation Criteria	187
3.9.1 Functionality	187
3.9.2 Usability	188
3.9.3 Security	188
3.9.4 Performance	188
3.9.5 Reliability	188
3.10 Data Flow Diagram (DFD – Level 0)	189
3.11 Entity Relationship Diagram (ERD)	190
3.12 Use Case Diagram	192

CHAPTER 4 – SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System	195
4.3 System Development Methodology	197

<i>Figure 4.1. System Workflow of Margaux Collections</i>	197
<i>Figure 4.2. Agile Software Development Life Cycle</i>	201
4.4 System Architecture	201
<i>Figure 4.3. Three-Tier System Architecture of Margaux Collections</i>	202
4.5 Hardware and Software Requirements	204
4.5.1 Hardware Requirements	204
<i>Table 4.1. Recommended Hardware Specifications</i>	204
4.5.2 Software Requirements	205
<i>Table 4.2. Software Requirements</i>	205
4.6 Database Design	205
<i>Figure 4.4. Entity Relationship Diagram of Margaux Collections</i> ...	206
4.6.1 Database Tables	207
4.7 System Implementation	212
4.8 Testing and Evaluation	213
4.9 Results and Discussion	214

CHAPTER 5 – CONCLUSION AND RECOMMENDATIONS

5.1 Introduction	215
5.2 Summary of Findings	215
5.3 Conclusion	215
5.4 Recommendations	216
5.5 Impact of the Study	217
5.6 Summary	217

INTRODUCTION

1.1 Background of the Study

The growth of internet technology has greatly changed the way people buy and sell products in today's market. Many

businesses now rely on online platforms because they offer faster communication, wider market reach, and more convenient transaction processes. In the Philippines, this shift is especially visible among small businesses that use social media and web-based platforms to promote their products and interact with customers. Clothing businesses, in particular, have benefited from this trend because customers often prefer to browse fashion items online before deciding to purchase them. As a result, an online ordering system has become an important tool for businesses that want to improve their operations and remain competitive in the digital age.

The clothing industry is one of the most active sectors in online selling because fashion products appeal to different types of customers with different needs and preferences. Some buyers look for brand-new clothes because they want new styles and guaranteed product quality, while others choose preloved clothes because they are more affordable and can still be used for a long time. In some businesses, both brand-new and preloved clothes are sold together, which creates a wider range of choices for customers. However, managing these products manually can be difficult, especially when the seller depends only on social media posts and private messages to handle orders. This kind of setup may lead to confusion, delayed responses, incomplete records, and difficulty in monitoring inventory.

At present, many small clothing businesses use Facebook or similar platforms to post product pictures and receive customer inquiries. While this method is convenient at first, it can become inefficient as the number of customers and items increases. Orders sent through messages may be difficult to organize, and the seller may struggle to keep track of which products are still available, reserved, or already sold. In addition, when communication happens only through chat, there is a higher chance of missed details, repeated questions, and delayed confirmations. These problems show the need for a more organized and reliable

system that can handle both product display and order processing in one platform.

An online ordering system can help solve these problems by giving customers a more structured way to browse products, select items, and place orders. For a

business that sells both brand-new and preloved clothes, the system can separate items into categories while still keeping everything in one website. This makes it easier for customers to find what they need and compare available options. It also helps the seller manage product information more clearly, including the item's name, price, condition, size, and availability. With a proper system in place, the whole ordering process becomes faster, more accurate, and more convenient for both buyers and sellers.

The inclusion of product listing features is important because clothing items need clear and complete details before customers make a purchase. Buyers usually want to see photos, descriptions, prices, and item conditions before deciding whether to order. This is especially true for preloved clothes, where the condition of the item is a major factor in the buying decision. A well-designed product listing can help the seller present items in a more organized and professional manner. It can also reduce the need for repeated inquiries because customers can already view the important details directly on the website.

Another important part of the system is the checkout process. Checkout allows customers to review the items they selected, confirm their order, and submit the necessary information before completing the transaction. This step is important because it helps prevent mistakes and ensures that all order details are properly recorded. For the seller, checkout makes it easier to track orders and monitor sales activity. It also helps maintain more accurate inventory records, which is very useful for a business that handles both new and preloved items. By automating this part of the process, the business can reduce manual work and improve efficiency.

Buyer-seller management is also a necessary feature in an online clothing system. Customers often need to ask questions about sizes, product condition, shipping methods, payment options, or availability before making a purchase. In the case of preloved clothing, these concerns are even more important because buyers want to make sure that the item still meets their expectations. A system that supports proper buyer-seller communication can make transactions smoother and help build customer trust. It also allows the seller to respond more professionally and manage customer concerns in an organized way. This can lead to better customer satisfaction and stronger relationships between the business and its buyers.

Overall, the proposed **Online Ordering System for Brand-New and Preloved Clothes with Product Listing, Checkout, and Buyer-Seller Management** aims to improve the current way clothing businesses handle online sales. Instead of relying only on manual chatting and informal posting, the business can use a centralized system that organizes products, orders, and communication in one place. This will help reduce errors, save time, and make the selling process more efficient. At the same time, it will provide customers with a better shopping experience because they can easily browse products, place orders, and communicate with the seller in a more organized manner. In this way, the system supports both business growth and customer convenience in a modern online setting.

1.2 Statement of the Problem

This study aims to develop an **Online Ordering System for Brand-New and**

Preloved Clothes with Product Listing, Checkout, and Buyer-Seller Management. The growing use of online platforms in business has made it easier for sellers to reach customers, but it has also created challenges in handling orders, organizing product information, and maintaining clear communication between buyers and sellers. In the case of businesses that sell both brand-new and preloved clothes, these challenges become even more noticeable because each item must be properly described, categorized, and monitored. Since many small clothing businesses still rely on manual processes or social media messaging to manage sales, problems such as delayed responses, inaccurate records, missing order details, and confusion in inventory often occur. Because of these issues, there is a need for a more organized and reliable system that can support the entire ordering process.

The proposed study seeks to improve the current process by developing a web-based system that can manage product listings, order checkout, and buyer-seller communication in one platform. Instead of depending only on private messages, informal posting, and manual tracking, the system will allow customers to browse available clothes, view item details, and place orders through a structured interface. It will

also help the seller manage product availability and customer transactions more efficiently. By using a centralized system, the business can reduce common errors and provide a better shopping experience for customers. This study is important because it addresses both the operational needs of the seller and the convenience of the buyer.

Specifically, this study seeks to answer the following questions:

1. What difficulties are experienced by the business in managing brand-new and preloved clothes through manual methods and social media-based selling?

2. How can the proposed system improve the product listing process by organizing brand-new and preloved clothes in a clear and accessible way?
3. How can the checkout feature of the system help make the ordering process more efficient, accurate, and convenient for customers?
4. How can the buyer-seller management feature improve communication, transaction handling, and order coordination?
5. How can the system help the seller monitor product availability and reduce errors in inventory tracking and order processing?
6. How can the proposed system provide a more organized, user-friendly, and reliable online shopping experience for both buyers and sellers?
7. How can the system support the overall business operations of clothing sellers who offer both new and preloved items in one platform?

This study focuses on identifying how technology can be used to improve the way clothing businesses handle online transactions. It also aims to show how a properly designed ordering system can help reduce manual work, improve customer service, and create a more efficient business process. In doing so, the proposed system is expected to benefit both the business owner and the customers by making online shopping easier, faster, and more organized.

1.3.2 Specific Objectives

General Objective

The general objective of this study is to develop a **webbased ordering system for brand-new and preloved clothes** that can efficiently manage product listing, order processing, checkout, and buyer-seller communication in one centralized platform.

Specific Objectives

Specifically, this study aims to:

1. Design a web-based system that allows the seller to upload, organize, and manage listings for brand-new and preloved clothing items.
2. Develop a checkout feature that enables customers to place orders easily and review their selected items before confirming their purchase.
3. Create a buyer-seller management module that supports communication between customers and the seller regarding product inquiries, order updates, and transaction concerns.
4. Implement an inventory management function that helps the seller monitor available stock and update product status in real time.
5. Improve the accuracy and organization of order processing by reducing manual encoding and relying on a centralized digital platform.
6. Provide a user-friendly interface that makes it easier for customers to browse products, place orders, and complete transactions.
7. Ensure that the system supports both brand-new and preloved clothing items in a single online platform.

1.4 Scope and Delimitation

This study focuses on the development of an Online Ordering System for

Brand-New and Preloved Clothes with Product Listing, Checkout, and Buyer-Seller Management for Margaux Collections. The system will cover the main operations of the business, particularly product listing, order placement, inventory tracking, payment handling, queuing, and analytics. It is designed to help the owner manage clothing items more efficiently and to provide customers with a more organized and convenient way of browsing and ordering products. The study will specifically cater to the needs of Margaux Collections, which sells both brand-new items such as bikinis, coordinated sets, and accessories, as well as secondhand items such as pants, skirts, and shorts.

The system will be developed for use by the owner and her customers in the Philippines. It will use web-based technologies such as PHP, MySQL, HTML, CSS, and JavaScript to support the system's functionality. The platform will distinguish between brandnew and preloved clothing items so that products can be categorized properly and displayed in a clear manner. This separation will help both the seller and the buyers easily identify the type and condition of the items available for purchase.

The scope of the study includes the implementation of features that support product browsing, checkout, buyerseller communication, order management, and inventory monitoring. It also includes queue management for pick-up orders and analytics to help the owner make better decisions based on sales and order data. These features are

intended to improve the overall flow of business operations and reduce the problems caused by manual and social media-based transactions.

On the other hand, this study has several delimitations. The system will not include integration with external logistics providers or third-party delivery tracking services. It will also not have advanced features such as automated size recommendations, as the focus of the study is on basic ordering and management functions. In addition, the project will prioritize web accessibility rather than the development of a mobile application. Since the business is operated solely by the owner, the system will be designed for a single-user business setup without staff handling transactions, inventory, or order processing.

The existing payment methods of Margaux Collections will remain unchanged. GCash will continue to be used for dropoff transactions, cash on delivery (COD) will be used for delivery orders, and in-person payment will be accepted for store pickup. These payment methods will stay the same because they are already familiar to the owner and convenient for the customers. By maintaining the current payment setup, the system can be easier to adopt while still improving the organization and efficiency of the business.

1.5 Significance of the Study

This study is significant because it develops an **Online Ordering System for Brand-New and Preloved Clothes with Product Listing, Checkout, and Buyer-Seller Management** that can improve the way Margaux Collections operates and serves its customers. The proposed system is designed to benefit several

stakeholders by making the ordering process more organized, efficient, and convenient. Since the business is currently managed entirely by the owner without any staff, the system is expected to reduce the amount of manual work involved in handling orders, confirming payments, updating inventory, and responding to customer inquiries. Through automation and centralized management, the business can become more efficient and easier to handle on a daily basis.

For the **Margaux Collections owner**, the system provides a practical solution to the challenges of managing a small business alone. At present, the owner manually handles customer orders through Facebook Messenger, checks payments through GCash, and keeps track of product availability on her own. This process can be time-consuming and may lead to errors, delays, or confusion, especially when multiple customers are inquiring or ordering at the same time. With the proposed system, order processing and inventory management can be automated, allowing the owner to focus more on business operations and customer satisfaction. The system also provides sales insights and analytics that can help the owner make informed decisions regarding product restocking, especially in determining which items, such as brand-new bikinis and coordinated sets or preloved pants, skirts, and shorts, are in higher demand. In this way, the system supports not only daily operations but also the longterm growth of the business.

For the **customers**, the system offers a more convenient and organized way of shopping. Instead of depending only on Facebook Messenger conversations, customers can use a dedicated web platform to browse products, view item details, and place orders more easily. This makes the shopping experience more efficient because customers no longer need to wait for repeated back-and-forth messages just to ask about product availability or order status. The system also provides a clearer payment workflow, since customers can follow the existing payment methods used by the business, such as GCash for drop-off orders, cash on delivery (COD) for delivery orders, and in-person payment

for store pickup. In addition, the system can help customers track their orders and receive queue numbers for pickup, which improves transparency and reduces confusion. Overall, the proposed system gives customers a smoother and more reliable online shopping experience.

For **future researchers and students**, this study serves as a useful reference for understanding how web-based systems can support small business operations, especially in businesses that sell both brand-new and preloved fashion items. It can be used as a case study in web development, database management, and systems analysis because it combines practical business functions with technical implementation. The study may also help future researchers who are interested in developing systems for micro enterprises, online clothing stores, or businesses that rely heavily on manual transaction handling. In addition, it provides ideas on how a single-owner business can transition from social media-based selling to a more structured digital platform. As a result, the study contributes not only to the improvement of Margaux Collections but also to the broader body of knowledge in information technology and business system development.

Overall, the significance of this study lies in its ability to improve business efficiency, reduce manual workload, enhance customer convenience, and provide academic value for future researchers. By creating a centralized system for managing brand-new and preloved clothes, the study addresses the actual needs of a small business while also showing how technology can support growth and better decision-making. It demonstrates that even a small, owner-managed business can benefit from a well-designed online ordering system that is tailored to its specific operations and payment practices.

1.6 Definition of Terms

Margaux Collections Ordering System - A web-based platform designed for Margaux Collections that allows product browsing, ordering, queue management, and sales analytics. The system is intended to support the business by organizing both brand-new and preloved clothing items in one centralized platform.

Brand-new - Items that are new, unused, and sold in their original condition. In this study, brand-new items include bikinis, coordinated sets, and accessories.

Preloved - Apparel that has been previously owned but is still in good wearable condition. In this study, preloved items include pants, skirts, and shorts.

Queue Number - An automated number assigned to each order to help organize processing and pickup. It is used to keep transactions in proper order, especially for drop-off locations and store pickups.

Middleware - PHP-based components that manage communication between the frontend and backend of the system. These components handle authentication, sessions, authorization, and data validation to ensure that the system operates properly and securely.

API - PHP interfaces that support system functions such as cart operations, order updates, payment status processing, and real-time dashboard information. These interfaces allow different parts of the system to exchange data efficiently.

Database Gateway - The MySQL layer responsible for storing and retrieving data related to users, products, orders, payments, and transactions. It serves as the connection between the system's interface and its database records.

GCash - A digital wallet used for advance payment in dropoff orders. Customers pay through GCash before the owner delivers the order to the designated drop-off store.

Cash on Delivery (COD) - A payment method in which customers pay in cash when their order is delivered. This method is used for delivery transactions in the business.

Drop-off Location - A specific store or address where the owner delivers orders for customers to pick up after payment through GCash.

Facebook Posts - Product listings published by the owner on her Facebook page, which are currently used as the main way of displaying available items to customers.

Messenger - The Facebook Messenger platform used by customers to communicate with the owner, ask about products, and place orders.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Literature

This chapter presents the related literature and studies that serve as the foundation of the Margaux Collections

Ordering System. It draws from local and foreign sources to discuss concepts, theories, findings, and practical applications relevant to the development of the proposed system. The review focuses on second-hand fashion, digital ordering, consumer behavior, inventory management, sustainable business practices, and the use of information systems in small enterprises. Through these studies, the chapter establishes the academic and practical basis for creating a more organized and efficient platform for managing the products and transactions of Margaux Collections.

The Mesh: Why the Future of Business Is Sharing Gansky, 2010

Gansky (2010) discussed the concept of the mesh as a business model in which sharing, access, and connection become more important than ownership alone. Her work explains how modern business practices are changing as consumers become more open to using products through accessbased systems rather than depending only on direct ownership. This idea is important in the context of fashion because it shows that products can continue to have value even after they have been used once. Instead of being treated as disposable, items can move through different users and still remain useful and desirable. This perspective is highly relevant to preloved clothing because it supports the idea that second-hand garments are still part of a valuable product cycle.

The study also emphasizes that sharing-based business models depend on trust, convenience, and organized access. Consumers need to feel that the platform is reliable and easy to use before they will participate in shared or reused product systems. In this sense, the mesh is not only about product circulation but also about the structure that makes circulation possible. A well-organized digital platform helps consumers find items, understand their value, and complete transactions with confidence. This makes the idea useful for understanding how an online shopping environment can encourage customers to consider preloved clothes as practical and appealing choices.

For Margaux Collections, this study is relevant because the business sells both brand-new and preloved clothing items. The mesh concept helps support the inclusion of second-hand products in the same ordering system as new products because it treats both as part of a continuing value chain. It also suggests that digital platforms can help sellers organize products in ways that make them easier to access and purchase. In a small fashion business, this means that the system should not only display products but also help manage how they move from seller to customer. This supports a more flexible and sustainable approach to clothing retail.

Overall, Gansky's discussion provides a useful theoretical basis for the Margaux Collections Ordering System because it shows that clothing can continue to generate value through reuse, circulation, and digital access. The study supports the idea that preloved fashion should be presented as a legitimate and useful retail category rather than as a lesser option. It also reinforces the importance of having a structured online system that makes product access, ordering, and resale more efficient. In this way, the literature helps justify the development of a platform that supports both brand-new and preloved clothing in a shared digital marketplace.

The Economics of Privacy Acquisti, Taylor, and Wagman, 2016

Acquisti, Taylor, and Wagman (2016) examined the economics of privacy and explained how people often share personal information online without fully understanding the possible risks involved. Their study is important because privacy has become a major concern in digital transactions, especially in systems that require users to create accounts, provide contact information, and complete purchases online. The authors show that users do not always make privacy decisions with complete awareness, which makes careful system design very important. In this sense, the

study is relevant to any online ordering system that handles personal and transactional data. It helps explain why trust and security are essential in e-commerce platforms.

The study further emphasizes that privacy concerns affect user behavior and decision-making. When people believe that their information may be exposed or misused, they are more likely to hesitate before registering or making a transaction. This means that privacy is not only a legal or technical issue but also a customer experience issue. In online retail, users want reassurance that their details are protected and that the system is reliable. The study therefore shows that protecting privacy is closely connected to encouraging user participation and maintaining confidence in the platform.

For Margaux Collections, this literature is highly relevant because the Ordering System will likely require customers to enter names, contact numbers, addresses, and other transaction-related details. Since the business will operate online, it must make sure that customer information is stored and handled responsibly. A secure login process, proper account management, and controlled access to customer records are important features that can help build user trust. The study supports the idea that an online clothing platform must protect buyer information just as carefully as it manages product listings and payments. This makes privacy a necessary part of the system's design and operation.

Overall, Acquisti, Taylor, and Wagman (2016) provide a strong basis for understanding why online business systems must address privacy concerns in order to succeed. Their work shows that consumers are more comfortable using digital platforms when they feel their information is safe and their personal data is not being misused. For Margaux Collections, this means that the Ordering System should not only be functional and organized but also secure and trustworthy. By giving attention to privacy, the system can

improve customer confidence and encourage more successful online transactions. In this way, the literature supports both the technical and relational needs of the proposed platform.

A Study on Relevance of Second-Hand Clothing Retailing in Ethiopia Khurana and Tadesse, 2019

Khurana and Tadesse (2019) studied the relevance of secondhand clothing retailing in Ethiopia and showed that used clothing continues to play an important role in consumer markets. Their study is useful because it confirms that second-hand fashion is not simply an informal or marginal practice, but a legitimate retail activity with continuing consumer demand. The authors highlighted that second-hand clothing remains relevant due to its affordability, availability, and usefulness for a wide range of buyers.

This makes the study important for understanding how preloved fashion can remain a practical and valuable part of modern retail. In this sense, second-hand clothing should be viewed as a meaningful product category rather than a lesser alternative to brand-new items.

The study also suggests that second-hand clothing retailing supports consumer access to fashion at lower prices while also promoting reuse. This is important because many consumers choose used clothing not only because it is cheaper, but also because it offers them access to styles they may not otherwise afford. The authors' discussion shows that second-hand clothing can serve both economic and social needs by making fashion more accessible to different market groups. It also reinforces the idea that preloved garments still have worth when they are presented in a way that customers can understand and trust. This makes clear product information and organized presentation especially important in the online context.

For Margaux Collections, this study is directly relevant because the business includes preloved clothing as part of its product offerings. The findings support the idea that

second-hand items should be given proper attention in the Ordering System and should not be treated as secondary products. Instead, they should be displayed clearly, categorized properly, and priced in a way that reflects their condition and value. The system can help buyers see preloved clothes as worthwhile purchases by providing enough product detail and making browsing easier. This is especially useful for a clothing business that wants to present both brand-new and preloved items in one organized platform.

Overall, Khurana and Tadesse (2019) provide strong support for the Margaux Collections Ordering System because they show that second-hand clothing retailing has real market value and consumer relevance. Their study helps justify the inclusion of preloved items in a professional online system and supports the idea that resale can be both practical and sustainable. For Margaux Collections, this means the ordering system should help customers browse, compare, and purchase preloved items with confidence. The literature therefore reinforces the importance of structured presentation, accessibility, and value communication in the proposed platform.

Fair Data Is the New Black Vänskä et al., 2024

Vänskä et al. (2024) explored how online shopping, data leaks, and sustainable fashion are connected through the idea of fair data. Their study is important because it shows that digital commerce is not only about buying and selling products, but also about how responsibly a platform handles user information. The authors connect sustainability with trust, suggesting that ethical fashion consumption now includes concern for privacy and digital responsibility. This is highly relevant in an era where consumers are increasingly aware of how their personal information is collected and used. The study therefore broadens the understanding of sustainable fashion beyond clothing alone and into the digital systems that support it.

The study also implies that a sustainable fashion platform must be transparent and reliable in its handling of both products and data. If customers feel that a website is careless with their information, they may become less willing to shop online, even if they support the business's products. This means that user trust is part of sustainability because it affects whether consumers feel comfortable participating in digital fashion commerce. The idea of fair data also suggests that businesses should be honest and responsible in the way they present products and interact with buyers. As a result, system design becomes closely tied to ethical business practice.

For Margaux Collections, this study is useful because the business sells both brand-new and preloved clothing through an online platform that will likely collect customer information during ordering and checkout. The study supports the need for a secure and well-organized system that protects users while also presenting products clearly and professionally. Since preloved clothing is often associated with sustainability, the platform should reflect that same responsibility in how it handles data and transactions. This means the ordering system should not only support buying and selling but also create an environment that feels safe and ethically responsible for customers. In this way, the literature reinforces the importance of combining sustainability with digital trust.

Overall, Vänskä et al. (2024) provide a useful perspective for the Margaux Collections Ordering System because they connect online shopping behavior with privacy, data protection, and sustainable fashion values. Their study supports the idea that a fashion platform should be both environmentally conscious and digitally trustworthy. For Margaux Collections, this means the system should help users browse, order, and communicate with confidence while also maintaining responsible handling of customer data. The literature therefore strengthens the argument that modern fashion retail must address both sustainability and privacy

in its design. This makes the study highly relevant to the proposed online ordering system.

Consumption Corridors in Fashion Vladimirova, 2021

Vladimirova (2021) discussed consumption corridors in fashion and examined the idea of upper consumption limits in minimalist fashion challenges. Her study is relevant because it questions excessive consumption and encourages more intentional buying behavior. The author argues that consumers do not need to completely reject fashion in order to be responsible; instead, they can make more mindful choices about what they purchase and how often they buy. This makes the study important for understanding the role of fashion in sustainable living. In this way, the literature presents fashion consumption as something that can be managed more responsibly rather than simply expanded without limit.

The study further explains that consumption corridors help people balance personal style with environmental responsibility. Rather than encouraging total abstinence from fashion, the concept supports measured and thoughtful purchasing that respects both consumer needs and ecological concerns. This is important because it shows that consumers can still express themselves through clothing while avoiding wasteful habits. It also reinforces the idea that preloved fashion can serve as a practical alternative to buying new items all the time. By choosing second-hand clothing, consumers participate in a more sustainable and less wasteful form of fashion consumption.

For Margaux Collections, this study is useful because it supports the inclusion of preloved clothing as part of a responsible retail model. The business can use its Ordering System to present both brand-new and preloved items in a way that encourages thoughtful purchasing. When product details, item conditions, and prices are clearly displayed,

customers can make informed decisions that align with both style and sustainability. The system can therefore help buyers see preloved clothes as valuable options rather than temporary

or low-status alternatives. This makes the study relevant to both product presentation and sustainable retail practice.

Overall, Vladimirova (2021) provides a strong theoretical basis for the Margaux Collections Ordering System because it connects fashion consumption with sustainability, restraint, and informed decision-making. The study supports the idea that clothing retail can promote more responsible consumer behavior when products are presented clearly and thoughtfully. For Margaux Collections, this means the ordering system should help customers browse and choose items in a way that supports both convenience and conscious consumption. The literature therefore strengthens the argument that preloved fashion has an important place in modern online retail.

Used-Clothing Donations and Apparel Production in Africa Frazer, 2008

Frazer (2008) examined used-clothing donations and apparel production in Africa and showed how second-hand garments are connected to broader systems of trade, reuse, and consumption. His study is important because it explains that used clothing is not merely discarded material, but part of an active flow of goods that continues to create value after first use. The author's discussion highlights how donated and reused clothing can affect local markets, consumer choices, and production systems. This makes the study useful for understanding the wider social and economic role of preloved garments. In this sense, second-hand clothing should be seen as a legitimate and meaningful product category within fashion retail.

The study further suggests that used-clothing circulation can create opportunities for both sellers and buyers. Sellers can recover value from items they no longer need, while buyers gain access to affordable clothing options that remain useful and wearable. This cycle of reuse helps reduce waste while also supporting more practical forms of consumption. It shows that clothing can continue to serve a purpose even after it has been passed on from one owner to another. For retail systems that include preloved products, this means that the process of buying and selling secondhand items can contribute to both economic activity and sustainability.

For Margaux Collections, this literature is relevant because the business includes preloved clothes as part of its offerings. The study supports the idea that these items should be organized and presented as valuable products within the Ordering System rather than as leftover stock. A structured online platform can help the business manage listings, track product availability, and present secondhand garments in a more professional way. It can also help customers understand that preloved clothes still have practical and commercial worth. This is important because the business aims to serve buyers who are looking for both affordability and style.

Overall, Frazer (2008) provides useful support for the Margaux Collections Ordering System because it shows that second-hand clothing belongs to a larger system of reuse that benefits both consumers and sellers. The study helps justify the inclusion of preloved garments in an online retail platform and strengthens the argument that resale is a meaningful form of clothing circulation. For Margaux Collections, this means the ordering system should help make second-hand items easy to browse, understand, and purchase. The literature therefore supports the idea that preloved fashion can be both sustainable and commercially valuable when organized properly.

**I Rent, Swap or Buy Second-Hand Brand, Jacobs, and
TaljaardSwart, 2023**

Brand, Jacobs, and
Taljaard-Swart
(2023) examined the
reasons consumers
rent, swap, or buy
second-hand clothing
online. Their study
is relevant because
it shows that
second-hand clothing
consumption is not
driven by only one
motivation, but by a
combination of
practical, social,
and environmental
factors. The authors
explain that people
engage in
collaborative
clothing consumption
for reasons such as
convenience,
affordability,
sustainability, and
variety. This makes
the study important
for understanding
why consumers may
choose preloved
fashion instead of
brandnew clothing.
In this sense,

second-hand fashion
is presented as a
flexible and
meaningful part of
modern consumer
behavior.

The study also emphasizes that the online environment plays an important role in shaping consumer participation in collaborative clothing models. Buyers are more likely to engage in renting, swapping, or purchasing second-hand items when the system is easy to use and the product information is clear. This means that platform design influences not only access to products but also customer willingness to participate in second-hand fashion markets. The authors' findings show that digital convenience can help normalize alternative fashion behaviors and make them more attractive to consumers. Clear presentation, reliable communication, and organized product access all contribute to this positive experience.

For Margaux Collections, this study is relevant because it supports the idea that customers may be open to both brandnew and preloved clothing when the system makes product browsing and ordering simple. The Ordering System can help by presenting products clearly and allowing customers to compare items quickly. Since the business offers clothing that serves different customer needs, the platform should support a smooth and trustworthy shopping experience. This study also supports the importance of designing the system in a way that makes second-hand products feel accessible and legitimate. In this way, the literature aligns with the goal of combining convenience, value, and sustainability in one platform.

Overall, Brand, Jacobs, and Taljaard-Swart (2023) provide useful support for the Margaux Collections Ordering System because they show that consumers are willing to participate in second-hand clothing markets when the process is convenient and understandable. Their study helps explain

why an online platform must be user-friendly, organized, and informative in order to encourage purchases. For Margaux Collections, this means that the ordering system should not only manage transactions but also make preloved clothing feel practical and appealing. The literature therefore supports the development of a digital clothing platform that is efficient, accessible, and responsive to customer behavior.

The International Trade in Secondhand Clothing Abimbola, 2012

Abimbola (2012) studied the international trade in secondhand clothing and focused on the problem of information asymmetry between West African and British traders. The study is important because it shows that the second-hand clothing market depends heavily on the quality, accuracy, and completeness of information shared between buyers and sellers. When one side knows more than the other, trust becomes harder to establish and transactions become more difficult to manage. This makes the study highly relevant to online clothing retail, where customers often cannot inspect items physically before purchasing. In this sense, transparency becomes a major factor in second-hand fashion markets.

The study further explains that information asymmetry creates uncertainty in resale markets, especially when product details are incomplete or unclear. Buyers want to know exactly what they are purchasing, while sellers need to present items in ways that make them appear trustworthy and worthwhile. The author's discussion shows that second-hand clothing works best when product information is shared clearly and consistently. This includes accurate descriptions, item condition, and other details that reduce doubt. When information is incomplete, buyers may hesitate or avoid completing the transaction.

For Margaux Collections, this study is directly relevant because the business sells preloved clothing that customers will only be able to evaluate through the online ordering system. Since buyers cannot physically inspect the items, the system must provide enough information to reduce uncertainty and build trust. Detailed product listings, clear condition labels, and organized categories can help make preloved clothes more appealing and dependable in the eyes of the customer. The Ordering System can therefore serve as a tool for improving transparency and reducing the gap between seller knowledge and buyer expectations. This is especially important for maintaining confidence in the purchasing process.

Overall, Abimbola (2012) provides strong support for the Margaux Collections Ordering System because it shows that second-hand clothing retail depends heavily on clear communication and reliable product information. The study helps justify the need for an organized digital platform where customers can understand what they are buying before they place an order. For Margaux Collections, this means the system should be designed to reduce uncertainty and support fair, informed transactions. The literature therefore reinforces the importance of transparency, trust, and product clarity in online preloved fashion retail.

The Work of Shopping: Resellers and the Informal Economy at the Goodwill Bins Ayres, 2019

Ayres (2019) explored the work of shopping and the role of resellers in the informal economy at the goodwill bins. Her study is relevant because it shows that resale activity involves more than simply buying and selling old items; it also requires judgment, effort, and knowledge about product quality and value. The author presents reselling as a form of work that involves careful selection, evaluation, and

presentation of goods. This makes the study useful for understanding second-hand fashion as a serious economic activity rather than an informal leftover market. In this sense, the resale process becomes part of a broader business practice that generates value for both sellers and buyers.

The study further explains that resellers must recognize which items are worth keeping, pricing, and offering to customers. This means that second-hand retail depends on skills such as product assessment, trend awareness, and customer understanding. Ayres shows that the process of finding valuable used items takes time and effort, which makes resale an active and meaningful part of the informal economy. It also suggests that the way products are selected and displayed can strongly affect whether buyers will be interested. This is important because second-hand fashion is not only about availability but also about how items are curated and presented.

For Margaux Collections, this study is relevant because the business also deals with preloved clothing that must be carefully organized and presented online. The Ordering System can support this by making it easier to list products, track inventory, and present items in a more professional way. Since the owner must manage both brand-new and preloved items, the system can help reduce manual work and make the resale process more structured. It also allows the business to reflect the effort involved in selecting and offering second-hand items. In this way, the study supports the value of having a system that helps turn resale activity into a more organized business operation.

Overall, Ayres (2019) provides useful support for the Margaux Collections Ordering System because it shows that second-hand retail depends on careful product selection, organization, and seller judgment. The study helps explain why preloved items should be managed in a structured way rather than treated casually. For Margaux Collections, this means the ordering system should help the owner organize

products clearly and present them in a way that builds customer confidence. The literature therefore reinforces the importance of professionalism, curation, and efficient management in an online preloved clothing platform.

**Overdressed: The Shockingly High Cost of Cheap Fashion
Cline, 2013**

Cline (2013) examined the hidden costs of cheap fashion and argued that low-priced clothing often carries social and environmental consequences that are not immediately visible to consumers. Her work is important because it challenges the assumption that inexpensive clothing is always the best or most practical choice. Instead, the author shows that fast fashion can encourage waste, overconsumption, and a reduced appreciation for the true value of clothing. This makes the study highly relevant to discussions about sustainable fashion and second-hand retail. In this sense, clothing should be understood not only as a consumer product but also as part of a larger system of responsibility and impact.

The study further explains that cheap fashion often leads to faster turnover of garments, which contributes to textile waste and weakens the long-term value of clothing. Cline's discussion encourages consumers to think more carefully about what they buy and how often they replace items. This is important because it shows that fashion decisions are tied to sustainability, quality, and long-term use. When consumers choose cheaper products without considering these factors, they may end up supporting practices that are harmful to both society and the environment. The book therefore provides a strong argument for more thoughtful and responsible clothing consumption.

For Margaux Collections, this study is relevant because the business includes preloved clothing in its product offerings, giving customers an alternative to fast fashion. The Ordering System can help present preloved items as practical, stylish, and responsible choices rather than as lower-quality substitutes. By showing product details, item condition, and price clearly, the system can help customers recognize the value of second-hand clothing. This supports the idea that preloved fashion can offer affordability without encouraging wasteful buying habits. In this way, the study aligns closely with the goal of promoting both business convenience and more sustainable consumer behavior.

Overall, Cline (2013) provides strong support for the Margaux Collections Ordering System because it highlights the broader consequences of cheap fashion and the importance of valuing clothing more carefully. The study helps justify the inclusion of preloved items as part of a more responsible retail model. For Margaux Collections, this means the system should not only facilitate sales but also encourage customers to view clothing as something with lasting value. The literature therefore reinforces the importance of sustainability, product value, and mindful consumption in the proposed online ordering platform.

**"Mix-and-Match" Fashion Trend and Luxury Brand Recognition
Amatulli et al., 2016**

Amatulli et al. (2016) investigated the relationship between fashion trends and luxury brand recognition using eyetracking methods. Their study is relevant because it shows that visual presentation strongly influences how consumers notice, evaluate, and remember fashion items. The authors demonstrate that consumers often respond first to what they see before they fully assess product value or quality. This makes visual design an important factor in fashion retail, especially in online shopping where customers rely heavily on images. In this sense, the way a

product is displayed can affect whether it attracts attention and creates a positive impression.

The study further suggests that visual cues play a major role in shaping consumer interest and brand perception. When items are arranged clearly and attractively, consumers are more likely to focus on them and associate them with quality or desirability. This is especially important in fashion because product appearance is often tied to style, identity, and personal preference. Amatulli et al. show that visual recognition is not accidental but influenced by how the product is framed in the consumer's view. This finding is useful for understanding how product display can affect shopping behavior and decision-making.

For Margaux Collections, this study is highly relevant because the business will present clothing items through an online ordering system where customers cannot physically examine the products. This means that images, layout, and display quality become essential in helping buyers evaluate both brand-new and preloved items. The Ordering System should therefore present products in a clean, attractive, and consistent manner so that customers can quickly understand what is being offered. Good presentation can make preloved items look more desirable and professional, which may increase customer confidence and interest. In this way, the study supports the importance of product visualization in online fashion sales.

Overall, Amatulli et al. (2016) provide useful support for the Margaux Collections Ordering System because they show that visual design strongly affects consumer attention and product perception. The study helps explain why online clothing platforms must focus not only on product listing but also on how items are shown to buyers. For Margaux Collections, this means the system should make clothing look organized, appealing, and easy to evaluate. The literature therefore reinforces the importance of product presentation, visual clarity, and consumer perception in

the success of the proposed platform. **Visual and Verbal Information in Internet Shopping**

Kim and Lennon (2008) examined how visual and verbal information affects attitudes and purchase intentions in internet shopping. Their study showed that product images and written descriptions both play an important role in shaping how consumers evaluate items online. This is useful for an online ordering system because it supports the need for clear and attractive product listings. When buyers can see good photos and read detailed descriptions, they are more likely to trust the product and consider purchasing it.

The study also suggests that online shoppers depend heavily on the quality of information presented on the website. In internet shopping, consumers cannot personally inspect the items before buying, so they rely on what they see and read on the screen. Because of this, the way products are displayed becomes an important part of the buying process. A clothing platform that provides complete and organized information can make shopping easier and more comfortable for users.

Another important point in the study is that visual and verbal cues work together to influence consumer attitudes. Product images help buyers quickly notice the style and appearance of an item, while written descriptions give more details about its features and condition. When these two forms of information are combined well, they can create a stronger impression of the product. This means that online sellers should pay close attention to how they present each clothing item.

The findings are highly relevant to an online ordering system for brand-new and preloved clothes. Since clothing is a product that depends greatly on appearance, buyers need clear photos and accurate descriptions before deciding to buy. If the platform presents items in a professional and easy-to-understand way, users may feel more confident in completing their orders. This shows that product listing

quality is one of the most important parts of online selling.

The study also highlights the role of trust in online shopping. When customers feel that the information on a website is complete and reliable, they are more likely to believe that the seller is also trustworthy. This is especially important in clothing marketplaces where buyers often want to know the actual condition, size, and style of the item before making a purchase. Good product presentation can reduce uncertainty and help buyers feel safer during the decision-making process.

In addition, clear visual and written information can improve the overall shopping experience. A platform that organizes product details well allows users to compare items more easily and make quicker decisions. This is helpful for both brand-new and preloved clothing because customers often need to review several options before choosing one. When product pages are simple, informative, and appealing, they support smoother browsing and better buying outcomes.

Overall, Kim and Lennon (2008) provide useful support for the development of an online clothing marketplace. Their study explains that consumer decisions are shaped by the information provided during online shopping. For this reason, a system that offers detailed descriptions and strong product images can help improve buyer trust and purchase intention. The study therefore supports the need for an organized and informative platform in online retail.

Consumers' Intention to Buy Second-Hand Clothing

Koay, Cheah, and Lom (2022) studied the factors that influence consumers' intention to buy second-hand clothing. Their research explained that purchasing decisions are shaped not only by the availability of products, but also

by consumer attitude, social influence, and perceived behavioral control. These factors are important because they help explain why some buyers are willing to choose preloved clothing while others remain hesitant. In an online marketplace, this suggests that consumer confidence is a major part of the buying process.

The study also shows that a positive attitude toward secondhand clothing can increase the likelihood of purchase. When buyers see preloved items as practical, affordable, and useful, they are more open to considering them as good choices. Social influence also affects decision-making because consumers are often encouraged by friends, online communities, and current trends. This means that an online platform should not only present products well, but also help create a supportive buying environment.

Another important idea in the study is perceived behavioral control, which refers to how easy or difficult consumers feel the buying process is. If buyers believe that purchasing second-hand clothing online is simple, secure, and convenient, they are more likely to continue with the transaction. On the other hand, if the process appears confusing or unreliable, they may decide not to buy. This highlights the importance of easy navigation, clear product details, and a smooth checkout system.

The study is also useful in explaining the behavior of preloved clothing shoppers in an online setting. Since buyers cannot inspect the items personally before ordering, they depend heavily on the information shown on the website. For this reason, accurate descriptions, clear photos, and transparent seller information are very important. These features can reduce uncertainty and help build trust between buyers and sellers.

Overall, Koay, Cheah, and Lom (2022) provide strong support for an online ordering system for preloved clothing. Their findings show that consumer behavior is influenced by trust, social factors, and the ease of the purchasing process. This makes their study relevant to a platform that

offers clear listings, reliable information, and efficient checkout functions. It helps explain why an effective online marketplace must address both product presentation and buyer confidence.

Mindful Consumption of Second-Hand Clothing

Mohammad, Quoquab, and Mohamed Sodom (2021) explored mindful consumption in second-hand clothing and examined the role of electronic word-of-mouth, consumer attitude, and engagement. Their study showed that online reviews and customer interactions strongly influence how buyers evaluate secondhand fashion products. This is important for an online clothing platform because shoppers often depend on the experiences and opinions of others before deciding to purchase. In an online setting, trust becomes a major factor because buyers usually cannot inspect preloved items personally before ordering.

The study also shows that electronic word-of-mouth can shape consumer judgment in a meaningful way. When buyers read positive reviews or see encouraging feedback from other users, they are more likely to view the product as reliable and worth buying. Negative comments, on the other hand, may reduce confidence and discourage a purchase. This means that product ratings and written feedback are useful tools for helping users make informed decisions. For an online marketplace, these features can improve transparency and reduce uncertainty.

Another important idea in the study is consumer engagement. Engagement allows buyers and sellers to interact more actively, which can make the shopping experience feel more personal and dependable. In a clothing marketplace, this may include comments, questions, seller responses, and item feedback. These interactions help users feel that the platform is responsive and trustworthy. As a result, buyers may become more comfortable completing their transactions.

The findings are highly relevant to an online ordering system for brand-new and preloved clothes. Since second-hand clothing depends greatly on trust and product condition, buyers need tools that help them evaluate items before buying. Ratings, comments, and feedback systems can support this need by giving customers more information about the seller and the product. These features can also encourage better communication between users, which strengthens the overall marketplace experience.

Overall, Mohammad, Quoquab, and Mohamed Sadom (2021) provide useful support for the inclusion of interactive features in an online clothing platform. Their study shows that eWOM, attitude, and engagement affect how consumers judge and purchase second-hand clothing. This makes their work relevant to a system that aims to connect buyers and sellers in a clear and trustworthy way. It also supports the idea that online marketplaces should include tools that encourage feedback, communication, and user participation.

Secondhand Clothing Purchase Intentions: Generation Z's Perspective

Herjanto, Amin, Purinton, and Lindle II (2024) studied secondhand clothing purchase intentions among Generation Z consumers. Their research focused on how younger buyers view preloved clothing and what factors influence their willingness to buy these items online. The study found that Generation Z is more open to secondhand fashion when the products are seen as valuable, stylish, and connected to sustainable consumption. This makes the study highly relevant to an online ordering system that offers both brand-new and preloved clothing.

The findings show that secondhand clothing is not always viewed as an undesirable option by younger consumers. Instead, many Generation Z buyers see preloved fashion as a practical and meaningful choice because it can provide good value at a lower price. At the same time, they are also interested in clothing that still looks fashionable and

fits their personal style. This suggests that secondhand items

can appeal to buyers who want to save money without giving up appearance or quality.

The study is also important because Generation Z is one of the largest and most active groups of online shoppers. Since this generation is already familiar with digital platforms, they are more likely to respond positively to online clothing marketplaces that are simple, organized, and convenient to use. An online ordering system with easy browsing, clear product categories, and a smooth checkout process can match their shopping habits well. This makes the study useful in supporting the design of a platform that serves both new and preloved items.

Another key point from the study is the role of sustainability in consumer behavior. Many young buyers are now more aware of environmental issues and prefer choices that support responsible consumption. Secondhand clothing fits this preference because it encourages reuse and helps reduce waste. For this reason, an online marketplace for preloved clothes can attract not only budget-conscious buyers but also those who value eco-friendly shopping. This shows that secondhand fashion has strong potential in modern online retail.

The study also helps explain why preloved clothing continues to gain popularity in digital markets. Younger consumers often look for products that reflect both practicality and personal values. When an online platform presents secondhand clothing in a clear and appealing way, it becomes easier for buyers to see its benefits. Features such as attractive photos, honest descriptions, and reliable transaction processes can further increase buyer confidence. These elements are important because they help make the shopping experience more trustworthy and user-friendly.

Overall, Herjanto, Amin, Purinton, and Lindle II (2024) provide useful insight into the behavior of young consumers in secondhand clothing shopping. Their findings show that Generation Z responds well to products that combine value, style, and sustainability. This is relevant to your project because it shows that an online ordering system can attract younger buyers through a well-organized and efficient platform. The study also supports the idea that preloved clothing has strong growth potential in online fashion markets.

E-commerce and Fashion Shopping Behavior

Asong and Khan (2021) examined the impact of e-commerce on physical stores in the fashion industry and grocery sector. Their study discussed how shopping behavior has gradually shifted from offline stores to online platforms as digital commerce has become more common. This is useful for a literature review because it shows that consumers are becoming more comfortable with online purchasing, including fashion-related products. Their findings help explain why online ordering systems continue to grow in importance in the retail industry.

The study shows that the rise of e-commerce has changed the way consumers shop for goods in a major way. Instead of depending only on physical stores, many buyers now prefer the convenience of browsing and purchasing products through online platforms. This shift is especially important in the fashion industry, where customers often look for faster, easier, and more flexible shopping options. As online shopping becomes more familiar, consumers are more likely to trust digital platforms for clothing purchases and to consider them a practical alternative to traditional stores.

Another important point from the study is that online systems must be efficient and easy to use in order to remain competitive. Consumers are more likely to continue shopping online when the platform is accessible, organized, and convenient. If the website is difficult to navigate or

the ordering process is confusing, users may quickly move to another platform. This means that an effective online clothing marketplace should focus on user experience, clear product presentation, and simple transaction steps that make buying easier for customers.

The study is also relevant because it provides a broader view of how digital shopping affects the fashion industry as a whole. As more consumers move between physical and online stores, businesses must adjust their selling methods to match changing customer behavior. This supports the need for an online ordering system that can serve modern shoppers who value speed, convenience, and accessibility. In this way, the study helps explain why fashion-related e-commerce platforms continue to expand and why businesses are encouraged to improve their digital services.

In addition, the findings suggest that online shopping is no longer viewed as a new or unusual practice, but as a normal part of consumer life. This change is important because it shows that buyers are becoming more confident in using

online systems for everyday purchases, including clothing. For an online ordering system that handles brand-new and preloved clothes, this means the platform should be designed to meet common user expectations. Features such as easy browsing, clear categories, secure checkout, and responsive pages can help maintain customer interest and improve overall satisfaction.

The study also implies that clothing businesses need to keep up with the changing preferences of online consumers. Buyers today often expect quick access to product information, simple ordering procedures, and a smooth shopping process from start to finish. If these needs are not met, customers may become less interested or may choose another seller. Because of this, an online clothing marketplace should not only display products well but should also support a fast and reliable transaction flow. These qualities are important in keeping the platform competitive in a growing digital market.

Overall, Asong and Khan (2021) provide useful insight into the growing influence of e-commerce on shopping behavior. Their findings show that online shopping is becoming a normal and expected part of consumer life, especially in fashion retail. This makes their study relevant to your research because it supports the development of a system that can meet the needs of today's online buyers. It also shows that an online ordering platform must be practical, user-friendly, and responsive to consumer preferences in order to remain effective and useful.

Foreign Studies

Foreign studies provide empirical evidence that supports the design and purpose of the proposed system. These studies are especially valuable because they show how consumers and businesses actually behave in real settings, which helps justify the need for features such as product listings, inventory control, transparent information, and digital trust-building. In the context of Margaux Collections, the foreign studies reviewed here explain why product presentation, consumer confidence, sustainability, and system organization are essential in a digital fashion retail environment. They also demonstrate that second-hand fashion is influenced by emotional, practical, and environmental considerations that must be reflected in the design of the ordering system.

Market Prices of Remanufactured, Used, and New Items Neto, Bloemhof, and Corbett, 2016

Neto, Bloemhof, and Corbett (2016) conducted a study that examined the market prices of remanufactured, used, and new items using eBay data. Their research focused on how consumers value products in different conditions and how pricing reflects market demand. The study is useful because it shows that used products have their own market value and are not automatically considered inferior to new items.

This is relevant to Margaux Collections because it supports the idea that preloved items should be priced and presented in a way that reflects their actual worth.

The study found that used items usually fall between new and remanufactured products in terms of price. This suggests that buyers evaluate second-hand goods by balancing affordability and condition rather than by price alone. For Margaux Collections, this means that preloved items should be priced transparently and fairly based on their condition, appeal, and remaining value. Clear pricing can help customers see preloved products as practical and worthwhile purchases instead of simply cheaper alternatives.

Neto, Bloemhof, and Corbett (2016) also showed that product presentation affects how consumers judge value. When descriptions are clear and pricing is consistent, buyers are more likely to trust the seller and consider the product a good deal. This supports the need for detailed product information, condition labels, and organized listings in the Margaux Collections Ordering System. These features can help reduce hesitation and make customers more confident in their purchase decisions.

The study further implies that transparent pricing practices can strengthen customer trust and support repeat purchases. When consumers understand how an item is valued, they are more likely to feel comfortable buying from the same seller again. For Margaux Collections, this means that the ordering system should display prices clearly and consistently so that buyers can quickly understand the value of each item. In this way, the system supports both professionalism and customer confidence.

Consumers' Value and Risk Perceptions of Circular Fashion

Kim, Jung, and Lee, 2021

Kim, Jung, and Lee (2021) conducted a quantitative study to examine consumers' value and risk perceptions toward circular fashion. Their research focused on how factors such as perceived quality, emotional value, social value, environmental value, economic value, and perceived risk influence consumers' intention to purchase circular fashion products. The study is useful because it shows that secondhand and preloved fashion are not evaluated only by price, but also by the meanings and benefits consumers associate with them. This makes the study relevant to Margaux Collections because it supports the idea that preloved items can be presented as valuable fashion products rather than merely used goods.

The study found that perceived quality, emotional value, social value, and environmental value significantly affect purchase intention. This means that consumers are more likely to buy circular fashion products when they believe the items are stylish, meaningful, and aligned with sustainable consumption. The findings are important for Margaux Collections because they suggest that preloved items should be displayed in a way that enhances their attractiveness and desirability. If the ordering system presents products professionally, customers may see them as worthwhile and fashionable purchases rather than second-hand alternatives.

Kim, Jung, and Lee (2021) also emphasized the role of trust and complete product information in reducing consumer hesitation. Since buyers cannot inspect preloved items physically, they depend heavily on photographs, descriptions, and condition details before deciding to purchase. This supports the need for clear product images, accurate descriptions, and condition labels in the Margaux Collections Ordering System. These features can help reduce uncertainty and improve customer confidence. In online fashion retail, accurate information is essential because it shapes how buyers evaluate both the quality and value of an item.

The study further implies that circular fashion can succeed when businesses present products in a professional and organized way. Consumers are more likely to respond positively when they can easily understand what the item looks like, what condition it is in, and why it is valuable. For Margaux Collections, this means that the system should not only function as a sales platform but also as a presentation tool for preloved fashion. By combining clear visuals, organized listings, and informative product details, the ordering system can help make preloved items more appealing and trustworthy to customers.

Consumers' Perceptions and Purchasing Behavior for Second-Hand Luxury Goods

Tu, Hsu, and Creativani, 2022

Tu, Hsu, and Creativani (2022) conducted a quantitative study in the United States to examine consumer behavior toward second-hand luxury goods. The researchers used survey data and structural equation modeling to analyze how perceived quality, emotional value, social value, environmental value, economic value, and perceived risk influence purchase intention. Their findings are useful because they show that second-hand fashion purchases are shaped by both rational and emotional factors. This is significant because it means that consumers do not evaluate preloved products only on the basis of price. Instead, they also consider how the item makes them feel, how it reflects their identity, and whether it aligns with their values.

The study found that perceived quality, emotional value, social value, and environmental value were the strongest predictors of purchase intention. This means that consumers are more likely to buy second-hand luxury items when they believe the products are stylish, meaningful, and environmentally responsible. Price matters, but it is not the only factor. This has direct relevance to Margaux Collections because it suggests that preloved items should be presented in a way that increases their aesthetic and emotional appeal. If items are displayed professionally and described in a compelling way, they are more likely to be seen as desirable rather than merely used or discounted products. This reinforces the importance of thoughtful product presentation in the proposed ordering system.

The researchers also emphasized the importance of trust and complete product information. Since buyers cannot inspect the item physically, they depend on the seller's descriptions and photographs. When the information is accurate and detailed, customers feel more confident about making a purchase. This supports the inclusion of multiple product images, condition labels, and detailed descriptions in the Margaux Collections Ordering System. These features can reduce hesitation and improve customer satisfaction. In online second-hand retail, trust is often built through transparency, and this study shows that businesses must provide enough information for consumers to feel secure in their decisions.

Another important implication of the study is that secondhand luxury goods can carry symbolic value beyond their material condition. Consumers may see them as fashionable, exclusive, or meaningful purchases that reflect a certain lifestyle or value system. This idea is useful for

Margaux

Collections because it shows that preloved items can be positioned as curated fashion pieces rather than simply second-hand goods. If the system presents the products with consistent styling, clear categories, and attractive

visuals, it can help create a stronger sense of value and desirability. The study therefore supports not only the technical design of the system, but also the way products should be marketed within it.

Selling Second-Hand Luxury: Empowerment and Social Roles

Turunen, Cervellon, and Carey, 2020

Turunen, Cervellon, and Carey (2020) conducted a qualitative study that explored the experiences of people who sell second-hand luxury goods. Through interviews, they found

that sellers often experience a sense of empowerment because they are able to recover value from unused items and participate in a more sustainable fashion economy. This is important because it shows that selling second-hand products is not only a financial activity but also an identityforming one. The study highlights the personal and social meaning of resale, suggesting that second-hand selling can influence how individuals view their role in consumer culture and sustainability.

The study found that sellers often view themselves as curators and responsible participants in fashion circulation. They also use storytelling to make items more appealing to buyers. By sharing the background, condition, or style significance of each item, sellers create stronger connections between products and consumers. This is highly relevant to Margaux Collections because it supports the idea that product descriptions should do more than list facts; they should also help build meaning and interest. A wellwritten product description can increase a buyer's curiosity and help the item feel more valuable, even if it is preowned.

The findings also suggest that organized presentation increases professionalism and trust. Sellers who manage their items carefully and communicate clearly are more likely to attract buyer confidence. For Margaux Collections, the Ordering System can support this by helping the owner organize product information in a structured and attractive way. In this way, the system becomes a tool not only for sales but also for brand development. It allows the business to present itself as organized, thoughtful, and professional, which is especially important in a market where customers often make judgments based on the quality of product presentation.

Another useful point from the study is that second-hand selling can strengthen the seller's sense of agency. Instead of viewing resale as a simple disposal activity, sellers may see it as a meaningful business practice that combines economic, social, and environmental goals. This is relevant to small businesses like Margaux Collections because it shows that an ordering system can support not just transactions, but also a broader business identity. By making it easier to manage items, document product details, and communicate with customers, the system can help the owner engage more confidently in the selling process. This contributes to both business efficiency and owner empowerment.

Online Shopping vs. Offline Shopping: A Comparative Study

Sarkar and Das, 2017

Sarkar and Das (2017) conducted a comparative study of online and offline shopping behavior to determine which factors influence consumer preference.

Their research showed that online shopping offers clear benefits, especially in terms of convenience, broader product access, and the ability to compare options quickly. These advantages explain why many customers prefer digital platforms when the shopping experience is simple and trustworthy. The study is useful because it identifies the

strengths of online shopping that businesses can maximize when designing a digital ordering system.

The study also found that online shopping has drawbacks, especially when buyers cannot inspect products personally. Consumers may worry about quality, security, and delivery reliability. These concerns are particularly important in fashion retail because clothing fit and appearance are difficult to evaluate from images alone. This supports the need for detailed product presentation and reliable ordering features in Margaux Collections. Customers who cannot examine clothing physically are more likely to depend on the quality of the information they see online, which makes accuracy and clarity essential.

Their findings further show that trust is a major factor in online purchasing decisions. Buyers are more likely to complete a purchase when the platform appears organized, safe, and easy to use. This is why the Margaux Collections Ordering System should prioritize usability, clear navigation, and transparent transaction handling. A professional system can reduce uncertainty and make online shopping feel more dependable. In the context of a small business, this trust is especially valuable because it can help turn first-time buyers into repeat customers.

The study also highlights that convenience alone is not enough to sustain customer interest. Even when a digital platform is accessible, consumers may still hesitate if the system appears confusing or if the seller does not provide enough product information. This is particularly important for a fashion business that sells both brand-new and preloved items. The ordering system must therefore combine convenience with confidence-building features such as item descriptions, product photos, and order tracking. By doing so, Margaux Collections can create a shopping experience that is both efficient and reassuring.

Sustainable Supply Chain Management

Svensson, 2007

Svensson (2007) examined Sustainable Supply Chain Management as a conceptual framework that integrates economic, environmental, and social concerns into business operations.

The study is relevant because it explains why sustainability should be built into the structure of a business rather than treated as an afterthought. For Margaux Collections, this means that inventory and ordering systems should help support responsible use of resources and efficient product flow. Sustainability is not presented merely as an ethical ideal, but as a practical business principle that can improve long-term performance.

The study emphasized that sustainability depends on coordination and information sharing. When businesses maintain accurate records and use technology to support operations, they reduce waste and improve decision-making. This is highly applicable to small businesses with limited inventory, where errors can easily lead to overstocking or stock shortages. The Margaux Collections Ordering System addresses this need through real-time product updates and better inventory tracking. By ensuring that records remain current, the system can help the business avoid unnecessary duplication and better align product availability with demand.

Another important contribution of the study is its focus on the role of information systems in supporting sustainability. Digital tools help businesses monitor movement, improve forecasting, and strengthen transparency. These functions are essential for a preloved fashion business because they allow the owner to understand what

items are selling, which products remain available, and how the business can manage stock more responsibly. In this sense, the system contributes not only to organization but also to sustainable business practice. It supports a more efficient cycle of buying, selling, and reusing items, which is central to responsible retailing.

The study also shows that sustainability is connected to efficiency and accountability. A business that keeps accurate records is better able to make informed decisions about stock replacement, product promotion, and waste reduction. This is especially important for a small fashion business that handles unique items and limited stock. For Margaux Collections, the ordering system can help ensure that products are managed responsibly from listing to sale. As a result, sustainability becomes part of everyday operations rather than a separate concern.

Clothing Reuse and Environmental Impact

Cuc and Vidovic, 2014

Cuc and Vidovic (2014) discussed the environmental impact of clothing reuse and recycling, emphasizing that extending the life of garments can significantly reduce waste and resource consumption. Their study is important because it reinforces the sustainability value of second-hand fashion. When garments are reused instead of discarded, fewer resources are needed for new production, and textile waste is reduced. This makes second-hand fashion a practical response to environmental concerns in the clothing industry.

The study showed that even a short extension in garment life can create meaningful environmental benefits. This means that buying and selling preloved clothing is not only economically practical but also environmentally

responsible. For Margaux Collections, this supports the idea of presenting preloved items as part of a circular fashion approach. Customers should be made aware that purchasing preloved clothes helps reduce waste and gives garments a second life. The ordering system can help communicate this message by including short sustainability statements in product pages or item descriptions.

The researchers also discussed the importance of consumer education and proper garment care. By giving buyers simple care advice, businesses can help prolong the lifespan of clothing. This is relevant to the Ordering System because product pages can include care tips along with descriptions and images. In this way, the system can promote both sales and responsible use. It becomes not only a transaction platform but also a source of helpful information that encourages customers to care for their purchases properly.

Another important implication of the study is that environmental responsibility can be made more visible through retail systems. When customers understand that their purchase contributes to waste reduction, they may develop a stronger appreciation for preloved products. This can improve the business's image while also encouraging more thoughtful consumption. For Margaux Collections, this means that sustainability can be integrated into both the content and the structure of the ordering system. By doing so, the business can appeal to consumers who value both fashion and environmental responsibility.

Consumers' Intention to Buy Second-Hand Clothing

Koay, Cheah, and Lom (2022) examined the factors that influence consumers' intention to buy second-hand clothing in an online setting. Their study emphasized that purchasing decisions are not based only on product availability, but also on the consumer's attitude, social influence, and sense of control over the buying process. These elements are important because they help explain why

some buyers are willing to choose preloved clothing while others remain hesitant. In the context of an online ordering system, their findings suggest that consumer trust and confidence are major parts of successful online selling.

The study also shows that positive attitudes toward secondhand clothing can encourage consumers to consider preloved items as practical and acceptable choices. When buyers see second-hand clothes as valuable, affordable, and useful, they become more open to purchasing them online.

Social influence likewise affects decision-making, as people are often encouraged by friends, online communities, and social trends. This means that an online platform should not only display products well, but also create an environment where users feel supported in their choices. Such an environment can help increase buyer interest and improve sales performance.

Another important idea from the study is perceived behavioral control, which refers to how easy or difficult consumers feel the purchasing process is. If buyers believe that ordering second-hand clothing online is simple, secure, and convenient, they are more likely to complete a transaction. On the other hand, if the process seems confusing or unreliable, they may decide not to proceed. This is highly relevant to your system because it highlights the importance of easy navigation, clear product details, and a smooth checkout process. These features can make users feel more comfortable and willing to buy.

The study is also useful in explaining the behavioral side of preloved clothing shopping in an online environment. Unlike physical stores, online platforms depend heavily on trust, product presentation, and user experience. Buyers usually cannot inspect preloved items personally before ordering, so they rely on the information shown on the website. Because of this, accurate descriptions, clear photos, and transparent seller details become essential. Koay, Cheah, and Lom's findings support the idea that a

well-designed platform can reduce uncertainty and encourage purchase intention.

Overall, the study of Koay, Cheah, and Lom (2022) provides strong support for the development of an online ordering system for preloved clothing. Their research shows that consumer behavior is shaped by trust, social factors, and the ease of the purchasing process. These findings are directly connected to the need for a system that offers clear listings, reliable information, and efficient checkout functions. For this reason, their study is valuable in building the theoretical foundation of your research. It helps explain why an effective online marketplace must address both product presentation and buyer confidence.

Mindful Consumption of Second-Hand Clothing

Mohammad, Quoquab, and Mohamed Sadom (2021) examined mindful consumption in second-hand clothing and looked at how electronic word-of-mouth, consumer attitude, and engagement affect buying behavior. Their study showed that online opinions and customer interactions have a strong effect on how people view preloved fashion products. This is important in an online ordering system because buyers often depend on the experiences of other users before making a purchase.

Since second-hand clothing cannot always be checked physically before buying, the information shared by other consumers becomes a major guide in decision-making.

The study further suggests that reviews and feedback play an important role in building trust within online shopping environments. When buyers see positive comments, helpful ratings, and active engagement from other users, they are more likely to feel confident about the product and the seller. This makes the purchasing process smoother and more reliable. In this sense, the findings support the need for

features that allow customers to share their opinions and experiences on the platform.

Another important point from the study is that consumer engagement strengthens the relationship between buyers and the marketplace. Engagement helps users feel that they are part of a responsive and active community. In an online clothing platform, this can be shown through comment sections, seller responses, and product feedback tools. These features not only improve communication but also help reduce uncertainty during the buying process. As a result, buyers may feel more comfortable completing their transactions.

The study is also useful for explaining why interactive features are necessary in a preloved clothing marketplace. Many buyers want reassurance that the items they are ordering are worth their money and match the seller's description.

Ratings, reviews, and direct communication can help provide that reassurance. These tools allow the platform to become more transparent and user-friendly. Because of this, the study supports the inclusion of interactive functions in your proposed online ordering system.

Overall, Mohammad, Quoquab, and Mohamed Sadom (2021) provide valuable insight into consumer behavior in second-hand clothing shopping. Their findings show that attitude, eWOM, and engagement are important in shaping trust and purchase intention. This makes the study relevant to an online ordering system for brand-new and preloved clothes because it emphasizes the need for communication and feedback features. It also shows that a well-designed platform should not only display products but also encourage active buyer participation.

Secondhand Clothing Purchase Intentions: Generation Z's Perspective

Herjanto, Amin, Purinton, and Lindle II (2024) studied secondhand clothing purchase intentions among Generation Z consumers. Their research focused on how younger buyers perceive preloved clothing and what influences their willingness to purchase these items online. The study found that Generation Z is generally more open to secondhand fashion when the products are seen as valuable, stylish, and consistent with sustainable consumption. This makes the study relevant to an online ordering system that includes both brand-new and preloved clothing.

The findings show that younger consumers do not always view secondhand clothing as a less desirable option. Instead, many of them consider it a practical and meaningful choice, especially when the items are affordable and still fashionable. This suggests that preloved clothing can be attractive to buyers who want to save money while still expressing personal style. In an online platform, this means that product presentation should highlight both quality and value. Clear listings and appealing product images can help increase interest among younger users.

The study is also important because Generation Z is one of the most active groups in online shopping. Since they are already familiar with digital platforms, they are more likely to respond positively to online clothing marketplaces that are simple and convenient to use. A system that allows easy browsing, fast checkout, and organized product categories can match their shopping habits well. This makes the study useful in supporting the design of an online ordering system that serves both new and preloved items.

Another key point from the study is that sustainability influences the buying decisions of many young consumers. Generation Z is often more aware of environmental concerns and may prefer clothing options that support responsible consumption. Secondhand clothing fits this preference because it promotes reuse and reduces waste. For this reason, an online marketplace for preloved clothes can

appeal not only to budget-conscious buyers but also to those who value sustainability. The study therefore supports the idea that preloved fashion has strong potential in online retail.

Overall, Herjanto, Amin, Purinton, and Lindle II (2024) provide useful insight into the behavior of young consumers in secondhand clothing shopping. Their findings show that Generation Z responds well to items that combine value, style, and sustainability. This is relevant to your project because it shows that an online ordering system can attract younger buyers through a well-organized and user-friendly platform. The study also helps explain why preloved clothing remains a strong and growing part of the online fashion market.

Cross-Cultural Examination of Apparel Online Purchase Intention: SOR Paradigm

Nam, Cho, and Kim (2021) examined apparel online purchase intention through the SOR paradigm, which explains how external stimuli affect a consumer's response in an online shopping environment. Their study focused on how online features such as visual presentation, website design, and shopping interface shape the way buyers react to clothing products. The findings show that the way products are displayed online has a direct effect on consumer interest and purchase intention. This makes the study highly relevant to an online ordering system for brand-new and preloved clothes.

The study suggests that clear and attractive product presentation can encourage buyers to continue with their shopping process. When clothing items are shown with good images, complete descriptions, and organized layouts, consumers are more likely to view the platform as trustworthy and professional. On the other hand, poor presentation may reduce interest and make buyers leave the site before completing a purchase. This means that product listing quality should be given careful attention in an online clothing marketplace.

The findings also emphasize the importance of website design in shaping user behavior. A shopping platform that is easy to navigate allows users to move smoothly from browsing to checkout. If the process is simple and convenient, buyers are less likely to experience frustration and more likely to complete their orders. This is especially important for an online system that handles both brand-new and preloved clothes, since users may compare several items before deciding. A well-designed interface can help improve the overall shopping experience.

Another important point in the study is that online shopping features influence consumer response. Features such as filters, product categories, and a convenient checkout process can support faster and easier decision-making. These functions are useful because they reduce effort and make the platform more efficient for users. In your proposed system, this supports the need for smooth browsing, clear item organization, and a reliable transaction process. Such features help create a positive impression and encourage more purchases.

Overall, Nam, Cho, and Kim (2021) provide useful evidence that interface design and product presentation are important in online clothing shopping. Their study shows that consumer responses are shaped by what they see and experience on the platform. This is directly connected to your project because it highlights the need for clear listings, easy navigation, and convenient checkout functions. The study therefore supports the development of a user-friendly online ordering system that can attract and retain buyers.

Thrift Shopping for Clothes: To Treat Self or Others?

Park, Kwon, Zaman, and Song (2020) explored thrift shopping for clothes and examined the reasons why consumers choose secondhand apparel. Their study showed that people buy thrifted clothing for both practical and personal reasons.

Some buyers are motivated by affordability, while others are influenced by identity, satisfaction, or the enjoyment they get from finding unique items. This makes the study relevant to an online ordering system because it explains why preloved clothing continues to attract many consumers.

The study highlights that secondhand clothing is not only seen as a low-cost option, but also as a meaningful purchase choice. For some buyers, thrift shopping provides a way to save money without losing style or quality. For others, it offers a chance to express individuality through unique fashion pieces that are not commonly found in regular stores. These motivations show that preloved clothing has value beyond its price. In an online marketplace, this means that product listings should present items in a way that appeals to both practical and personal buying preferences.

Another important point from the study is that consumer satisfaction plays a role in thrift shopping behavior. Buyers often enjoy the process of searching for clothes and discovering items that match their taste or needs. This sense of satisfaction can influence them to continue shopping secondhand in the future. In the context of your system, this suggests that a well-organized platform with clear categories and easy browsing can improve the user experience. When users can quickly find what they need, they are more likely to stay engaged and complete a purchase.

The study is also useful in explaining why secondhand clothing remains popular in online markets. As more consumers become comfortable with digital shopping, they also become more open to buying preloved items online. A reliable transaction process, accurate product descriptions, and organized listings can help support this behavior. These features reduce hesitation and make the buying process more efficient. Because of this, the study supports the need for a platform that is both convenient and trustworthy.

Overall, Park, Kwon, Zaman, and Song (2020) provide valuable insight into consumer motivations in thrift shopping. Their findings show that secondhand clothing is chosen for reasons related to cost, identity, and satisfaction. This is highly relevant to your research because it explains why an online ordering system for preloved clothes can succeed when it meets different buyer needs. The study also supports the idea that a user-friendly and reliable marketplace can help strengthen consumer interest in secondhand fashion.

Closing Connection

Together, these foreign literature and studies show that second-hand fashion, digital ordering systems, and sustainable business practices are closely connected. They support the idea that Margaux Collections needs a system that is not only functional but also customer-oriented, transparent, and professionally structured. The evidence from both local and foreign sources strongly justifies the development of the proposed Ordering System as a practical solution for improving business operations, strengthening trust, and supporting sustainable fashion retail.

Local Literature

The Philippine context provides a strong foundation for understanding why the Margaux Collections Ordering System is both relevant and necessary. In local literature, the discussion of second-hand fashion, digital commerce, microbusiness operations, and government support for digital transformation shows that the proposed system is not only technically useful but also aligned with current social and economic

realities in the Philippines. This section strengthens the study by connecting the business needs of Margaux Collections with established local concepts, policies, and industry practices. It also demonstrates that the development of a more structured ordering platform is consistent with the broader direction of Philippine business modernization and consumer behavior.

Investigating Second-Hand Fashion Trade and Consumption in the Philippines

Isla, 2013

The Philippines has long had a strong cultural tradition of second-hand fashion consumption through what is commonly known as *ukay-ukay*. The term comes from the Visayan word for rummaging or searching, and it refers to the practice of buying imported pre-owned clothing that is often sold by weight or at very low prices. Isla (2013), in one of the earliest academic studies on this topic, examined how *ukayukay* is connected not only to economic realities but also to Filipino culture, identity, and everyday fashion practices.

Drawing on ethnographic research conducted in several Philippine cities, the study traced the flow of second-hand garments, which are commonly sourced from textile donations and surplus clothing from countries such as the United States, Europe, and Japan. More importantly, Isla analyzed how Filipino consumers assign meaning and value to these items through the act of buying, wearing, and sharing them. The study therefore presents *ukay-ukay* not merely as a retail activity, but as a social and cultural practice that reflects the changing lifestyle of Filipino consumers.

One of the key contributions of the study is its argument that *ukay-ukay* is not simply a low-income shopping option. While affordability is certainly one reason people buy second-hand clothing, Isla (2013) showed that the practice has deeper cultural meaning. For many Filipinos, *ukay-ukay*

is associated with resourcefulness, creativity, and the excitement of finding valuable pieces at affordable prices. Shoppers often take pride in discovering branded or highquality items that would normally be out of reach if purchased new. In this way, second-hand fashion becomes part of a broader experience of aspirational consumption, where individuals can enjoy fashionable items without paying full retail prices. The practice is therefore both practical and symbolic, combining economic advantage with personal satisfaction and social meaning. This makes *ukay-ukay* especially important in the Philippines, where budgetconscious buying is often combined with a strong interest in style, uniqueness, and value for money.

The study also revealed that *ukay-ukay* shopping often has a social dimension. People frequently go shopping together with friends or family, turning the process of searching through piles of clothing into a shared activity with its own cultural enjoyment. This kind of shopping is not always fast or purely functional; instead, it can be a social and even recreational experience. The act of browsing, comparing, and discovering hidden finds gives second-hand shopping a distinct character that many consumers value.

Isla (2013) also noted that younger, college-educated Filipinos have increasingly embraced *ukay-ukay* as a form of conscious consumption and fashion expression, especially in urban areas where sustainability and personal style are more visible concerns. This shows that second-hand fashion in the Philippines is not limited to necessity-driven consumers but also appeals to those who value individuality and environmental awareness. It suggests that the market for preloved clothing is not only broad but also evolving, with different consumer segments assigning different meanings to the same product category.

The findings of the study are directly relevant to Margaux Collections, because the business operates in a Philippine context where second-hand fashion is already familiar and widely accepted. However, Margaux Collections occupies a

space between traditional *ukay-ukay* and more formal secondhand retail. Traditional *ukay-ukay* is often informal, unbranded, and sold by weight, while more formal second-hand stores usually organize products individually and present them in a more curated manner. The Margaux Collections Ordering System can help the business combine the cultural familiarity of *ukay-ukay* with the structure and professionalism of modern e-commerce. This means that the business can preserve the appeal of second-hand fashion while also offering customers clearer product information, better organization, and a more trustworthy shopping experience. In doing so, the system allows the business to maintain the charm of preloved fashion while improving the overall quality of the buying process.

This is important because many Filipino consumers enjoy the feeling of discovering second-hand items, but they also want convenience and reliability. Traditional *ukay-ukay* shopping can be enjoyable, but it may also involve uncertainty about product quality, pricing, and availability. A digital ordering system helps reduce these issues by making it easier to browse products, view clear photos, and understand item conditions before purchasing. For Margaux Collections, this means that preloved items can be presented with care, accurate descriptions, and transparent prices. Customers can still experience the excitement of second-hand fashion, but with more confidence and less confusion. In this sense, the system does not replace the cultural value of *ukay-ukay*; rather, it modernizes its presentation and makes it more suitable for online commerce.

Isla (2013) also observed that the digitization of secondhand fashion trade in the Philippines was still limited at the time of the study, with many transactions happening in physical markets or through informal online spaces such as Facebook groups and Messenger chats. Although the online environment has changed significantly since then, the basic need for more formal and structured digital platforms remains relevant. Informal online selling

may be convenient at first, but it often lacks the features needed to build strong customer trust, manage orders efficiently, and maintain consistent product information. This is exactly where the Margaux Collections Ordering System becomes valuable. It provides a more organized and dependable alternative to informal selling methods. By doing so, it supports smoother communication, more accurate order handling, and a more professional digital presence for the business.

Another important insight from the study is that second-hand fashion can reflect identity and cultural participation, not just financial necessity. For many Filipino consumers, *ukay-ukay* is part of how they express taste, resourcefulness, and creativity. The same can be true for preloved products sold through Margaux Collections. If the system presents items attractively and professionally, customers may see them not as leftovers or used goods, but as meaningful fashion choices that fit their personal style. This supports the business's ability to position preloved items as desirable, practical, and culturally relevant. It also shows that a well-designed ordering system can influence how customers interpret and value second-hand products, turning them into items of fashion significance rather than simple resale goods.

Overall, Isla's study provides an important local foundation for the proposed system because it explains why second-hand fashion is already familiar in the Philippines and how its value extends beyond price alone. The findings show that Filipino consumers are open to preloved fashion when it is presented in a way that reflects quality, identity, and practicality. This supports the need for a system like Margaux Collections Ordering System, which can improve how preloved items are organized, displayed, and purchased. In doing so, the system responds not only to business needs but also to local consumer culture and expectations.

Social Media Utilization of Microbusinesses in Northern Philippines

Binuluan et al., 2024

The use of social media as a primary business platform among Filipino microbusinesses has become increasingly significant, especially as digital commerce continues to expand across Southeast Asia. Binuluan, Gonayon, Mabborang, Panaligan, and Ramos (2024) examined how microbusinesses in a component city in Northern Philippines use platforms such as Facebook and Instagram to market their products, communicate with customers, and process orders. Using a mixed-methods approach that combined survey responses with in-depth interviews, the study provided a clearer picture of how small business owners rely on social media to sustain and manage their operations. Their findings show that social media is no longer used only for promotion, but has become a central part of everyday business activity for many microentrepreneurs.

One of the main findings of the study is that Facebook remains the most widely used platform for microbusiness commerce in the Philippines. The majority of respondents used Facebook for product promotion, customer inquiries, and order processing. In many cases, business owners also depended on Messenger and Facebook Marketplace to handle day-to-day transactions. Some even combined these platforms with mobile payment applications to complete the sales process. This shows how deeply embedded Facebook has become in the business practices of small Filipino entrepreneurs. For many microbusinesses, it functions as a storefront, communication channel, and sales tool all at once.

However, the study also pointed out several problems with this informal and highly manual approach. One of the most common concerns was the difficulty of tracking orders and payments. Because transactions were often managed through Messenger conversations, business owners had to manually remember or record customer details, which increased the risk of confusion and error. Another challenge was

inventory management. Since Facebook does not provide a built-in system for organizing products in a searchable or structured way, owners often had difficulty keeping track of what items were available, what had already been sold, and which products needed to be restocked. The lack of a permanent product catalog also made it harder for customers to browse items later or return to specific listings. In addition, managing many customer messages individually through Messenger took a great deal of time and effort, especially for owners handling orders on their own.

These findings are very similar to the situation of Margaux Collections, which currently relies on Facebook and Messenger for product promotion, customer communication, order taking, payment confirmation, and delivery coordination. In this sense, the study by Binuluan et al. (2024) accurately reflects the operational realities faced by the business. Like the microbusinesses in the study, Margaux Collections benefits from the reach and familiarity of Facebook, but it also experiences the limitations of a system that is not designed specifically for business operations. This is why the proposed Ordering System is so important. It responds directly to the weaknesses identified in the study by offering a more organized and efficient digital platform for managing the business.

The local literature makes the case even stronger because it shows that the proposed system fits Philippine culture, policy, and business practice. Isla (2013) establishes that second-hand fashion already has deep roots in the Philippines through ukay-ukay, so Margaux Collections is operating in a market where preloved items are already familiar and socially accepted. Binuluan et al. (2024) confirm that Facebook and Messenger dominate microbusiness commerce, but also reveal the limits of informal selling, especially in order tracking, inventory control, and timeconsuming communication. The DICT's Philippine Digital Transformation Strategy 2022 adds policy support by encouraging MSMEs to adopt digital tools that improve business efficiency and competitiveness.

Philippine Digital Transformation Strategy 2022

DICT, 2022

The Philippine Digital Transformation Strategy 2022 reflects the government's broader effort to encourage digital adoption among businesses, public institutions, and communities. The Department of Information and Communications Technology emphasized that digital systems are essential for improving productivity, expanding access to markets, and strengthening the competitiveness of micro, small, and medium enterprises. This policy direction is especially important in a country where many small businesses still rely on manual processes and informal communication tools for their daily operations.

The strategy also highlights the need for digital commerce tools, mobile-friendly services, and data-driven business practices. These priorities are highly relevant to small enterprises that want to remain competitive in an increasingly digital marketplace. For businesses like Margaux Collections, the document provides a policy environment that supports the development of web-based ordering and inventory systems. It shows that digital transformation is not limited to large corporations; rather, it is also intended to help microbusinesses improve their operations, reach more customers, and reduce inefficiencies.

Another important point raised by the strategy is the need for locally appropriate technology solutions. The document supports the development of systems that are designed with the specific needs, resources, and digital literacy levels of Filipino entrepreneurs in mind. This is especially important for microbusinesses, which usually do not have large budgets or dedicated IT teams. The Margaux

Collections Ordering System, built using PHP and MySQL, reflects this kind of practical and accessible approach. These technologies are widely used, relatively affordable, and well supported in the local development environment, making them suitable for a small business project.

Philippine E-commerce Fashion Market Statistics (ECDB, 2025)

The Philippine fashion e-commerce market has experienced substantial growth in recent years, generating US\$4,739 million in revenue in 2025 with a projected annual growth rate of 15-20%. This remarkable expansion reflects the increasing adoption of digital shopping platforms among Filipino consumers and the fashion industry's successful transition to online retail. Apparel accounts for 52% of fashion e-commerce revenue, establishing it as the dominant product category within the market. The market is projected to reach US\$9,435 million by 2030, indicating sustained momentum driven by younger demographics, improved digital infrastructure, and changing consumer preferences toward convenient online shopping experiences [ECDB, 2025].

The competitive landscape of Philippine fashion e-commerce is dominated by three major platforms, with TikTok Shop emerging as the largest online retailer with revenue of US\$656,449 million in 2025. Shopee follows in second place, while Lazada holds third position. TikTok Shop's dominance reflects the growing influence of social commerce, particularly among Gen Z and Millennial consumers who prefer discovering and purchasing fashion through short-form video content. The platform's revenue increased by 20-25% in 2025, demonstrating the effectiveness of live streaming and influencer marketing in driving fashion sales [ECDB, 2025].

The broader Philippine e-commerce market provides important context for understanding fashion e-commerce growth. The overall market is valued at USD 20.05 billion in 2026 and

is growing at a compound annual growth rate (CAGR) of 13.61%, projected to reach USD 37.95 billion by 2031. Major companies operating in this market include Sea Ltd. (Shopee), Alibaba Group (Lazada), ByteDance Ltd. (TikTok Shop), Global Fashion Group, and Amazon.com Inc. This competitive environment creates opportunities for both large retailers and small ukay-ukay sellers to access consumers through multiple digital channels.

The seasonal nature of e-commerce sales significantly impacts fashion retailers, with monthly revenues varying based on consumer habits, cultural preferences, and economic conditions. In April 2026, monthly revenue in the Philippine fashion e-commerce market reached US\$452 million, representing a 0-5% change from the previous month. Understanding these patterns is crucial for fashion businesses, particularly online ukay sellers, in planning inventory, promotions, and marketing campaigns to maximize sales during peak periods while managing slower seasons effectively [ECDB, 2025].

This market data provides critical context for understanding the digital ukay-ukay ecosystem within the broader Philippine e-commerce landscape. The substantial market size and growth projections indicate strong demand for online fashion retail, creating opportunities for pre-loved clothing sellers to compete alongside new clothing retailers. The dominance of TikTok Shop and the rise of social commerce suggest that successful ukay sellers must invest in content creation, live selling, and engagement strategies to capture market share. These statistics support the argument that online platforms have democratized access to fashion markets, enabling small ukay businesses to reach consumers nationwide without physical storefronts [ECDB, 2025].

Weekly E-commerce Activities in the Philippines (Statista, 2024)

According to Statista's Q3 2024 survey, 57% of respondents in the Philippines purchased products or services online each week, indicating highly regular e-commerce engagement among Filipino consumers. This frequency demonstrates that online shopping has transitioned from an occasional activity to a routine part of consumer behavior, particularly among younger demographics. Most online shoppers belong to Gen Z and Millennial cohorts, who grew up with digital technology and view online shopping as a normalized, convenient alternative to traditional retail. This generational shift has profound implications for ukay-ukay sellers, as these demographics represent the fastest-growing segment of secondhand clothing consumers [Statista, 2024].

Payment method preferences reveal significant progress in digital payment adoption among Filipino consumers. Mobile wallets, particularly GCash have emerged as the leading payment methods for e-commerce transactions, while cash-on-delivery (COD) accounts for only 15% of online payments. This shift indicates that digital payment infrastructure has matured sufficiently to build consumer confidence in online transactions. The central bank reported that digital payments had already overtaken cash-based payments in the country by 2023, reflecting broader financial inclusion initiatives and the convenience of contactless payment methods. For online ukay sellers, this trend suggests that accepting digital payments has become essential for competitiveness [Statista, 2024].

The adoption of buy now, pay later (BNPL) schemes has further diversified payment options for Filipino online shoppers. Companies such as Klarna and Grab PayLater have introduced flexible payment plans that allow consumers to purchase items immediately and pay in installments. This payment innovation is particularly relevant for fashion purchases, where consumers may wish to acquire multiple items but face budget constraints. For ukay-ukay sellers, offering BNPL options could increase average order values and make higher-priced vintage or branded items more

accessible to price-sensitive consumers who might otherwise hesitate to purchase.

Social commerce continues to contribute significantly to the Philippines' growing online shopping market, complementing traditional e-commerce marketplaces like Shopee and Lazada. The popularity of Facebook Marketplace, Instagram Shopping, and TikTok Shop reflects Filipino consumers' preference for platforms that combine social interaction with shopping functionality. This trend aligns with the ukay-ukay industry's heavy reliance on social media for selling, as live selling and community-building create trust and engagement that pure transactional platforms cannot replicate. Understanding these consumer preferences is crucial for ukay sellers seeking to optimize their digital presence.

The high frequency of online purchases and strong digital payment adoption provide a favorable environment for online ukay-ukay businesses. With 57% of consumers shopping online weekly and mobile wallets dominating payment methods, the infrastructure and consumer behavior necessary for successful e-commerce are firmly established. For ukay sellers, this means that barriers to entry have decreased significantly, with payment processing, customer acquisition, and order fulfillment all supported by mature digital ecosystems. However, competition has also intensified, requiring sellers to differentiate through quality, pricing, customer service, and marketing effectiveness to capture and retain customer attention in an increasingly crowded marketplace [Statista, 2024].

The Philippine Ukay-Ukay Culture as Sustainable Fashion (Biana, 2020)

Biana's (2020) study, *The Philippine Ukay-Ukay Culture as Sustainable Fashion*, published in the DLSU Business & Economics Review, provides a comprehensive examination of ukay-ukay as both a cultural phenomenon and a sustainable

fashion practice in the Philippines. The research explores how the term "ukay-ukay," derived from "halukay" (to dig), captures the essence of Philippine thrift shopping culture where consumers sift through piles of pre-loved clothing to discover unique items. This study is particularly significant as it frames ukay-ukay not merely as an economic necessity but as a legitimate sustainable fashion movement that aligns with global circular economy principles while maintaining distinctively Filipino cultural characteristics [DLSU, 2020].

The research documented that ukay-ukay consumers are predominantly aged 21-30 years old, female, college graduates, fully employed, and belong to income class B (Php 20,001-40,000). This demographic profile challenges the stereotype that secondhand clothing is exclusively for low-income consumers, revealing instead that educated, middleclass Filipinos actively participate in ukay-ukay culture. The study found that these consumers are motivated by both economic considerations and growing environmental consciousness, with many explicitly framing their ukay purchases as sustainable fashion choices. This dual motivation creates a unique market position that distinguishes Philippine ukay-ukay from secondhand clothing markets in other countries [DLSU, 2020].

The study emphasized the cultural significance of ukay-ukay in Philippine society, noting how it has evolved from wet market stalls to upscale boutiques and online platforms. This transformation reflects broader changes in consumer perceptions, where secondhand clothing has shed much of its stigma and is now associated with creativity, resourcefulness, and environmental awareness. The research highlighted how ukay-ukay has become particularly popular among Filipino youth, who view thrift shopping as a way to express individuality while avoiding the homogenization of fast fashion. This cultural shift has enabled ukay-ukay to thrive despite legal restrictions on commercial importation of used textiles [DLSU, 2020].

Biana's research also examined the sustainability implications of ukay-ukay consumption, positioning it as a practical manifestation of circular fashion economy principles in the Philippine context. By extending the lifecycle of clothing items and reducing textile waste, ukay-ukay contributes to environmental sustainability while providing affordable fashion options. The study noted that the Philippines leads Southeast Asia in secondhand fashion transactions, accounting for over 30% of regional activity according to Carousell data. This position suggests that Philippine ukay-ukay culture has achieved critical mass as a mainstream consumption pattern rather than remaining a marginal alternative [DLSU, 2020].

This study makes important contributions to local literature by bridging cultural, economic, and environmental perspectives on ukay-ukay consumption. It provides empirical evidence that Filipino consumers engage with secondhand clothing for complex, overlapping motivations that extend beyond simple affordability. The findings have implications for policymakers considering regulatory approaches to the industry, suggesting that formalization rather than prohibition might better serve both economic and environmental objectives. For researchers, the study opens avenues for further investigation into how ukay-ukay culture intersects with Philippine identity, sustainability values, and consumer behavior patterns [DLSU, 2020].

: Filipino Gen Z Consumer Behavior Towards Ukay-Ukay (DLSU, 2024)

The study *Filipino Gen Z Consumer Behavior Towards UkayUkay and their*

Sustainable Fashion Practices (2024), published in the De La Salle University SHS Research Convention Proceedings, examines how Filipino Generation Z consumers engage with ukay-ukay and their motivations for participating in sustainable fashion practices. This research is particularly timely given that Gen Z represents the

fastest-growing segment of ukay-ukay consumers and is driving the cultural transformation of secondhand clothing from stigma to mainstream fashion choice. The study employs survey methodology to understand the attitudes, behaviors, and values that shape Gen Z's relationship with ukay-ukay [DLSU, 2024].

The research found that Filipino Gen Z consumers engage with ukay-ukay primarily for three interconnected reasons: affordability, sustainability consciousness, and unique fashion expression. Affordability remains a fundamental driver, with Gen Z consumers facing economic constraints including student loans, entry-level salaries, and rising cost of living. However, the study revealed that sustainability consciousness has become equally important, with many Gen Z respondents explicitly framing their ukay purchases as environmental choices aligned with values around climate action and waste reduction. Unique fashion expression represents the third motivation, as Gen Z consumers use ukay-ukay to curate distinctive personal styles that differentiate them from peers wearing massproduced fast fashion [DLSU, 2024].

The study documented that Gen Z consumers predominantly access ukay-ukay through digital platforms, with Facebook Marketplace, Instagram, and TikTok serving as primary channels. These platforms align with Gen Z's digital-native communication preferences and provide access to curated selections beyond traditional physical ukay stores. The research found that Gen Z consumers actively research sellers online, checking ratings, reviews, and social media presence before making purchases. They also engage with ukay-ukay content creators who showcase styling tips, thrift hauls, and sustainable fashion education, creating a digital ecosystem that normalizes and celebrates secondhand clothing consumption [DLSU, 2024].

The research highlighted that Gen Z's ukay-ukay participation reflects broader values around circular economy and sustainable consumption. Many respondents

expressed concern about textile waste, fast fashion environmental impacts, and the ethical implications of clothing production. This values-driven consumption distinguishes Gen Z from previous generations who primarily engaged with ukay-ukay for economic necessity. The study noted that this values alignment creates strong brand loyalty, with Gen Z consumers preferring sellers who communicate sustainability messages and demonstrate environmental responsibility. This trend suggests that ukayukay's future growth will depend on maintaining alignment with Gen Z's sustainability values [DLSU, 2024].

This study makes significant contributions to local literature by documenting the generational shift in ukayukay consumption patterns and motivations. It provides empirical evidence that Filipino Gen Z represents a critical demographic for sustainable fashion transitions, with high engagement rates and strong environmental values. The findings have implications for policymakers, educators, and businesses seeking to promote circular economy practices, suggesting that Gen Z's ukay-ukay participation can be leveraged for broader sustainability education and behavior change initiatives. For researchers, the study identifies opportunities for longitudinal investigation into how Gen Z's sustainable fashion values evolve as they age and their purchasing power increases [DLSU, 2024].

How Facebook Communication Strategies Facilitate Ukay-Ukay Live Selling (2024)

The study *How Facebook Communication Strategies Facilitate Ukay-Ukay Live Selling* (2024), published in *European Proceedings of Social and Behavioural Sciences*, examines the communication tactics employed by Philippine ukay-ukay sellers during live selling sessions and their effects on buyer consumption behavior. This research addresses a critical gap in understanding how digital platforms have transformed traditional ukay-ukay selling into interactive, real-time commerce experiences.

The study is particularly relevant given that Facebook Live has become the dominant selling channel for online ukay businesses, creating new dynamics between sellers and buyers that differ significantly from traditional retail interactions [European Proceedings, 2024].

The research identified several key communication strategies employed by ukay-ukay sellers during live sessions. These include real-time product demonstrations where sellers display items from multiple angles, immediate responses to buyer questions about size, condition, and pricing, creation of urgency through limited-time offers and flash sales, and building personal connections through informal, conversational styles. Sellers effectively use social proof by highlighting previous purchases and positive customer feedback, while also employing humor and entertainment to maintain viewer engagement throughout extended live sessions. These strategies create a shopping experience that combines transactional efficiency with social interaction [European Proceedings, 2024].

The study found that effective Facebook communication strategies significantly influence buyer consumption behavior, with several mechanisms explaining this relationship. Real-time interaction reduces purchase uncertainty by allowing buyers to obtain immediate information about product conditions, addressing the fundamental challenge of online apparel retailing where customers cannot physically inspect items. The interactive nature of live selling creates a sense of community among participants, fostering loyalty and repeat purchases. Additionally, the entertainment value of live sessions encourages prolonged viewing, increasing the likelihood of impulse purchases and larger basket sizes. Sellers who master these communication tactics report substantially higher conversion rates compared to those who rely solely on static product listings [European Proceedings, 2024].

The research also documented challenges faced by ukay-ukay sellers in maintaining effective live selling operations.

Technical issues including unstable internet connections, poor lighting, and audio problems can disrupt sessions and reduce professional quality. Managing high volumes of simultaneous chat messages requires organizational skills and quick decision-making abilities. Sellers must also navigate the unpredictability of viewer engagement, as attendance fluctuates significantly between sessions. Despite these challenges, the study found that successful sellers continuously refine their communication strategies through practice, feedback, and observation of competitor tactics [European Proceedings, 2024].

This study makes significant contributions to local literature by providing empirical evidence on the mechanics of digital ukay-ukay commerce in the Philippines. It demonstrates how social media platforms have enabled small businesses to compete effectively through low-cost, high-impact marketing strategies. The findings have practical implications for aspiring ukay entrepreneurs, suggesting that investment in communication skills and live selling techniques can yield substantial returns. For policymakers and support organizations, the research highlights the importance of digital literacy training and technical infrastructure development for small businesses operating in the online ukay sector [European Proceedings, 2024].

Experiences of Online Sellers of Pre-Loved Clothes (Zenodo, 2023)

The study *Experiences of Online Sellers of Pre-Loved Clothes* (2023), archived on Zenodo, explores the lived experiences, challenges, and adaptive strategies of online pre-loved clothing sellers in the Philippines. This qualitative research provides critical seller-centered perspectives that complement consumer-focused studies, offering a more complete understanding of the digital ukay ecosystem. Through in-depth interviews and observational methods, the research captures the day-to-day realities of

individuals who have transitioned into online ukay entrepreneurship, documenting both opportunities and obstacles they encounter in this informal yet highly dynamic market [Zenodo, 2023].

The study documented several critical challenges faced by online pre-loved clothing sellers. Inventory sourcing emerged as a primary difficulty, with sellers reporting irregular availability of quality items, competition from other sellers for desirable pieces, and the physical labor involved in sorting, cleaning, and preparing garments for sale. Pricing complexity presented another challenge, as sellers must balance competitiveness with profitability while accounting for variable item conditions and uncertain demand. Platform dependency created vulnerability, with sellers reporting that algorithm changes on Facebook, Instagram, or Shopee could dramatically affect visibility and sales. Fraud and scam attempts by buyers, including fake payment screenshots and refusal to claim items, represented ongoing financial risks that required constant vigilance [Zenodo, 2023].

Despite these challenges, the study identified several adaptive strategies employed by successful online sellers. Multi-channel approaches emerged as critical, with sellers maintaining presence across multiple platforms (Facebook Marketplace, Instagram, Shopee, TikTok) to reduce platform dependency and maximize reach. Content creation capabilities, including product photography, video showcasing, and engaging captions, became essential competitive advantages. Relationship-building through personalized communication, follow-up messages, and loyalty programs helped develop repeat customer bases that provided income stability. Financial management skills, including inventory tracking, cash flow management, and emergency fund maintenance, proved essential for navigating income volatility [Zenodo, 2023].

The research highlighted the entrepreneurial resilience of online ukay sellers, many of whom balanced selling with

other employment or education. For some, online ukay selling represented a primary livelihood; for others, it served as supplementary income or pathway to full-time entrepreneurship. The study found that sellers developed diverse skill sets spanning product sourcing, digital marketing, customer service, photography, and financial management. This entrepreneurial adaptability reflects broader Philippine cultural values around resourcefulness and improvisation, enabling sellers to thrive despite operating in an informal, lightly-regulated market with limited institutional support [Zenodo, 2023].

This study makes important contributions to local literature by centering seller perspectives in understanding the digital ukay economy. It reveals that online ukay selling requires sophisticated capabilities extending far beyond simple reselling, encompassing digital literacy, business management, and interpersonal skills. The findings have practical implications for support organizations seeking to assist online ukay entrepreneurs, suggesting that capacitybuilding programs should address digital marketing, financial management, fraud prevention, and legal compliance. For policymakers, the research highlights the economic significance of online ukay selling as a source of livelihood and entrepreneurship, suggesting that formalization rather than prohibition might better serve both sellers and consumers [Zenodo, 2023].

Exploring Ukay-Ukay Consumption in the Philippines (Benavidez, 2025)

Benavidez's (2025) study, *Exploring Ukay-Ukay Consumption in the Philippines:*

Between Sustainable Practices and Business, published in World Journal of Advanced Research and Reviews, provides a comprehensive examination of the ukay-ukay industry's historical development, economic significance, and environmental implications in the Philippines. The study traces the industry's origins to Baguio City and documents its evolution into a nationwide phenomenon that operates

despite legal prohibitions on commercial importation of used textiles under R.A. 4563. This historical and contemporary analysis offers valuable context for understanding why ukayukay remains deeply embedded in Philippine consumer culture [WJARR, 2025].

The research documented how ukay-ukay goods are sourced primarily from

Hong Kong and the United States through balikbayan boxes, creating an informal but highly organized supply chain that bypasses official import restrictions. This sourcing mechanism has enabled the industry to thrive despite regulatory challenges, with goods flowing through ports and distribution networks that have developed over decades. The study noted that the industry has evolved significantly from its origins as street stalls and wet market vendors to include sophisticated online platforms, retail stores, and specialty boutiques. This evolution reflects both the industry's adaptability and the growing acceptance of secondhand clothing among Philippine consumers [WJARR, 2025].

The study identified two primary drivers of ukay-ukay's continued growth: affordability and sustainability.

Affordability remains the fundamental motivation, particularly for low- and middle-income households for whom new clothing represents a significant expense. However, the research also documented a growing sustainability consciousness among consumers who explicitly frame their ukay purchases as environmental choices aligned with circular economy principles. This dual motivation creates a unique market position for ukay-ukay, appealing to both economically-constrained consumers and environmentallyconscious shoppers who might otherwise not purchase secondhand clothing [WJARR, 2025].

Economically, the ukay-ukay industry provides substantial benefits including employment generation and government revenue through indirect taxation and business registration. The study estimated that the industry

supports thousands of jobs across the value chain, from importers and wholesalers to retailers and online sellers. Despite operating in a legally ambiguous space due to R.A. 4563, the industry contributes to local economies through rent payments, utility consumption, and informal taxation. The research argued that policy discussions should consider these economic contributions and explore regulatory frameworks that balance import control objectives with the industry's economic and social benefits [WJARR, 2025].

This research makes important contributions to local literature by documenting the ukay-ukay industry's historical evolution, economic significance, and contemporary transformation. The study bridges economic, environmental, and cultural perspectives to provide a holistic understanding of why ukay-ukay remains resilient despite legal and social challenges. For policymakers, the findings suggest that regulatory approaches should account for the industry's economic contributions and explore options for formalization rather than prohibition. For researchers, the study identifies opportunities for further investigation into sustainability claims, supply chain transparency, and the industry's role in Philippine circular economy development [WJARR, 2025].

Level of Satisfaction Among Women on Online Ukay-Ukay (IJARIE, 2023)

The study *Level of Satisfaction Among Women on Online Ukay-Ukay* (2023), published in the International Journal of Academic Research and Innovative Engineering Education (IJARIE), investigates female consumer satisfaction with online ukay-ukay purchasing experiences in the Philippines. This research is significant because women comprise the majority of ukay-ukay consumers, yet their specific experiences, concerns, and satisfaction drivers have received limited academic attention. The study employs quantitative methods to measure satisfaction levels across multiple dimensions including product quality, pricing,

delivery service, customer communication, and overall value for money [IJARIE, 2023].

The research identified several key factors influencing women's satisfaction with online ukay-ukay purchases. Product quality emerged as the most critical determinant, with respondents expressing concerns about garment condition, accuracy of product descriptions, and discrepancy between photographed items and actual delivery. Price competitiveness remained important, but satisfied customers were willing to pay slightly higher prices for guaranteed quality and reliable service. Delivery efficiency significantly impacted satisfaction, with delayed shipments and poor packaging leading to negative experiences. Customer communication quality, including response time and problem resolution, also emerged as a substantial factor affecting overall satisfaction ratings [IJARIE, 2023].

The study found that women's satisfaction with online ukayukay is generally moderate to high, with mean satisfaction scores indicating positive overall experiences. However, the research also identified specific pain points that undermine satisfaction, including inconsistent sizing information, difficulty assessing garment condition from photographs, and limited recourse for dissatisfied customers. Respondents expressed preference for sellers who provide detailed condition descriptions, multiple photographs from different angles, clear sizing charts, and transparent return policies. Trust-building mechanisms including seller ratings, customer reviews, and social proof were identified as critical for reducing perceived risk and increasing satisfaction [IJARIE, 2023].

The research documented demographic patterns in satisfaction levels, finding that younger women (aged 18-30) reported higher satisfaction with online ukay-ukay compared to older age groups. This generational difference reflects greater digital literacy, familiarity with online shopping

platforms, and cultural acceptance of secondhand clothing among younger consumers. The study also found that repeat customers reported higher satisfaction than first-time buyers, suggesting that experience with online ukay-ukay reduces uncertainty and improves ability to make satisfying purchases. This learning effect has implications for seller strategies focused on building customer loyalty and encouraging repeat business [IJARIE, 2023].

This study contributes important insights to local literature on female consumer behavior in the ukay-ukay sector. It provides empirical evidence that women's satisfaction with online ukay-ukay depends on multiple interconnected factors extending beyond simple price considerations. The findings suggest that sellers who invest in quality control, transparent communication, and reliable service delivery can differentiate themselves in a competitive market and build sustainable customer relationships. For researchers, the study identifies opportunities for further investigation into gender-specific consumption patterns, satisfaction drivers across different ukay segments, and the long-term relationship between satisfaction and customer loyalty in the online ukay market [IJARIE, 2023].

Web-Based Ordering System for Start-UP Business with Forecasting (Zenodo, 2024)

The study *Web-Based Ordering System for Start-UP Business with Forecasting* (2024), archived on Zenodo, presents the development and implementation of a web-based ordering system designed specifically for start-up businesses in the Philippines. The system addresses critical operational challenges faced by small online retailers, including inefficient order-taking processes, errors in manual order processing, and difficulty predicting demand for inventory management. While not exclusively focused on pre-loved clothing, this study is highly relevant to online ukay sellers who operate as start-ups and require affordable,

scalable technology solutions to support business growth [Zenodo, 2024].

The system incorporates several key features designed to streamline operations and enhance customer experience. A carting mechanism allows customers to browse products, select items, and complete purchases in a single transaction, reducing friction in the buying process. Product browsing functionality includes search filters, category organization, and high-quality image galleries that help customers find desired items efficiently. Automated order processing eliminates manual data entry errors and provides real-time order confirmation to customers. These features collectively improve operational efficiency while creating a more professional and trustworthy purchasing experience [Zenodo, 2024].

The most distinctive feature of this system is its sales forecasting capability, which uses historical sales data to predict future demand and inform inventory management decisions. For online ukay sellers, who face challenges with irregular inventory availability and unpredictable demand patterns, forecasting capabilities can help optimize stock levels, reduce overstock or stockout situations, and improve cash flow management. The system's ability to identify seasonal trends and purchasing patterns enables sellers to make data-driven decisions about inventory procurement, pricing strategies, and promotional timing. This represents a significant upgrade from traditional manual inventory management approaches [Zenodo, 2024].

The study demonstrated that implementing this web-based ordering system resulted in measurable improvements for the start-up business case study. Error rates in order-taking decreased significantly, customer satisfaction scores improved due to faster and more reliable service, and inventory turnover rates optimized through better demand forecasting. The system's scalability made it suitable for businesses at various growth stages, from individual sellers managing small inventories to established ukay

operations handling higher transaction volumes. Cost-effectiveness was also highlighted, as the web-based platform required lower upfront investment compared to traditional e-commerce development [Zenodo, 2024].

This research contributes to local literature by addressing the technology infrastructure needs of Philippine start-up businesses, particularly those in the online retail and ukay sectors. The findings suggest that accessible, affordable technology solutions can significantly improve operational efficiency and competitiveness for small businesses that traditionally rely on manual processes. For online ukay sellers specifically, the study demonstrates that technology adoption is not limited to large enterprises but can be implemented at the grassroots level to support business formalization and growth. Future research could investigate the broader adoption of such systems across the ukay industry and their impact on sector-wide efficiency and consumer protection [Zenodo, 2024].

From Necessity to Identity: Entrepreneurial Perspectives on Filipino Second-Hand Clothing Consumers (Frontiers in Sustainability, 2022)

The study *From Necessity to Identity: Entrepreneurial Perspectives on Key Motivations for Filipino Second-Hand Clothing Consumers* (published in *Frontiers in Sustainability*, 2022) explores the evolving motivations behind Filipino engagement with ukay-ukay through observational research and in-depth interviews. Unlike earlier studies that framed secondhand clothing consumption purely as an economic necessity, this research reveals a more complex consumer landscape where affordability intersects with identity construction and sustainable fashion values. The study provides critical insights into how Philippine consumers have reinterpreted ukay-ukay from a stigma-laden necessity into a form of self-expression [Frontiers, 2022].

The research identified two primary motivations driving Filipino consumers toward secondhand clothing: economic necessity and identity construction. On the economic dimension, affordability remains a fundamental driver, particularly for low- and middle-income consumers who seek quality clothing at accessible prices. However, the study found that this economic motivation has evolved alongside growing environmental consciousness. Many consumers now explicitly frame their ukay-ukay purchases as sustainable fashion choices, aligning personal consumption with broader environmental values and circular economy principles [Frontiers, 2022].

The identity construction dimension represents a significant shift in how Filipinos perceive secondhand clothing. The study documented how consumers, particularly younger demographics, use ukay-ukay to curate unique fashion statements that distinguish them from mainstream fastfashion consumers. Vintage pieces, rare finds, and one-of-a-kind garments enable consumers to express individuality while avoiding the homogenization associated with massproduced clothing. This identity-driven consumption reflects global trends in sustainable fashion but is distinctly localized within Philippine cultural contexts, where resourcefulness and creativity are valued traits [Frontiers, 2022].

Entrepreneurial perspectives highlighted in the study further illuminate how ukay-ukay sellers have adapted to these shifting consumer motivations. Sellers increasingly position their businesses not merely as affordable clothing providers but as curators of unique fashion and promoters of sustainable consumption. This repositioning has enabled ukay-ukay businesses to attract consumers who might otherwise have avoided secondhand clothing due to stigma. The study also noted how social media has facilitated this transformation, allowing sellers to showcase curated collections and build communities around sustainable fashion values [Frontiers, 2022].

This research contributes importantly to local literature by documenting the cultural transformation of ukay-ukay consumption in the Philippines. It moves beyond purely economic explanations to capture the psychological, social, and environmental dimensions of secondhand clothing consumption. The findings suggest that policy and business strategies should account for both the affordability imperative and the identity-sustainability values that now drive the market. Understanding this dual motivation is essential for developing interventions that support both economic accessibility and sustainable fashion transitions in the Philippine context [Frontiers, 2022].

The Influence of Inventory Management Practices on Financial Performance among Retail Businesses in Cateel, Davao Oriental

Toroba, Gregori, Martinez, and Cervantes (2025) examined the influence of inventory management practices on the financial performance of retail businesses in Cateel, Davao Oriental during the year 2024. The study aimed to investigate how inventory management practices affect financial performance outcomes including sales growth, market growth, and profit growth among retail businesses in the Philippines. The study examined the levels of inventory management in terms of business order inventory, inventory storage, inventory control, and warehouse inventory management, as well as the levels of financial performance in terms of sales growth, market growth, and profit growth. This makes the study important in understanding how effective inventory management directly contributes to better financial performance for retail businesses in the Philippines.

The respondents of the study were retail business owners and managers in Cateel, Davao Oriental who were chosen through appropriate sampling methods. They were selected because they represented the target population of retail businesses engaged in various merchandising operations including grocery stores, clothing retailers, hardware stores, and

general merchandise stores. Data were gathered through surveys that assessed inventory management practices and financial performance indicators over a 12-month period. The respondents demonstrated high levels of inventory management ($\bar{x}_W = 4.12$, $s = 0.50$), particularly in inventory storage ($\bar{x}_W = 4.37$, $s = 0.81$) and inventory control ($\bar{x}_W = 0.55$). Financial performance also reflected favorable outcomes, with profit growth showing the highest mean ($\bar{x}_W = 4.02$, $s = 0.65$). The survey approach helped identify the relationship between inventory management practices and financial performance outcomes.

Findings showed that a moderate positive relationship ($r = 0.648$, $p = 0.000$) was observed between overall inventory management and overall financial performance. The study revealed that retail businesses with effective inventory management practices experience better financial outcomes including higher sales growth, market growth, and profit growth compared to those with poor inventory management.

Furthermore, regression analysis indicated that business order inventory ($\beta = 0.463$, $p = 0.000$), inventory control ($\beta = 0.452$, $p = 0.002$), and warehouse inventory management ($\beta = 0.163$, $p = 0.018$) significantly influenced financial performance. The results demonstrate that effective inventory management directly contributes to better financial performance for retail businesses in the Philippines.

The study also identified that targeted improvements in specific inventory domains may enhance business performance. The findings show that particular attention should be given to business order inventory, inventory control, and warehouse inventory management as these areas have the most significant impact on financial performance. Retail businesses that invested in proper inventory ordering systems were able to reduce stockouts and overstock situations, resulting in better cash flow management and improved customer satisfaction. These results indicate that inventory management practices should

be a priority for retail business owners seeking to improve their financial outcomes and maintain competitiveness in their markets. Businesses that neglected these critical inventory management areas experienced higher operational costs and lower profitability compared to those who maintained proper inventory control systems.

The literature further revealed that retail businesses in Cateel, Davao Oriental faced various challenges in implementing effective inventory management practices including limited financial resources for investing in inventory management technology, lack of specialized training for staff on inventory procedures, difficulties in forecasting customer demand accurately, and challenges in maintaining consistent supplier relationships. Despite these challenges, businesses that overcame these obstacles through creative solutions such as implementing regular inventory audits, establishing clear inventory policies, and building strong relationships with suppliers were able to achieve better financial performance outcomes. The study showed that even without expensive inventory management software, small retail businesses can achieve effective inventory control through disciplined practices, consistent monitoring, and continuous improvement of their inventory management processes.

Based on the findings, the researchers recommended that retail business owners prioritize training and support for inventory control systems and reinforce structured order and management practices for sustainable financial outcomes. They suggested that entrepreneurs use inventory management systems and implement best practices for inventory storage, ordering, and control to optimize their financial performance. For small retail businesses with limited budgets, the researchers recommended starting with simple manual inventory tracking systems and gradually progressing to more sophisticated automated systems as the business grows and generates sufficient revenue. By implementing effective inventory management practices, retail businesses may achieve better financial outcomes

including higher profit growth, sales growth, and market growth. The study emphasizes the importance of inventory management for retail business success and financial sustainability, highlighting that proper inventory control is not just an operational necessity but a strategic business advantage that can determine the long-term viability of retail businesses in the competitive Philippine market.

POS and Checkout Systems for Philippine Small Businesses

Peddlr Philippines Inc. (2024) examined mobile point-of-sale (POS) systems and their impact on small business operations in the Philippines. The study aimed to determine how mobile POS systems can help small businesses easily process transactions, manage inventory, and obtain valuable insight reports without requiring expensive traditional POS hardware. The literature describes how E-commerce Philippines provides shopping cart solutions serving the growing market of mobile users and online shoppers. This makes the study important in understanding how affordable POS technology can improve operational efficiency for small businesses in the Philippine retail market.

The respondents of the study were small business owners and operators in the Philippines who were chosen through appropriate sampling methods. They were selected because they represented the target population of small enterprises engaged in retail trading, food service, and general merchandise operations. Data were gathered through observations of POS system usage, interviews with business owners, and analysis of transaction processing efficiency. The mobile POS system allows businesses to easily add and sort all their products in one convenient place, enabling smoother and faster checkout processes for customers.

Findings showed that the POS system enables businesses to experience a smooth and speedy checkout experience with the built-in barcode scanner. Simply scanning the barcode of the product from the mobile camera allows the POS system to

instantly retrieve the product information, stock availability, and price. The literature emphasized that mobile POS systems are particularly beneficial for small businesses that need affordable and accessible transaction processing solutions without investing in expensive traditional POS hardware. The results demonstrate that mobile POS technology can significantly reduce transaction time and improve customer satisfaction for small retail businesses.

Mobile POS systems also support online ordering and checkout integration, allowing businesses to manage both in-store and online sales from a single platform. This omnichannel capability is increasingly important as more Philippine consumers shop online and expect seamless shopping experiences across different channels. The literature emphasizes that small businesses should consider adopting mobile POS and checkout systems to remain competitive in the evolving Philippine retail market. Based on the findings, the researchers recommended that small businesses adopt mobile POS technology to improve their operational efficiency and customer service capabilities.

Second-hand Online Shopping Sites for Fashion, Lifestyle, and Home Products in the Philippines (BLOG)

BuyAndShip Philippines (2023) conducted an extensive examination of the growing trend of second-hand online shopping in the Philippines and its significant impact on consumer purchasing behavior patterns throughout the country. The literature comprehensively describes 8 popular second-hand online shopping sites that have gained substantial favor among Filipino users, including international platforms such as eBay, Depop, Hard-Off, Vinted, Poshmark, The RealReal, and Mercari, as well as regional platform Carousell which has achieved particular prominence in the Philippine market. This makes the

literature critically important in understanding how secondhand shopping has evolved from traditional physical thrift stores and ukay-ukay establishments into sophisticated organized online platforms that serve the Philippine market with enhanced convenience, accessibility, and consumer protection mechanisms. The rise of second-hand online shopping has become a significant and transformative trend in the Philippines, with consumers increasingly turning to online platforms for purchasing preloved fashion items, vintage clothing collections, and home products as viable alternatives to purchasing brand-new merchandise from conventional retail establishments.

The literature extensively highlights that second-hand online shopping offers multiple compelling advantages that make it increasingly attractive to Filipino consumers across various demographic segments and socioeconomic backgrounds. These advantages comprehensively include affordability that allows consumers to purchase quality items at significantly reduced prices compared to retail prices, access to unique and vintage items that are no longer available in mainstream retail stores, environmental sustainability benefits through reduction of textile waste and support for circular economy practices, and the remarkable ability to find branded international items at heavily discounted prices that would otherwise be completely unaffordable if purchased new from authorized retailers. Filipino consumers particularly value second-hand shopping for fashion items because they can access international luxury brands, designer pieces, and premium quality clothing that would be prohibitively expensive if purchased new, thereby enabling them to maintain fashionable appearances while staying within their budget constraints. The literature strongly emphasizes that online second-hand shopping platforms provide exceptional convenience through comprehensive home delivery services that eliminate the need for consumers to travel to physical stores, detailed product descriptions with multiple photographs that provide thorough visual information about item conditions, and customer review systems that help

buyers make well-informed purchasing decisions based on previous buyers' experiences and seller reliability ratings without needing to physically inspect items before purchasing.

The study also identified and thoroughly documented that second-hand online shopping platforms have implemented various sophisticated features and security mechanisms specifically designed to build buyer trust and ensure comprehensive transaction security for all parties involved in the marketplace. These essential features include comprehensive buyer protection programs that offer refunds or replacements for items that do not match their descriptions, detailed seller rating and review systems that provide transparent information about seller reliability and product quality, secure payment processing systems that protect financial information and prevent fraud, and formal dispute resolution mechanisms that provide structured processes for addressing conflicts between buyers and sellers when issues arise during transactions. The literature clearly shows that platforms like Carousell have become exceptionally popular in the Philippines because they allow local Filipino buyers and sellers to connect directly within the same geographic region without international shipping fees, customs duties, or extended delivery times that often accompany cross-border transactions on international platforms. These significant findings demonstrate that second-hand online shopping platforms effectively address common consumer concerns about online purchasing including product quality uncertainty, seller reliability doubts, payment security fears, and delivery reliability issues, thereby providing reliable and trustworthy alternatives to traditional retail shopping for budget-conscious Filipino consumers who seek quality merchandise at affordable prices.

The literature further revealed through comprehensive analysis that second-hand online shopping has made substantial contributions to the growing sustainability movement among Filipino consumers who are becoming

increasingly environmentally conscious and aware of the negative environmental impacts of fast fashion and overconsumption. By purchasing preloved items instead of brand-new merchandise, consumers actively reduce clothing waste that would otherwise end up in landfills, extend the useful life of garments and other products, and support circular fashion economy practices that prioritize reuse and recycling over disposal and replacement. The platform features also effectively enable sellers to earn supplemental income from items they no longer need or want, creating mutually beneficial economic opportunities for both buyers seeking affordable fashion and sellers looking to declutter their belongings while generating additional revenue. This literature comprehensively demonstrates that second-hand online shopping serves both critical economic purposes by providing affordable access to quality merchandise and important environmental purposes by supporting sustainable consumption practices and reducing the environmental footprint of the fashion industry in the Philippine consumer market.

Based on the comprehensive findings and thorough analysis presented throughout this literature, the researchers strongly recommended that Filipino consumers actively explore second-hand online shopping platforms as legitimate and viable alternatives to traditional retail shopping for fashion, lifestyle, and home products. They specifically suggested that buyers should conduct thorough research on seller ratings and reputations before making purchases, read product descriptions carefully and examine all provided photographs meticulously, take full advantage of buyer protection programs and secure payment methods when making online purchases, and maintain clear communication with sellers regarding item conditions, shipping arrangements, and return policies. By using second-hand online shopping platforms responsibly and strategically, Filipino consumers may successfully access quality branded items at significantly affordable prices while simultaneously supporting sustainable consumption practices that benefit both individual consumers and the broader

environment. The literature conclusively emphasizes the critical importance of second-hand online shopping as a rapidly growing, legitimate, and increasingly dominant segment of the Philippine e-commerce market that will continue to expand as more consumers recognize its economic and environmental benefits.

Local Studies

Local studies provide the empirical basis for the proposed system because they show how Filipino businesses actually use digital tools in practice. These studies are especially valuable because they reveal the operational benefits, limitations, and adoption patterns of systems designed for small enterprises in the Philippines. In the case of Margaux Collections, the local studies reviewed here support the need for a more structured ordering platform that can improve efficiency, accuracy, and customer trust. They also demonstrate that even small-scale businesses can benefit significantly from digital solutions when these are designed according to their actual workflow, resources, and operational challenges.

Developing a Point-of-Sale Inventory Management System for Lords Grace General Merchandising

Antonio, Reyes, and Torres, 2017

The development of web-based point-of-sale and inventory management systems has become an important area of applied computing research in the Philippines, especially for small businesses that need affordable and practical digital tools.

Antonio, Reyes, and Torres (2017) developed a PHP and MySQLbased inventory management system for Lords Grace

General Merchandising, a small retail business in the country, as part of an undergraduate capstone project at the Polytechnic University of the Philippines. Their work demonstrates how local businesses can benefit from digital systems that help improve inventory control, sales tracking, and overall operational efficiency. It also shows that even small enterprises can successfully adopt technology when the system is designed according to their actual needs and limitations. In this sense, the study serves as a practical example of how information systems can be applied in a real business environment to solve everyday operational problems.

The system developed by Antonio et al. (2017) included several important modules that supported day-to-day retail operations. These modules covered product management, which allowed users to add, edit, and delete product records; inventory tracking, which monitored stock levels and generated low-stock alerts; sales recording, which logged transactions and produced receipts; and basic reporting, which summarized sales performance over specific periods and identified top-selling products. Together, these features created a more organized and efficient workflow compared to manual recordkeeping. Instead of relying on notebooks, memory, or scattered paper records, the business could use a centralized digital system that stored data more accurately and made it easier to access when needed. This kind of structure is especially useful in small business settings where the owner or staff may need to manage several responsibilities at the same time and cannot afford delays or mistakes in recordkeeping.

The study also showed that the adoption of a digital system helped reduce common operational problems. Manual methods often led to misplaced records, delayed updates, and inconsistencies in stock counts. By contrast, the computerbased system allowed information to be updated more quickly and consistently. This is important because inventory accuracy affects not only internal management but also customer satisfaction. When stock records are wrong,

businesses may promise products that are no longer available or fail to notice items that need replenishment. Antonio et al. (2017) therefore demonstrate that a well-designed system can improve reliability and reduce preventable errors. Their findings also suggest that digital recordkeeping supports better decision-making because owners can monitor sales trends more effectively and respond to changes in product demand with greater confidence.

Another important contribution of the study is that it highlights the practicality of using widely accessible technologies such as PHP and MySQL for local business applications. These technologies are commonly used in academic and small-scale commercial system development because they are relatively affordable, flexible, and easy to maintain. For small enterprises in the Philippines, this is significant because budget limitations often prevent them from adopting expensive proprietary software. The system developed by Antonio et al. (2017) shows that a business does not need a complex or costly platform to improve its operations. What it needs is a system that is functional, user-friendly, and aligned with its actual processes. This makes the study a useful model for the development of similar systems in other small retail businesses.

These findings are highly relevant to Margaux Collections, since the business also needs a reliable way to manage products, sales, and inventory. Like the business in the study, Margaux Collections currently faces the challenge of handling product information and transactions in a more informal way. The proposed Ordering System is designed to address these same concerns by providing a structured platform for product management, sales recording, and inventory monitoring. This makes Antonio et al. (2017) a strong local precedent for the system, because it proves that a PHP and MySQL-based solution can work effectively in a small Philippine business setting. More importantly, it demonstrates that local businesses can benefit from a

formal digital system that organizes their operations, reduces errors, and supports more efficient customer service.

The relevance of this study to Margaux Collections goes beyond technical similarity. It also supports the idea that an ordering system can serve as a practical management tool that improves the overall business process. By centralizing product records and monitoring stock movements in real time, the system can help the owner avoid duplication, confusion, and manual errors. It can also make it easier to update product availability, track sales performance, and review which items are most frequently purchased. In a small fashion business that offers both brand-new and preloved products, these functions are especially important because product conditions, categories, and availability must be carefully monitored. Antonio et al. (2017) therefore provide a strong foundation for justifying the need for a similar system in Margaux Collections.

In addition, the study emphasizes that digital systems are most effective when they are designed around the actual workflow of the business. This is a crucial point for Margaux Collections because the proposed Ordering System is not intended to be a generic e-commerce platform. Instead, it is being developed to match the specific needs of the business, including managing fashion items, separating brand-new from preloved products, and tracking customer orders more systematically. The study by Antonio et al. (2017) supports this approach by showing that practical, business-specific systems are more useful than broad or overly complicated applications. As a result, the study reinforces the importance of tailoring the system to the real operational needs of the business rather than forcing the business to adjust to an unsuitable digital tool.

Analysis of Inventory Management Systems of Small-Sized Restaurants in Quezon Province

Santos, Dela Cruz, and Lopez, 2021

Santos, Dela Cruz, and Lopez (2021), in the *Philippine Journal of Hospitality and Service Management*, examined the inventory management practices of small-sized restaurants in Quezon Province, Philippines, in order to identify effective methods and common weaknesses that could guide the development of better inventory systems. Their study used a mixed-methods approach that combined inventory audits, interviews with business owners, and a comparison of manual and digital inventory management methods. Although the setting of the study was the restaurant industry, the research offers valuable insights that extend beyond food service because it focuses on the practical challenges of tracking inventory accurately, minimizing waste, and improving daily operational efficiency. The study is especially relevant to small enterprises because it reflects the realities of businesses that often operate with limited staff, limited time, and limited resources, yet still need to maintain accurate and dependable records.

One of the central findings of the study was that manual inventory methods frequently resulted in delays, errors, and incomplete records. Small restaurant owners found it difficult to monitor their stock consistently because inventory levels changed often and updates were sometimes made only after problems had already occurred. This kind of reactive recordkeeping created confusion and made it harder to maintain accurate stock visibility. In many cases, manual methods also increased the risk of overordering, underordering, and wasting supplies, especially when the business depended on memory or handwritten logs. These challenges show that inventory management is not merely an administrative task, but a critical operational function

that directly affects business continuity and service quality.

In contrast, the study found that digital inventory systems provided better control because they allowed owners to monitor stock movement more systematically and respond more quickly to changes in demand. With a digital system, updates could be made in a more organized and timely way, which improved the accuracy of records and reduced the likelihood of errors. This is important because inventory accuracy influences not only internal operations but also customer satisfaction. When a business knows what is available and what has already been used or sold, it can serve customers more efficiently and avoid disappointing them with unavailable items. For small businesses, this level of control can make a significant difference in daily performance and long-term reliability.

The study also identified three important factors that influenced successful digital inventory management adoption among small Philippine businesses: simplicity of the user interface, reliability of the system, and perceived usefulness compared with the old method. These factors are particularly important for microbusinesses because owners often have multiple responsibilities and cannot afford to spend much time learning complicated software. A system must therefore be easy to understand, dependable in actual use, and clearly better than the manual process it replaces. If a system is too complex, business owners may avoid using it consistently, which defeats the purpose of digitalization. This finding suggests that the success of an inventory system depends not only on its technical features, but also on how practical and user-friendly it is in everyday business operations.

Another significant implication of the study is that digital inventory systems are more likely to be adopted when business owners can immediately see their value. If the system helps save time, improves accuracy, and reduces

stress in managing stock, then it becomes easier to justify its use. This is especially relevant in small-scale enterprises where owners often make decisions based on direct usefulness and observable outcomes. In this sense, the study demonstrates that technology adoption among small businesses is strongly shaped by practical benefits rather than abstract innovation alone. A system that clearly simplifies work and improves recordkeeping is more likely to be embraced and used consistently.

Although the restaurant industry is different from fashion retail, the core inventory management tasks are very similar. Both types of businesses must track stock levels, record sales, identify fast-moving and slow-moving items, and determine when products need to be replenished. Both also face the challenge of ensuring that records match actual stock conditions so that business decisions can be made accurately. Because of these shared operational needs, the findings of Santos et al. (2021) are highly relevant to Margaux Collections. The proposed Ordering System should support the same basic inventory functions that the study found to be effective in restaurants, especially in terms of stock monitoring, system reliability, and ease of use.

For Margaux Collections, this means that the Ordering System should help the business owner monitor what items are available and which ones have already been sold in real time. This function is important because the business handles both brand-new and preloved fashion items, each of which must be tracked carefully to avoid confusion. Since preloved items are often unique or limited in quantity, the risk of duplication or incorrect availability is higher if inventory is not managed properly. A digital system can reduce this risk by keeping product records organized and updated. It can also help the owner identify which items are selling quickly and which are remaining in stock for longer periods, which supports better planning and decision-making.

The study by Santos et al. (2021) also supports the idea that a good inventory system should be designed with the actual needs of the business in mind. For Margaux Collections, this means that the Ordering System should not be overly complicated or filled with unnecessary features that do not contribute to daily operations. Instead, it should focus on practical functions such as product listing, stock updating, sales tracking, and order monitoring. These features can help the business operate more smoothly and professionally while reducing the burden of manual recordkeeping. In doing so, the system becomes a useful tool for both operational control and business growth.

Overall, Santos, Dela Cruz, and Lopez (2021) provide a strong local basis for the development of the Margaux Collections Ordering System. Their study shows that small businesses benefit from digital inventory systems because these improve accuracy, reduce errors, and make stock monitoring more efficient. More importantly, the study demonstrates that successful system adoption depends on simplicity, reliability, and usefulness, which are all essential qualities for a small fashion business. These insights support the need for an organized, user-friendly, and dependable ordering system that can help Margaux Collections manage its products more effectively and respond better to customer needs.

Impact of Inventory Management on Financial Performance of MSMEs

Gaoat, Cabungcal, and Esteban, 2023

Gaoat, Cabungcal, and Esteban (2023) examined how inventory management practices affect the financial performance of MSMEs in Santiago City, Philippines. Their study used a quantitative research design and surveyed MSME owners and managers to determine how inventory management quality relates to financial performance indicators such as

revenue, profit margin, and return on investment. The results provided strong evidence that inventory control is not merely an administrative task, but a business function with direct financial consequences. This makes the study highly relevant to small enterprises because it shows that the way a business handles stock can influence not only daily operations but also long-term profitability and sustainability.

The study found that businesses with better inventory practices tended to perform more effectively because they could avoid overstocking, reduce losses, and respond more accurately to customer demand. This is an important result because it shows that poor inventory control can lead to unnecessary costs, while efficient inventory management can support profitability and long-term sustainability. When stock is tracked properly, business owners are able to make better decisions about which items should be reordered, which products should be promoted, and which items are not moving quickly enough. In this way, inventory management becomes more than a background process; it becomes a strategic part of business planning and financial control.

Another important implication of the study is that inventory data should be used to guide decision-making rather than being treated as a simple recordkeeping task. The

researchers emphasized that businesses can achieve better financial results when they rely on accurate and up-to-date stock information. This means that inventory records should not only show what is available but should also help owners understand sales trends, purchasing patterns, and product performance. For MSMEs, this is especially useful because they often operate with limited capital and cannot afford mistakes such as overbuying items that do not sell or failing to restock products that customers want. Accurate inventory data therefore supports smarter planning and more efficient use of resources.

The study also suggests that good inventory management helps businesses respond more effectively to customer demand.

When owners know which products are popular and which ones are slow-moving, they can adjust their sales and restocking strategies accordingly. This improves the business's ability to meet customer needs while minimizing waste and unnecessary storage costs. For small businesses, this kind of responsiveness is essential because customer preferences can change quickly, and financial margins are often tight. By maintaining control over stock levels, MSMEs can improve both customer satisfaction and operational efficiency at the same time.

For Margaux Collections, this study reinforces the importance of developing an ordering system that includes accurate stock tracking, sales summaries, and reporting features. Since the business sells both brand-new and preloved items, the owner must keep track of different item conditions, categories, and availability. A digital system can help ensure that the stock database is updated in real time, which reduces the risk of selling unavailable items or failing to notice slow-moving products. These improvements can contribute to better financial performance by helping the business owner make informed decisions about pricing, restocking, and promotion. In other words, the system can help the business not only stay organized but also operate more profitably.

The study is also relevant because it highlights the connection between operational efficiency and financial stability. When inventory is managed properly, businesses can reduce losses caused by stock mismanagement, product damage, or unnecessary purchasing. This is especially significant for a small fashion business like Margaux Collections, where every product matters and inventory decisions directly affect cash flow. A digital ordering system can support this by giving the owner a clearer overview of product movement and helping identify which items are contributing most to sales. As a result, the business can allocate attention and resources more effectively.

Another useful point from the study is that inventory management should be viewed as a continuing business process rather than a one-time task. MSMEs need systems that can adapt as their operations grow and their product lines change. For Margaux Collections, this means that the Ordering System should be flexible enough to handle new products, preloved items, and updated customer orders without creating confusion. The ability to monitor and analyze inventory consistently can help the business remain responsive and financially stable over time.

Overall, the local studies reviewed here show that digital systems are highly valuable for small Philippine businesses. Antonio et al. (2017) demonstrate that PHP and MySQL-based systems can improve retail operations; Santos et al. (2021) show that digital inventory systems are more reliable than manual methods; and Gaoat et al. (2023) confirm that effective inventory management supports stronger financial performance. Together, these studies justify the development of the Margaux Collections Ordering System as a practical and research-based solution for a small fashion business in the Philippines. They also show that a well-designed system can strengthen both operational control and financial decision-making, which are essential for the long-term success of any MSME.

Digital Ukaynomics (Tomoling, 2024)

Tomoling's (2024) undergraduate thesis, *Digital Ukaynomics: Analysis on the Market for Secondhand Clothing Sold through Online Channels*, investigates the rapidly growing online market for secondhand clothing in the Philippines. The study focuses on consumers who purchase pre-loved garments through Facebook groups, a platform that has become central to the digital ukay ecosystem. By examining socio-demographic profiles and purchase-influencing factors, the research provides a localized understanding of how e-

commerce has reshaped traditional ukay-ukay consumption patterns in the country.

The study employed a quantitative design using a Google Forms questionnaire distributed across multiple Facebook groups dedicated to secondhand clothing. A total of 248 respondents participated, comprising 200 females and 48 males, with the majority either employed or currently studying. This gender-skewed sample reflects broader trends in online thrift shopping, where female consumers dominate the market. The focus on Facebook groups is particularly relevant given that these platforms serve as informal yet highly organized marketplaces for ukay sellers and buyers across the Philippines.

Using ordinary least squares (OLS) regression, the study identified several key factors that significantly affect consumer demand for secondhand clothes sold online. These include age, sex, income, willingness to pay for both brandnew and secondhand clothing, delivery fees, buying experience, and browsing time. The findings suggest that online ukay consumption is not merely driven by affordability but is also influenced by logistical considerations such as delivery costs and the consumer's familiarity with online shopping platforms. This nuanced understanding helps explain why certain demographic groups are more likely to engage in digital ukay transactions.

The logarithmic regression analysis further revealed that secondhand clothes sold online are normal goods with inelastic demand. This means that as consumer income increases, demand for online ukay clothing also increases, but at a slower proportional rate. The inelastic nature of demand indicates that once consumers enter the online ukay market, they tend to remain consistent buyers regardless of minor price fluctuations. This finding has important implications for sellers strategies and for policymakers considering regulations on secondhand clothing imports.

Overall, Tomoling's study contributes significantly to the growing body of local literature on secondhand clothing

consumption in the Philippines. It demonstrates how digital platforms have transformed ukay-ukay from a physically bound, necessity-driven market into a sophisticated online consumer behavior pattern. The research also highlights the tension between the legal framework (R.A.

4653, which prohibits commercial importation of used textiles) and the widespread, accepted practice of online ukay trading, suggesting that policy discussions must account for the reality of digital marketplace dynamics.

Pre-Loved Clothing and Local Fashion Industry in Kalibo (Itubay, 2025)

Itubay's (2025) qualitative study, *Pre-Loved Clothing and Local Fashion Industry: Its Marketability, Sustainability and Challenges*, examines the coexistence of pre-loved clothing businesses and local fashion enterprises in Kalibo, Aklan. The research addresses a critical gap in local literature by exploring how these two distinct sectors navigate competition, sustainability concerns, and market dynamics within a single geographic context. Using focus group discussions with 15 business owners (seven from preloved clothing and eight from local fashion), the study provides rich, context-specific insights into the Philippine fashion economy.

The study identified several key marketing strategies employed by both pre-loved and local fashion businesses. Social media platforms, particularly Facebook, Instagram, and TikTok, emerged as primary tools for visibility and customer engagement. Pre-loved clothing sellers leveraged live selling, multiple Facebook pages, and integration with e-commerce platforms like Shopee to attract tech-savvy, price-conscious consumers. Meanwhile, local fashion businesses relied more heavily on word-of-mouth referrals and personal relationships, though they also increasingly adopted digital channels. Both sectors emphasized quality

control, flexible pricing, and personalized customer service as critical success factors. Sustainability strategies were another central theme, with both industries demonstrating adaptability to changing fashion trends and consumer preferences. Pre-loved clothing businesses catered to younger, trend-conscious buyers by continuously updating their inventory based on social media influences and celebrity culture. Local fashion businesses maintained consistency in operations through regular posting schedules and customer engagement activities. Both sectors also implemented sustainable pricing strategies, loyalty programs, and promotions such as Buy 1 Take 1 offers to retain customers and move inventory effectively.

Despite their differences, both industries faced significant shared challenges. Sourcing reliable suppliers proved difficult for pre-loved clothing sellers due to the irregular nature of bulk secondhand clothing supply. Quality inconsistencies in pre-loved items required meticulous inspection and often repairs before sale. Local fashion businesses struggled with workforce management issues, including inconsistent staff performance, motivation, and recruitment difficulties. Both sectors also faced pressure to meet customer expectations regarding product quality, pricing transparency, and delivery timelines.

The study's findings underscore the resilience and adaptability of small fashion businesses in Kalibo. Coping mechanisms included forming strategic partnerships with suppliers, seeking financial support through family loans, providing personalized services to strengthen customer relationships, and leveraging community engagement through word-of-mouth marketing. Importantly, the research challenges the narrative that ukay-ukay necessarily harms local fashion industries, instead showing how both sectors can coexist through differentiation, complementary market positioning, and shared commitment to sustainability. This has important implications for local policymakers and

industry stakeholders seeking to support both sectors simultaneously.

From Necessity to Identity (Frontiers in Sustainability, 2022)

The study *From Necessity to Identity: Entrepreneurial Perspectives on Key Motivations for Filipino Second-Hand Clothing Consumers* (published in *Frontiers in Sustainability*, 2022) explores the evolving motivations behind Filipino engagement with ukay-ukay through observational research and in-depth interviews. Unlike earlier studies that framed secondhand clothing consumption purely as an economic necessity, this research reveals a more complex consumer landscape where affordability intersects with identity construction and sustainable fashion values. The study provides critical insights into how Philippine consumers have reinterpreted ukay-ukay from a stigma-laden necessity into a form of self-expression .

The research identified two primary motivations driving Filipino consumers toward secondhand clothing: economic necessity and identity construction. On the economic dimension, affordability remains a fundamental driver, particularly for low- and middle-income consumers who seek quality clothing at accessible prices. However, the study found that this economic motivation has evolved alongside growing environmental consciousness. Many consumers now explicitly frame their ukay-ukay purchases as sustainable fashion choices, aligning personal consumption with broader environmental values and circular economy principles .

The identity construction dimension represents a significant shift in how Filipinos perceive secondhand clothing. The study documented how consumers, particularly younger demographics, use ukay-ukay to curate unique fashion statements that distinguish them from mainstream fastfashion consumers. Vintage pieces, rare finds, and one-of-a-kind garments enable consumers to express individuality

while avoiding the homogenization associated with massproduced clothing. This identity-driven consumption reflects global trends in sustainable fashion but is distinctly localized within Philippine cultural contexts, where resourcefulness and creativity are valued traits .

Entrepreneurial perspectives highlighted in the study further illuminate how ukay-ukay sellers have adapted to these shifting consumer motivations. Sellers increasingly position their businesses not merely as affordable clothing providers but as curators of unique fashion and promoters of sustainable consumption. This repositioning has enabled ukay-ukay businesses to attract consumers who might otherwise have avoided secondhand clothing due to stigma. The study also noted how social media has facilitated this transformation, allowing sellers to showcase curated collections and build communities around sustainable fashion values .

This research contributes importantly to local literature by documenting the cultural transformation of ukay-ukay consumption in the Philippines. It moves beyond purely economic explanations to capture the psychological, social, and environmental dimensions of secondhand clothing consumption. The findings suggest that policy and business strategies should account for both the affordability imperative and the identity-sustainability values that now drive the market. Understanding this dual motivation is essential for developing interventions that support both economic accessibility and sustainable fashion transitions in the Philippine context .

The Philippine Ukay-Ukay Culture as Sustainable Fashion (Biana, 2020)

Biana's (2020) study, *The Philippine Ukay-Ukay Culture as Sustainable Fashion*, published in the DLSU Business & Economics Review, provides a comprehensive examination of ukay-ukay as both a cultural phenomenon and a sustainable fashion practice in the Philippines. The research explores

how the term "ukay-ukay," derived from "halukay" (to dig), captures the essence of Philippine thrift shopping culture where consumers sift through piles of pre-loved clothing to discover unique items. This study is particularly significant as it frames ukay-ukay not merely as an economic necessity but as a legitimate sustainable fashion movement that aligns with global circular economy principles while maintaining distinctively Filipino cultural characteristics [DLSU, 2020].

The research documented that ukay-ukay consumers are predominantly aged 21-30 years old, female, college graduates, fully employed, and belong to income class B (Php 20,001-40,000). This demographic profile challenges the stereotype that secondhand clothing is exclusively for low-income consumers, revealing instead that educated, middle-class Filipinos actively participate in ukay-ukay culture. The study found that these consumers are motivated by both economic considerations and growing environmental consciousness, with many explicitly framing their ukay purchases as sustainable fashion choices. This dual motivation creates a unique market position that distinguishes Philippine ukay-ukay from secondhand clothing markets in other countries [DLSU, 2020].

The study emphasized the cultural significance of ukay-ukay in Philippine society, noting how it has evolved from wet market stalls to upscale boutiques and online platforms. This transformation reflects broader changes in consumer perceptions, where secondhand clothing has shed much of its stigma and is now associated with creativity, resourcefulness, and environmental awareness. The research highlighted how ukay-ukay has become particularly popular among Filipino youth, who view thrift shopping as a way to express individuality while avoiding the homogenization of fast fashion. This cultural shift has enabled ukay-ukay to thrive despite legal restrictions on commercial importation of used textiles [DLSU, 2020].

Biana's research also examined the sustainability implications of ukay-ukay consumption, positioning it as a practical manifestation of circular fashion economy principles in the Philippine context. By extending the lifecycle of clothing items and reducing textile waste, ukay-ukay contributes to environmental sustainability while providing affordable fashion options. The study noted that the Philippines leads Southeast Asia in secondhand fashion transactions, accounting for over 30% of regional activity according to Carousell data. This position suggests that Philippine ukay-ukay culture has achieved critical mass as a mainstream consumption pattern rather than remaining a marginal alternative [DLSU, 2020].

This study makes important contributions to local literature by bridging cultural, economic, and environmental perspectives on ukay-ukay consumption. It provides empirical evidence that Filipino consumers engage with secondhand clothing for complex, overlapping motivations that extend beyond simple affordability. The findings have implications for policymakers considering regulatory approaches to the industry, suggesting that formalization rather than prohibition might better serve both economic and environmental objectives. For researchers, the study opens avenues for further investigation into how ukay-ukay culture intersects with Philippine identity, sustainability values, and consumer behavior patterns [DLSU, 2020].

How Facebook Communication Strategies Facilitate Ukay-Ukay Live Selling (2024)

The study *How Facebook Communication Strategies Facilitate Ukay-Ukay Live Selling* (2024), published in *European Proceedings of Social and Behavioural Sciences*, examines the communication tactics employed by Philippine ukay-ukay sellers during live selling sessions and their effects on buyer consumption behavior. This research addresses a critical gap in understanding how digital platforms have transformed

traditional ukay-ukay selling into interactive, real-time commerce experiences. The study is particularly relevant given that Facebook Live has become the dominant selling channel for online ukay businesses, creating new dynamics between sellers and buyers that differ significantly from traditional retail interactions [European Proceedings, 2024].

The research identified several key communication strategies employed by ukay-ukay sellers during live sessions. These include real-time product demonstrations where sellers display items from multiple angles, immediate responses to buyer questions about size, condition, and pricing, creation of urgency through limited-time offers and flash sales, and building personal connections through informal, conversational styles. Sellers effectively use social proof by highlighting previous purchases and positive customer feedback, while also employing humor and entertainment to maintain viewer engagement throughout extended live sessions. These strategies create a shopping experience that combines transactional efficiency with social interaction [European Proceedings, 2024].

The study found that effective Facebook communication strategies significantly influence buyer consumption behavior, with several mechanisms explaining this relationship. Real-time interaction reduces purchase uncertainty by allowing buyers to obtain immediate information about product conditions, addressing the fundamental challenge of online apparel retailing where customers cannot physically inspect items. The interactive nature of live selling creates a sense of community among participants, fostering loyalty and repeat purchases. Additionally, the entertainment value of live sessions encourages prolonged viewing, increasing the likelihood of impulse purchases and larger basket sizes. Sellers who master these communication tactics report substantially higher conversion rates compared to those who rely solely on static product listings [European Proceedings, 2024].

The research also documented challenges faced by ukay-ukay sellers in maintaining effective live selling operations. Technical issues including unstable internet connections, poor lighting, and audio problems can disrupt sessions and reduce professional quality. Managing high volumes of simultaneous chat messages requires organizational skills and quick decision-making abilities. Sellers must also navigate the unpredictability of viewer engagement, as attendance fluctuates significantly between sessions. Despite these challenges, the study found that successful sellers continuously refine their communication strategies through practice, feedback, and observation of competitor tactics [European Proceedings, 2024].

This study makes significant contributions to local literature by providing empirical evidence on the mechanics of digital ukay-ukay commerce in the Philippines. It demonstrates how social media platforms have enabled small businesses to compete effectively through low-cost, high-impact marketing strategies. The findings have practical implications for aspiring ukay entrepreneurs, suggesting that investment in communication skills and live selling techniques can yield substantial returns. For policymakers and support organizations, the research highlights the importance of digital literacy training and technical infrastructure development for small businesses operating in the online ukay sector [European Proceedings, 2024].

Level of Satisfaction Among Women on Online Ukay-Ukay (2023)

The study *Level of Satisfaction Among Women on Online Ukay-Ukay* (2023), published in the International Journal of Academic Research and Innovative Engineering Education (IJARIE), investigates female consumer satisfaction with online ukay-ukay purchasing experiences in the Philippines. This research is significant because women comprise the majority of ukay-ukay consumers, yet their specific experiences, concerns, and satisfaction drivers have

received limited academic attention. The study employs quantitative methods to measure satisfaction levels across multiple dimensions including product quality, pricing, delivery service, customer communication, and overall value for money [IJARIE, 2023].

The research identified several key factors influencing women's satisfaction with online ukay-ukay purchases. Product quality emerged as the most critical determinant, with respondents expressing concerns about garment condition, accuracy of product descriptions, and discrepancy between photographed items and actual delivery. Price competitiveness remained important, but satisfied customers were willing to pay slightly higher prices for guaranteed quality and reliable service. Delivery efficiency significantly impacted satisfaction, with delayed shipments and poor packaging leading to negative experiences. Customer communication quality, including response time and problem resolution, also emerged as a substantial factor affecting overall satisfaction ratings [IJARIE, 2023].

The study found that women's satisfaction with online ukayukay is generally moderate to high, with mean satisfaction scores indicating positive overall experiences. However, the research also identified specific pain points that undermine satisfaction, including inconsistent sizing information, difficulty assessing garment condition from photographs, and limited recourse for dissatisfied customers. Respondents expressed preference for sellers who provide detailed condition descriptions, multiple photographs from different angles, clear sizing charts, and transparent return policies. Trust-building mechanisms including seller ratings, customer reviews, and social proof were identified as critical for reducing perceived risk and increasing satisfaction [IJARIE, 2023].

The research documented demographic patterns in satisfaction levels, finding that younger women (aged 18-30) reported

higher satisfaction with online ukay-ukay compared to older age groups. This generational difference reflects greater digital literacy, familiarity with online shopping platforms, and cultural acceptance of secondhand clothing among younger consumers. The study also found that repeat customers reported higher satisfaction than first-time buyers, suggesting that experience with online ukay-ukay reduces uncertainty and improves ability to make satisfying purchases. This learning effect has implications for seller strategies focused on building customer loyalty and encouraging repeat business [IJARIE, 2023].

This study contributes important insights to local literature on female consumer behavior in the ukay-ukay sector. It provides empirical evidence that women's satisfaction with online ukay-ukay depends on multiple interconnected factors extending beyond simple price considerations. The findings suggest that sellers who invest in quality control, transparent communication, and reliable service delivery can differentiate themselves in a competitive market and build sustainable customer relationships. For researchers, the study identifies opportunities for further investigation into gender-specific consumption patterns, satisfaction drivers across different ukay segments, and the long-term relationship between satisfaction and customer loyalty in the online ukay market [IJARIE, 2023].

Filipino Gen Z Consumer Behavior Towards Ukay-Ukay (DLSU, 2024)

The study *Filipino Gen Z Consumer Behavior Towards UkayUkay and their Sustainable Fashion Practices* (2024), published in the De La Salle University SHS Research Convention Proceedings, examines how Filipino Generation Z consumers engage with ukay-ukay and their motivations for participating in

sustainable fashion practices. This research is particularly timely given that Gen Z represents the fastest-growing segment of ukay-ukay consumers and is driving the cultural transformation of secondhand clothing from stigma to mainstream fashion choice. The study employs survey methodology to understand the attitudes, behaviors, and values that shape Gen Z's relationship with ukay-ukay [DLSU, 2024].

The research found that Filipino Gen Z consumers engage with ukay-ukay primarily for three interconnected reasons: affordability, sustainability consciousness, and unique fashion expression. Affordability remains a fundamental driver, with Gen Z consumers facing economic constraints including student loans, entry-level salaries, and rising cost of living. However, the study revealed that sustainability consciousness has become equally important, with many Gen Z respondents explicitly framing their ukay purchases as environmental choices aligned with values around climate action and waste reduction. Unique fashion expression represents the third motivation, as Gen Z consumers use ukay-ukay to curate distinctive personal styles that differentiate them from peers wearing massproduced fast fashion [DLSU, 2024].

The study documented that Gen Z consumers predominantly access ukay-ukay through digital platforms, with Facebook Marketplace, Instagram, and TikTok serving as primary channels. These platforms align with Gen Z's digital-native communication preferences and provide access to curated selections beyond traditional physical ukay stores. The research found that Gen Z consumers actively research sellers online, checking ratings, reviews, and social media presence before making purchases. They also engage with ukay-ukay content creators who showcase styling tips, thrift hauls, and sustainable fashion education, creating a digital ecosystem that normalizes and celebrates secondhand clothing consumption [DLSU, 2024].

The research highlighted that Gen Z's ukay-ukay participation reflects broader values around circular economy and sustainable consumption. Many respondents expressed concern about textile waste, fast fashion environmental impacts, and the ethical implications of clothing production. This values-driven consumption distinguishes Gen Z from previous generations who primarily engaged with ukay-ukay for economic necessity. The study noted that this values alignment creates strong brand loyalty, with Gen Z consumers preferring sellers who communicate sustainability messages and demonstrate environmental responsibility. This trend suggests that ukayukay's future growth will depend on maintaining alignment with Gen Z's sustainability values [DLSU, 2024].

This study makes significant contributions to local literature by documenting the generational shift in ukayukay consumption patterns and motivations. It provides empirical evidence that Filipino Gen Z represents a critical demographic for sustainable fashion transitions, with high engagement rates and strong environmental values. The findings have implications for policymakers, educators, and businesses seeking to promote circular economy practices, suggesting that Gen Z's ukay-ukay participation can be leveraged for broader sustainability education and behavior change initiatives. For researchers, the study identifies opportunities for longitudinal investigation into how Gen Z's sustainable fashion values evolve as they age and their purchasing power increases [DLSU, 2024].

Experiences of Online Sellers of Pre-Loved Clothes (Zenodo, 2023)

The study *Experiences of Online Sellers of Pre-Loved Clothes* (2023), archived on Zenodo, explores the lived experiences, challenges, and adaptive strategies of online

pre-loved clothing sellers in the Philippines. This qualitative research provides critical seller-centered perspectives that complement consumer-focused studies, offering a more complete understanding of the digital ukay ecosystem. Through in-depth interviews and observational methods, the research captures the day-to-day realities of individuals who have transitioned into online ukay entrepreneurship, documenting both opportunities and obstacles they encounter in this informal yet highly dynamic market [Zenodo, 2023].

The study documented several critical challenges faced by online pre-loved clothing sellers. Inventory sourcing emerged as a primary difficulty, with sellers reporting irregular availability of quality items, competition from other sellers for desirable pieces, and the physical labor involved in sorting, cleaning, and preparing garments for sale. Pricing complexity presented another challenge, as sellers must balance competitiveness with profitability while accounting for variable item conditions and uncertain demand. Platform dependency created vulnerability, with sellers reporting that algorithm changes on Facebook, Instagram, or Shopee could dramatically affect visibility and sales. Fraud and scam attempts by buyers, including fake payment screenshots and refusal to claim items, represented ongoing financial risks that required constant vigilance [Zenodo, 2023].

Despite these challenges, the study identified several adaptive strategies employed by successful online sellers. Multi-channel approaches emerged as critical, with sellers maintaining presence across multiple platforms (Facebook Marketplace, Instagram, Shopee, TikTok) to reduce platform dependency and maximize reach. Content creation capabilities, including product photography, video showcasing, and engaging captions, became essential competitive advantages. Relationship-building through personalized communication, follow-up messages, and loyalty programs helped develop repeat customer bases that provided

income stability. Financial management skills, including inventory tracking, cash flow management, and emergency fund maintenance, proved essential for navigating income volatility [Zenodo, 2023].

The research highlighted the entrepreneurial resilience of online ukay sellers, many of whom balanced selling with other employment or education. For some, online ukay selling represented a primary livelihood; for others, it served as supplementary income or pathway to full-time entrepreneurship. The study found that sellers developed diverse skill sets spanning product sourcing, digital marketing, customer service, photography, and financial management. This entrepreneurial adaptability reflects broader Philippine cultural values around resourcefulness and improvisation, enabling sellers to thrive despite operating in an informal, lightly-regulated market with limited institutional support [Zenodo, 2023].

This study makes important contributions to local literature by centering seller perspectives in understanding the digital ukay economy. It reveals that online ukay selling requires sophisticated capabilities extending far beyond simple reselling, encompassing digital literacy, business management, and interpersonal skills. The findings have practical implications for support organizations seeking to assist online ukay entrepreneurs, suggesting that capacitybuilding programs should address digital marketing, financial management, fraud prevention, and legal compliance. For policymakers, the research highlights the economic significance of online ukay selling as a source of livelihood and entrepreneurship, suggesting that formalization rather than prohibition might better serve both sellers and consumers [Zenodo, 2023].

Consumer Behavior Towards Online Clothes Shopping in San Fernando City, La Union

Torres (2021) conducted a local study to describe how consumers in the urban barangays of San Fernando City, La Union behave when buying clothes online. The research

mainly aimed to present the demographic profile of online clothing shoppers and to identify the factors that influence their decision to purchase, such as convenience, website features, time considerations, and security of online transactions. A descriptive research design using a survey questionnaire was applied to gather data from individuals who already had experience purchasing clothes through online platforms. This approach helped capture the actual situation of online clothes shopping in the area during the time of the study.

The study involved 382 respondents who were online clothes shoppers residing in selected urban barangays of San Fernando City, La Union. They were chosen based on their engagement in online clothing purchases so that the results would reflect real consumer behavior rather than hypothetical opinions. The questionnaire asked about their demographic characteristics, access to computers and the internet, and their self-rated internet skills. Using descriptive analysis, the researcher summarized these pieces of information to present a clear picture of the local online shopping environment. Results indicated that most of the online clothing shoppers were female employees aged 21 to 25 years old, earning a monthly income of at least Php 5,000. Many respondents had access to a computer unit and internet connection either at home or at work, which made it easier for them to shop online. They also described their internet skills as good, suggesting that they were confident in navigating online shopping sites and completing online transactions. These conditions contributed to their active participation in online clothing purchases.

The findings further showed that several factors strongly influenced their online buying behavior, particularly convenience, website design and features, time-saving advantages, and security concerns. Convenience referred to the ability to shop anytime and anywhere without visiting physical stores, while attractive and user-friendly websites helped improve the overall shopping experience.

Time-saving was also an important consideration, because online shopping reduced the need to travel and spend long hours in malls. In addition, the perceived safety of personal and financial information played a major role in determining whether customers felt comfortable continuing with their online purchases. Based on these results, Torres (2021) recommended that entrepreneurs should consider engaging in online

selling as an alternative way of doing business in the clothing industry. By focusing on improving website design, ensuring secure payment systems, and emphasizing the convenience and time efficiency offered by online shopping, business owners can better respond to the needs and preferences of online consumers. The study suggested that there is a promising market for online clothing businesses in San Fernando City, especially among young, employed female consumers with good internet access and skills.

Emergence of Online Clothing Business and Its Effect on Local Businesses in Tuguegarao City

Banera, Binarao, De Castro, Furigay, and Mosquito (2022) examined how the rise of online clothing businesses has affected local clothing enterprises in Tuguegarao City. The study aimed to describe the experiences of local entrepreneurs and assess the impact of online competitors on their business performance. A descriptive research design was used to present the current situation of physical clothing stores in the face of growing online competition. This makes the study important in understanding how traditional businesses are challenged by changes in technology and consumer behavior.

The respondents of the study were 30 local entrepreneurs or owners of clothing businesses in Tuguegarao City who were chosen through purposive sampling.

They were selected because they directly experienced the effects of online clothing competitors on their own businesses. Data were gathered using survey questionnaires that asked about the financial performance of their stores and the strategies they used to cope with competition. The descriptive approach helped organize and summarize the owners' perceptions and the changes they observed in their businesses over time.

Findings showed that the emergence of online clothing businesses had a significantly negative effect on the financial performance of local clothing stores. The respondents reported notable decreases in sales, profit, working capital, customer demand, and their ability to retain existing customers. The overall category mean of 2.7267 was interpreted as "significantly decreased," meaning that the decline in performance was considerable rather than slight. This suggests that many customers shifted their purchases from physical stores to online platforms, putting pressure on local clothing businesses.

The study also identified several coping strategies used by local entrepreneurs to respond to this challenge. All of the respondents (100%) used sales talk to persuade and maintain the interest of customers. Most of them (96.67%) offered discounts as an incentive to attract and retain buyers, while 70% provided exclusive bargains to compete with online promotions. These strategies show that local clothing businesses are trying to adjust to the competitive environment, even if their operations are primarily based on physical stores.

Based on the findings, the researchers recommended that local business owners consider venturing into online clothing business themselves. They suggested that entrepreneurs use various social media platforms to promote and sell their products so they can reach more customers and keep up with changing shopping preferences. By combining traditional store operations with online selling, local clothing businesses may improve their financial

performance and remain competitive in the digital marketplace. The study emphasizes the need for adaptation and innovation among local enterprises in response to the growth of online businesses.

Effectiveness of Online Services Offered by Pre-Loved Clothes Sellers in Batangas City

Group 1 (2021) conducted a study on the effectiveness of online services offered by pre-loved clothes sellers in Batangas City. The research aimed to determine the factors that consumers consider when buying pre-loved clothing online and how these factors contribute to the success of online sellers. Specifically, the study focused on aspects such as fashion, brand, price competitiveness, delivery efficiency, and customer satisfaction. This local study is relevant because it shows how online pre-loved clothing businesses respond to customer needs and preferences in a specific city context.

The study examined how these different factors influenced consumers' decisions to purchase pre-loved clothes from online sellers. Fashion and brand appeal were considered in terms of how trendy or recognizable the clothing items were, while price competitiveness referred to whether the items were affordable compared to other sellers. Delivery efficiency was also studied, focusing on how fast and reliable the delivery process was. Customer satisfaction was treated as an overall measure of how buyers felt about their experience, from browsing to receiving the product.

Findings from the research showed that social media platforms, particularly Facebook Marketplace and Instagram, served as the main channels for selling pre-loved clothes. These platforms allowed sellers to easily post photos of their products, interact with potential buyers, and manage inquiries and orders. Because many consumers were active on these social media sites, they became convenient places to discover and purchase pre-loved clothing. This highlights

the importance of choosing appropriate online platforms in running a successful pre-loved clothing business.

The study also found that the effectiveness of online services offered by pre-loved clothes sellers depended greatly on having a strong online presence. This meant being active and consistent in posting items, responding to messages, and updating customers about new arrivals or promotions. Flexible pricing strategies were another key factor, as sellers often adjusted prices or offered deals to stay competitive and attract more buyers. These practices helped them stand out among many other online sellers offering similar products.

Finally, the research emphasized that maintaining good relationships with customers was essential for encouraging repeat purchases. When sellers provided clear communication, responded politely to inquiries, and handled issues properly, customers were more likely to trust them and buy again. Good service and positive interactions contributed to customer satisfaction and loyalty, which are important for the long-term success of online pre-loved clothing businesses. Overall, the study suggested that a combination of strong online presence, flexible pricing, and good customer relations made online services more effective for pre-loved clothes sellers in Batangas City.

Promotional Strategies and Consumers' Purchase Intention on Garment Bazaar Retailers in the National Capital Region

Torres, Santos, and Lim (2023) examined the relationship between the level of effectiveness of promotional strategies used by garment bazaar retailers and consumers' purchase intention in the National Capital Region (NCR), Philippines. The study aimed to determine the relationship between promotional strategy effectiveness and consumers' attitude, subjective norms, perceived control, and purchase intention since the onset of the COVID-19 pandemic when garment bazaar retailers shifted to online selling. A survey research design was used to gather data from

consumers across different selling platforms including physical store, virtual store, company website, and social media. This makes the study important in understanding how promotional strategies can be optimized for different selling platforms and how consumer purchase intention varies across these platforms.

The respondents of the study were 510 consumers from 16 cities and 1 municipality of NCR, Philippines who were chosen through appropriate sampling methods. They were selected because they had experience purchasing from garment bazaar retailers during and after the COVID-19 pandemic. Data were gathered using online survey questionnaires that asked about consumers' perceptions of promotional strategy effectiveness, their attitude toward garment purchases, subjective norms, perceived behavioral control, and purchase intention across different selling platforms. The survey approach helped organize and summarize the consumers' perceptions and the relationships between promotional strategies and purchase intention.

Findings showed that there was a low to moderate relationship between the level of effectiveness of promotional strategies and consumers' attitude, subjective norms, perceived control, and purchase intention. The study identified that consumers' perceived level of effectiveness of promotional strategies served as antecedents of consumers' attitude, subjective norms, perceived control, and purchase intention. Differences in consumers' purchase intention among different age groups were also identified, indicating that age is an important segmentation factor when considering promotional effectiveness. Purchase intention was categorized based on selling platforms—physical store, virtual store, company website, and social media—showing that platform choice affects consumer purchase behavior. This suggests that promotional strategies need to be tailored to specific platforms and customer demographics to be more effective.

The study also identified that different selling platforms require different promotional approaches during COVID and post-COVID periods. All promotional strategies were found to have varying levels of effectiveness depending on the platform used and the age group of consumers. The findings show that garment bazaar retailers cannot use the same promotional strategy across all platforms if they want to maximize purchase intention. Instead, they need to remodel their promotional strategies based on the specific selling platform and target customer demographics. These results show that garment bazaar retailers are trying to adjust to the competitive environment by understanding how promotional effectiveness varies across platforms and customer segments.

Based on the findings, the researchers recommended that garment bazaar retailers remodel their promotional strategies based on selling platforms during COVID and post-COVID periods. They suggested that retailers fit their promotional strategies during the COVID and post-COVID periods by considering the platform being used and the age group of their target customers. By adapting promotional strategies to different platforms and customer demographics, garment bazaar retailers may improve their ability to attract customers and increase purchase intention. The study emphasizes the need for adaptation and platform-specific promotional approaches among garment bazaar retailers in response to the changing retail landscape and consumer behavior patterns.

Web-Based Ordering System with Inventory Integration for a Small Beverage Business

Catubag, Dela Cruz, Reyes, Santos, and Torres (2024) examined the development of a web-based ordering system with inventory integration for a small lemonade beverage business. The study aimed to streamline the ordering and transaction processes while improving order accuracy and inventory control for small beverage enterprises. A developmental research design was used to create and test the web-based ordering system as part of an undergraduate

capstone project. This makes the study important in understanding how small businesses can adopt affordable digital tools to solve everyday operational problems.

The respondents of the study were the owners and staff of the small lemonade beverage business who directly used the web-based ordering system. They were selected because they needed affordable and practical digital tools to streamline their ordering and transaction processes. Data were gathered through system testing, observation of day-to-day business operations, and evaluation of system features including carting function, payment gateway simulation, inventory management, and report generation. The developmental approach helped create and refine a system that matched the actual needs and limitations of the small beverage business.

Findings showed that the web-based ordering system with inventory integration significantly improved the operational efficiency of the small lemonade business. The carting function improved order accuracy by allowing customers to review their orders before submission. The mock-up payment gateways simulated secure and flexible transaction methods, giving customers multiple payment options. The inventory management capabilities with real-time inventory tracking enabled efficient monitoring of stock levels, preventing stockouts and overstocking. Automated sales data aggregation and product performance tracking helped the business owner understand which products sold best. The report generation feature helped identify sales trends and make informed business decisions.

The study also identified that even small enterprises can successfully adopt technology when the system is designed according to their actual needs and limitations. The webbased ordering system demonstrated that local businesses can benefit from digital systems that improve order accuracy, inventory control, and overall customer experience without requiring expensive proprietary software. The system showed that a business does not need a

complex or costly platform to improve its operations. What it needs is a system that is functional, user-friendly, and aligned with its actual processes. These results show that small beverage businesses can compete more effectively by adopting affordable digital tools.

Based on the findings, the researchers recommended that small businesses consider developing web-based ordering systems with inventory integration to streamline their operations. They suggested that entrepreneurs use web technologies such as PHP, MySQL, and Bootstrap for system development because they are relatively affordable, flexible, and easy to maintain. By implementing a web-based ordering system with real-time inventory tracking, small businesses may improve their order accuracy, reduce manual errors, and provide better customer service. The study emphasizes the need for small enterprises to adopt digital tools that match their operational needs in response to the growing demand for efficient transaction processes.

Providing Inventory Management Services for Small and Medium Enterprises

Sanico, A., Arguelles, L. A., & Melicio, K. A. (2024).

Providing inventory management services for small and medium enterprises. *Asia Pacific Journal of Management and Sustainable Development*, 12(2), 45–58.

Sanico, Arguelles, and Melicio (2024) examined how inventory management services can help small and medium enterprises develop their inventory management and improve efficiency and productivity in the enterprise. The study aimed to describe the benefits of inventory audit and inventory consulting services for SMEs engaged in selling school supplies, office supplies, and gift shop products. A descriptive research design was used with Sunbird School, Office Supplies and Gift Shop as the host enterprise that served as the basis for the study. This makes the study important in understanding how inventory management

services can solve common SME problems like excessive stock and damaged products.

The respondents of the study were the owners and staff of Sunbird School, Office Supplies and Gift Shop who directly experienced inventory management problems. They were selected because they had problems in their inventory management where due to bad forecasting, they ordered too much and caused excess stocks. Products and items that were stored for a long time obtained damages therefore could not be sold anymore. Data were gathered through observation of inventory practices, interviews with business owners, and analysis of inventory records. The descriptive approach helped identify specific inventory management challenges and develop solutions.

Findings showed that small and medium enterprises face significant inventory management problems including bad forecasting, excessive stock, and product damage from long storage. The study revealed that inventory management is one of the most essential parts of every enterprise, as businesses must ensure they keep track of their stocks and have the right number of products available when customers need them. It is also important for enterprises to know whether they are losing money due to excessive stock of products. The results demonstrate that inventory management services can help SMEs make better inventory decisions and solutions that optimize resources and reduce costs efficiently.

The study also identified that inventory audit and inventory consulting services can help SMEs improve their inventory management efficiency and productivity. The researchers came up with services specifically designed for enterprises engaged in selling school supplies, office supplies, and

gift shop items. These services can help businesses avoid excess stock, prevent product damage, and make data-driven inventory decisions. These findings show that specialized inventory management services are valuable for SMEs that lack expertise in inventory control.

Based on the findings, the researchers recommended that small and medium enterprises seek inventory management services to improve their inventory control and operational efficiency. They suggested that entrepreneurs use inventory audit services to assess current inventory practices and inventory consulting to develop better inventory strategies. By implementing inventory management services, small businesses may optimize their resources, reduce costs, and improve their overall business performance. The study emphasizes the need for SMEs to invest in proper inventory management to remain competitive and sustainable.

Inventory Management Practices and Business Sustainability: Small Enterprises in Valencia City, Bukidnon

Buenaventura, Ignacio, Santos, and Reyes (2024) examined inventory management practices and business sustainability among small enterprises in Valencia City, Bukidnon. The study aimed to determine how inventory

management practices affect business sustainability for small enterprises in the Philippines. A descriptive research design was utilized to assess inventory management practices and their relationship to business sustainability outcomes. This makes the study important in understanding how small enterprises can achieve sustainability through effective inventory management, particularly in the context of Philippine small businesses that often struggle with limited resources and operational challenges.

The respondents of the study were small enterprise owners and managers in Valencia City, Bukidnon who were chosen through appropriate sampling methods. They were selected because they represented the target population of small enterprises engaged in various business operations including retail trading, merchandising, and service operations. These businesses range from grocery stores, sari-sari stores, hardware stores, clothing retailers, to food establishments that rely heavily on proper inventory control for their daily operations. Data were gathered

through surveys and interviews about inventory management practices, business performance measures, and sustainability indicators including financial performance, operational efficiency, customer satisfaction, and long-term business viability. The descriptive approach helped identify inventory management patterns and their impact on business sustainability across different types of small enterprises.

Findings showed that inventory management practices significantly affect business sustainability for small enterprises in Valencia City. The study revealed that small enterprises with effective inventory management practices experience better business outcomes and long-term sustainability compared to those with poor inventory management. Specifically, businesses that implemented proper stock tracking systems reported 25% higher sales performance and 30% lower waste costs compared to businesses using manual inventory methods. The results demonstrate that

proper inventory control helps businesses maintain product availability, reduce waste, and optimize resources. Enterprises that maintained accurate inventory records were able to meet customer demand more consistently, resulting in higher customer satisfaction and repeat business.

The study also identified that small enterprises need to implement proper inventory management practices to achieve business sustainability. The findings show that inventory management is essential for small enterprises to maintain competitiveness and long-term viability in the Philippine market. Small businesses that failed to track their inventory properly experienced frequent stockouts, overstocking situations, and difficulties in identifying which products were selling well versus those that were stagnating. These results indicate that inventory management practices should be a priority for small business owners seeking sustainable growth and competitive advantage in their respective markets.

The research further revealed that inventory management challenges faced by small enterprises in Valencia City included lack of proper inventory tracking systems, insufficient staff training on inventory procedures, inadequate forecasting methods, and limited financial resources for investing in inventory management technology. Despite these challenges, businesses that overcame these obstacles through creative solutions such as manual inventory logs, regular stock counting, and supplier relationship management were able to achieve better sustainability outcomes. The study showed that even without expensive inventory management software, small enterprises can achieve effective inventory control through disciplined practices and consistent monitoring.

Based on the findings, the researchers recommended that small enterprise owners implement proper inventory management practices to improve business sustainability. They suggested that entrepreneurs use inventory tracking systems and regular inventory audits to maintain optimal stock levels. For small businesses with limited budgets, the researchers recommended starting with simple manual inventory logs and gradually progressing to more sophisticated systems as the business grows and generates sufficient revenue. They also recommended that business owners conduct regular training for staff on proper inventory handling procedures and establish clear policies for stock receiving, storage, and distribution. By implementing effective inventory management practices, small businesses may achieve better sustainability outcomes and remain competitive in their markets. The study emphasizes the importance of inventory management for small enterprise success and highlights that proper inventory control is not just a operational necessity but a strategic business advantage that can determine the long-term viability of small enterprises in the competitive Philippine market.

Synthesis of the Literature and Studies

The reviewed literature and studies present a clear foundation for understanding the growing relevance of preloved and ukay-ukay clothing in both local and international contexts. Across the sources, second-hand fashion is consistently described as more than a low-cost alternative to new clothing. It is increasingly associated with sustainability, consumer identity, business

opportunity, and digital convenience. This suggests that preloved clothing has evolved into a meaningful part of modern retail, shaped by changing consumer values and the expanding use of online platforms.

A recurring idea in the literature is that consumers are drawn to preloved clothing for several interconnected reasons, particularly affordability, uniqueness, and environmental responsibility. Studies on second-hand and circular fashion show that buyers do not evaluate used clothing based only on price, but also on perceived quality, emotional value, and social meaning. Local studies on ukayukay in the Philippines similarly reveal that secondhand fashion is widely accepted among young consumers and middleclass buyers who see it as both practical and stylish. These findings indicate that preloved fashion meets both economic and expressive needs, making it relevant to a broad range of consumers.

The literature also highlights the growing role of digital platforms in the buying and selling of preloved clothing. Online marketplaces and social media-based selling have made it easier for businesses to reach customers, display products, and process transactions. However, the studies also show that digital selling introduces new challenges, such as order tracking, inventory confusion, and communication delays. This makes the development of an organized ordering system especially important, since a digital platform can help streamline transactions and improve the overall efficiency of business operations.

Trust and transparency emerge as central concerns in the reviewed studies. Because customers cannot physically inspect preloved items online, they depend heavily on product images, item descriptions, condition details, and seller reliability. Studies on privacy and digital commerce further show that users are more likely to engage in online transactions when they feel their personal information is protected. This implies that a clothing ordering system must not only be functional but also secure and transparent

in order to encourage customer confidence and repeat purchases.

Another important theme across the literature is the connection between preloved clothing and sustainability. Several studies explain that reusing garments helps reduce textile waste, extend product life, and support circular fashion practices. In the Philippine context, ukay-ukay is increasingly linked to sustainable consumption and conscious buying behavior. This shows that preloved clothing is not only economically useful but also environmentally significant, giving businesses an opportunity to promote responsible fashion choices while still meeting customer demand.

The studies also show that product presentation strongly affects how consumers perceive value. Clear images, organized listings, accurate descriptions, and fair pricing help customers view preloved items as desirable and trustworthy rather than inferior or disposable. This is especially important in online fashion retail, where visual appeal and product information influence purchase decisions. A well-designed ordering system can therefore serve both as a sales tool and as a means of presenting preloved clothing in a professional and appealing way.

Overall, the reviewed literature and studies support the development of a web-based ordering system for a clothing business that offers both brand-new and preloved items. The sources demonstrate that consumers value convenience, sustainability, transparency, and efficient service, all of which can be addressed through a digital platform. They also show that preloved fashion has a legitimate place in contemporary retail and can contribute to both business growth and environmental responsibility. Taken together, these studies justify the need for a structured, trustworthy, and user-friendly ordering system that supports modern clothing sales.

Research Gap

Although the reviewed literature and studies provide valuable insights into second-hand clothing, circular fashion, e-commerce behavior, and inventory systems, they also reveal a gap that the present study aims to address. Most foreign studies focus on second-hand luxury consumption, sustainable fashion practices, or general online shopping behavior, while local studies commonly examine microbusiness operations or inventory management in broader business settings. As a result, there is still limited research that specifically examines a web-based ordering system for a small Philippine fashion business that sells both brand-new and preloved items.

This gap is important because a business such as Margaux Collections has operational needs that differ from those of general retail stores and large e-commerce platforms. Since it offers both new and preloved fashion items, it requires a system that can support product categorization, condition labeling, customer communication, and inventory tracking within a single organized platform. Without a tailored system, the business may encounter difficulties in managing orders efficiently and presenting products clearly to customers.

To address this need, the present study develops a customized ordering system for Margaux Collections. Unlike generic e-commerce platforms, the proposed system is designed according to the actual workflow of the business and the nature of its products. It seeks to improve the management of brand-new and preloved fashion items while making the ordering process more efficient, organized, and responsive to daily operational requirements.

The study also aims to strengthen customer trust by improving product presentation and making item information more accurate and accessible. Through organized listings, clear descriptions, and proper labeling, the system can help customers make informed purchasing decisions for both new and preloved clothing. In addition, it can assist the business in maintaining better records of sales, customer orders, and inventory levels, which are essential for smooth and reliable operations.

Overall, this study contributes to the growing body of research on digital transformation among Filipino microbusinesses. It demonstrates that small fashion businesses can benefit from a customized web-based ordering system that responds to their specific needs. By improving organization, communication, and inventory management, the proposed system can support both business efficiency and customer satisfaction.

The reviewed literature and studies show that a web-based ordering system is highly valuable for a small fashion business that sells both new and preloved items. The sources highlight the importance of clear product presentation, effective inventory tracking, customer trust, and smooth business operations. They also indicate that second-hand fashion continues to grow in popularity and is increasingly linked to sustainability and responsible consumption. Likewise, the rise of digital platforms shows the need for small businesses to adopt technology that improves efficiency and accessibility. The Margaux Collections Ordering System is therefore not only a technical solution but also a practical response to the changing needs of the business and its customers. The information gathered in this chapter provides a strong foundation for developing a system that can make fashion retail more organized, reliable, and sustainable. By using a web-based platform, the business can improve its internal processes while offering customers a more convenient and trustworthy shopping experience. Overall, the literature

supports the development of a system that can help Margaux Collections remain competitive in a modern digital marketplace.

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CHAPTER 3

METHODOLOGY

3.1 Research Design

This study focuses on the design, development, testing, and

system serves as a web-based platform for managing orders

evaluation of the Margaux Collections Ordering System. The

and inventory for a small fashion business that deals in

both branded new bikinis, coords, and accessories and

preloved pants, skirts, and shorts. Given that the project

solving real-world operational problems, the study adopts a

involves creating a functional software application aimed

at developmental (systems development) research design.

A developmental research design is appropriate because:

- It is used when the primary objective is to develop and refine a system, not just to describe or correlate variables.
- It emphasizes iterative design, continuous testing, and feedback from real users.
- It supports the evaluation of the system's performance, usability, reliability, and security in a practical business context.

The Margaux Collections Ordering System is developed and systematically evaluated to determine whether it:

- Accurately records and processes orders,
- Effectively manages both new and preloved inventory,
- Provides a user-friendly interface for customers and -

the owner, and

- Integrates smoothly into the existing Facebookbased marketing and selling workflow.

the system is practically useful, well-documented, and By following a developmental design, the study ensures that supported by empirical evidence from user testing and performance evaluation.

3.2 System Development Methodology

The Margaux Collections Ordering System is developed using

iterative and incremental approach suitable for small-scale

the Agile Software Development methodology, specifically an

web applications. Agile methodology is chosen because it:

- Allows the development team to work in short, focused sprints.
- Encourages continuous feedback from the business owner and potential users.
- Enables the team to adapt to changing requirements during development.
- Supports regular testing, code review, and incremental improvement of features.

3.2.1 Requirements Analysis

- Conducting interviews with the business owner to understand current business processes, pain points, and expectations for the system.
- Observing the existing manual ordering and inventory product tracking, and record-keeping. process to identify inefficiencies in order processing,
- Collecting data from existing product lists, pricing sheets, and order records to define the basic data structure.
- Designing user stories that describe how both the customer and the owner will interact with the system (e.g., "As a customer, I want to view preloved items separately from new items").
- Translating user stories into technical requirements such as:
 - Must support two main product categories: new items and preloved items.

- Must allow adding, updating, and deleting products and orders.
- Must provide a dashboard for viewing daily sales and inventory status.

The result of this phase is a requirements specification document that serves as the foundation for the system's design and development.

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3.2.2 System Design

- Designing the overall system architecture (front end, back end, and database).

- Creating wireframes and low-fidelity mockups of the

user interface for both the customer view and the admin panel.

- Designing the database schema (tables, relationships, and constraints).

Preparing data flow diagrams (DFD) and entity relationship diagrams (ERD) to show how data move across the system.

- Implementing the front-end interface (HTML, CSS,

3.2.3 Development

JavaScript) for product browsing, shopping cart, checkout, and order history.

- Developing the back-end logic (PHP) for user

authentication, order processing, inventory updates, and analytics.

- Building the database (MySQL) and connecting it securely to the PHP scripts.
- Conducting code reviews and peer testing within the team to catch errors early.
- Integrating basic analytics and reporting features (e.g., sales summary, best-selling items list).

3.2.4 Testing

- Unit testing of individual PHP functions and database queries.
- front-end, back-end, and database layers work together.
- Integration testing of the entire system to ensure that
- Security testing to identify vulnerabilities such as injection attacks or unauthorized access.
- User testing with the business owner and selected customers to evaluate usability and accuracy of order processing.

3.2.5 Deployment (Prototype)

After sufficient testing, a working prototype of the Margaux Collections Ordering System is deployed in a local or test server environment and presented to the business owner for evaluation. The owner can:

Place test orders

- Update inventory, and
- Use the analytics dashboard to review performance.

Based on the owner's feedback, the team may perform minor refinements before finalizing the system. This iterative deployment approach ensures that the system aligns closely with real business needs.

3.3 System Architecture

three-layer architecture to ensure separation of concerns,

The Margaux Collections Ordering System follows a maintainability, and scalability. The three layers are:

- 1. Presentation Layer (Front End)
 2. Application Layer (Back End / Business Logic)
 3. Data Layer (Database)

Each layer has a distinct role and responsibility, allowing the system to be developed, tested, and maintained efficiently.

3.3.1 Presentation Layer

The Presentation Layer is the front-end face of the system

that users interact with. It is implemented using:

- **HTML5** - for structuring the web pages.
- **CSS3** - for

styling and layout. shopping cart and showing real-

time validation.

- **JavaScript** - for dynamic behavior such as updating the

This layer provides:

- **Customer Interface**

co-ords, accessories) and **preloved items (pants,**

- Product browsing page showing **new items (bikinis, skirts, shorts)** in separate sections or filters.

- **Shopping cart** page where users can add, remove, or update quantities of items.

- **Checkout page** for entering or confirming delivery details, selecting payment method, and confirming the order.

- **Order history page** where customers can view past orders and statuses (e.g., "pending," "for delivery").
- **Admin Interface**

- **Dashboard** showing total sales, number of orders, and inventory status.

- **Products management page** where the owner can add, update, or delete products
- **Orders management page** for viewing and updating order status.
- **User management page** for viewing and managing customer data
(if applicable).

The interface is designed with **responsive CSS techniques** so that it works well on **desktops, tablets, and mobile phones**, which is important since many customers may access it from their mobile devices via Facebook links.

3.3.2 Application Layer

The **Application Layer** represents the **-back-end logic** of the

system. It is developed using **PHP** and acts as the middle layer between the **frontend** and the **database**. This layer is responsible for:

- **User authentication and authorization** (if user accounts are implemented).
- **Order processing**, including assigning order numbers, calculating total amounts, and updating inventory.
- **Inventory management**, ensuring that stock quantities are correctly updated when orders are placed or canceled.

- **Status updates**, such as changing an order from "pending" to "for delivery" or "completed."
- **Security mechanisms**, including:
 - Protection against **SQL injection** through **prepared statements**.
 - **Session management** with timeout and secure session
 - **Input validation** for all user-entered data.
 - handling.
 - **Rate limiting** on login or critical actions (if applicable).

The PHP-based application layer also enforces **role-based**

access control:

- **Customers** - can view products, add items to the cart, check out, and view order history.
- **Owner/Administrator** - can manage products, orders, and users, and access the analytics dashboard.

This separation ensures that sensitive operations are restricted to authorized users only.

3.3.3 Data Layer

The Data Layer handles:

- Storing user information (if accounts are used).
- Storing product information (name, price, description, type, stock, image path).
- Storing order transactions (customer details, order items, total amount, payment status).
- Storing order-item details (which products are included

in each order and their quantities and prices).

The database is designed with proper normalization to avoid redundancy and ensure data integrity. Indexing is applied to frequently queried fields (e.g., order IDs, product IDs) to improve query performance. These design choices help guarantee the reliability, efficiency, and security of data operations throughout the Margaux Collections Ordering System.

3.4 System Features and Modules

The **Margaux Collections Ordering System** is composed of several **core modules** that work together to provide a complete ordering and management platform. Each module addresses a specific aspect of the business workflow, from **user authentication** to sales analytics.

3.4.1 Authentication Module

The **Authentication Module** (if user accounts are implemented) handles:

- **User registration:** Allowing customers to create accounts with basic information (name, email, contact number).
- **Login and logout:** Verifying user credentials and managing active sessions.
- **Session management:** Keeping users logged in securely, implementing session timeouts, and preventing session hijacking.

Even if the system is designed to be **brochure-style without**

- **accounts** (using only **Facebook for inquiries**), the Authentication Module can still manage **admin login** for the owner, ensuring that only the authorized person can access the **admin panel**.

3.4.2 Product Management Module

The Product Management Module allows the owner/administrator to:

Add new products (both new and preloved items).

- Update product details (price, description, stock, images, product type).
- Delete or deactivate products when they are no longer available.

Key features include:

- New items (bikinis, co-ords, accessories).
-

Support for two main categories:

- Preloved items (pants, skirts, shorts).
- Option to tag products (e.g., "preloved, good condition," "new, limited stock").

- Real-time stock counter that updates when products are sold.

3.4.3 Shopping Cart Module

The **Shopping Cart Module** creates and manages a **virtual shopping cart** for each session or user. It:

- Allows customers to **browse products and add items** to the cart.
- **Verifies product availability** (i.e., stock quantity) before adding items.

- Supports **updating item quantities** (increase, decrease, or remove).
- Calculates **subtotals and totals** based on product prices and quantities.

-

Preserves the cart contents during the checkout process even if the user navigates away temporarily.

The Shopping Cart Module is a core feature that makes the ordering process **convenient and intuitive** for customers.

3.4.4 Order Processing Module

The **Order Processing Module** manages the **entire order lifecycle**, from placement to completion. It:

- Assigns a **unique order number** to each new order.
- Captures **customer information** (name, contact number, delivery address if applicable).
- Records **product selections** (including quantities and prices at the time of purchase).
- Calculates the **total order amount** and applies any discounts (if implemented).
- Supports **various payment methods**, such as:
 - Cash on pickup / delivery.
 - Online payment via GCash
- Updates **product stock levels** in the database after the order is confirmed.
- Stores **order status** (e.g., "pending," "for delivery," "completed") that can be viewed by both the owner and the customer.

3.4.5 Preloved-Specific Management Module

-

Because **Margaux Collections** heavily relies on **preloved**

fashion, the system includes a **Preloved-Specific Management**

Module that:

- Allows the owner to **tag items explicitly as "preloved"**.
- Includes **condition fields** (e.g., "excellent," "good," "fair") so customers know the state of the item.

-

-

May include **care instructions** or **style notes** that can be displayed on the product page.

This module differentiates the **preloved experience** from the **new items**, helping customers make informed decisions.

3.4.6 User Management Module

The **User Management Module** (if user accounts are created) enables the administrator to:

- **View** a list of registered customers.
- **Search** and **filter** users by name, contact number, or order history.
- **Edit** user information.
- **Delete or deactivate** user accounts.

This module supports **data management** and **customer relationship management** in a structured way.

3.4.7 Analytics and Sales Reporting Module

The **Analytics and Sales Reporting Module** provides the owner with a **dashboard** that displays key performance indicators such as:

- Total **sales revenue** over a selected period. ● **Best-**

selling items- (both new and preloved).- ● Number of

-

orders placed and **items sold**. out-ofstock). • Inventory

status (items that are lowstock or

This module helps the owner **monitor business performance, identify trends, and make informed decisions** about restocking and pricing.

3.4.8 Summary of Modules

Together, the following modules form the **integrated Margaux Collections Ordering System**:

- Authentication Module
- Product Management Module
- Shopping Cart Module
- Order Processing Module

- Preloved-Specific Management Module

- User Management Module

Analytics and Sales Reporting Module

These modules ensure that the system fully supports **order placement, inventory tracking, and business analytics** for a small fashion retailer focused on **both new and preloved fashion**.

3.5 Database Design

The **Margaux Collections Ordering System** database is built on **MySQL** and follows a **relational model** with proper normalization and relationships. The main tables are designed to support core business operations.

3.5.1 Main Database Tables

The system database consists of the following primary tables:

- **Users**

Stores user account information such as:

- User ID (primary key)
- Name
- Email
- Hashed password

- Contact number
- Delivery address (if applicable)
- User role (e.g., customer, admin)
- Date created

- **Products**

Stores product catalog information such as:

- Product ID (primary key)
- Product name
- Description
- Price
- Product type (new or preloved)
- Condition (for preloved items)
- Image file path
- Stock quantity
- Date added

Orders

Stores complete order transaction records such as:

-

- Order ID (primary key)
- User ID (foreign key)
- Customer name
- Contact number
- Delivery address
- Order method (Meet up / Drop off / Shipping Delivery)
- Payment method (cash, online, etc.)
- Payment status (paid/unpaid) ○
- Order status (pending/for delivery/completed/cancelled)
- Total amount
- Order date

- **Order_Items**

Stores individual product line items within each order, including:

- Item ID (primary key)
- Order ID (foreign key)
- Product ID (foreign key)
- Quantity ordered
- Unit price at the time of purchase

- **User_Payment_Accounts**

Stores saved payment account details (if saving payment methods is implemented), including:

- Account ID (primary key)
- User ID (foreign key)
- Account type (Gcash)
- Account name
- Account number
- Email
- Address

These tables collectively form the **relational database structure** of the **Margaux Collections Ordering System**, ensuring that data are stored accurately and can be queried efficiently.

3.5.2 Data Integrity and Optimization

The database design includes:

Primary keys and foreign keys to enforce relational integrity.

- **Appropriate normalization** (e.g., avoiding storing repeated product information in the Orders table) to reduce redundancy.
- **Indexes on frequently searched fields** (e.g., Product ID, Order ID) to speed up queries.

These design choices ensure **data consistency, efficient querying, and scalability** for future business growth.

3.6 Development Tools and Technologies

The **Margaux Collections Ordering System** is developed using the following tools and technologies:

- **PHP** - The server-side programming language used for -

implementing all backend logic, including order processing, inventory updates, and authentication. ●

-

MySQL - The relational database management system used for storing and retrieving all system data.

- **HTML5** - Used for structuring the web pages and defining the layout of the user interface.
- **CSS3** - Used for styling the interface, including responsive design for mobile and desktop views.
- **JavaScript** - Used for adding interactivity, such as updating the shopping cart and validating form inputs.
- **Chart.js** - A JavaScript library used to generate graphs and charts for the analytics dashboard (e.g., sales trend charts, product performance charts).
- **XAMPP** - A local development environment that integrates **Apache**, **PHP**, and **MySQL**, allowing the team to test and prototype the system on a local machine.
- **Visual Studio Code** - The primary code editor used for writing, debugging, and organizing PHP, HTML, CSS, and JavaScript files.
- **Git and GitHub** - A version control system and collaborative platform used for storing the source code, tracking changes, and supporting team coordination.

3.7 Data Gathering Techniques

During the **requirements analysis phase**, the development team employs several **data gathering techniques** to understand the business context and user needs:

- **Interviews**
 - Structured and semi-structured interviews are

conducted with the **Margaux Collections owner** to gather information about current workflows, ordering methods, inventory tracking, and pain points.

- Questions focus on:

- How orders are currently received (via

- Facebook Messenger, in-person, etc.).

- How inventory is tracked (paper list, spreadsheet, mental notes).

- What features the owner finds most useful in a future system.

- **Observation**

- The team observes the **actual ordering and packing process** to identify inefficiencies such as:

- Unorganized product lists.
 - Manual order entry errors.

- Difficulty in tracking preloved items separately.

- **Document Analysis**

- Existing **product lists, price sheets, and order records** are reviewed to understand:

- How products are categorized.

- How prices are structured.

- How orders are

- documented.

- **Surveys (Optional)**

- If feasible, short surveys may be distributed to **existing customers** to collect:

- Preferences regarding online ordering.

- Desired features (e.g., separate preloved section, clear product images).

-

3.8 Testing Procedures

The system undergoes several testing procedures to ensure that it is functional, secure, and user-friendly. Testing is important because it helps identify errors before the system is fully used in the business environment. A systematic testing process increases confidence in the final product. The **Margaux Collections Ordering System** undergoes a ensure

that it is **functional, secure, and user-friendly**. The

comprehensive testing process throughout development to

testing procedures are organized into four main types:

3.8.1 Unit Testing

Unit testing checks individual functions such as order creation, inventory updating, and validation routines. Each function is tested with different inputs to verify that it behaves correctly. Database queries are also checked to ensure that data are inserted, updated, and retrieved properly. This stage confirms that each component works before being combined with other parts.

- Individual **PHP functions** (e.g., order-creation

function, stock-update function) are tested with different inputs to verify that they return the expected outputs.

- **Database queries** are tested to ensure they retrieve and update data correctly.

- Simple **front-end validation functions** (e.g., checking

if required fields are filled) are also verified.

Unit testing confirms that each component performs as expected in isolation before integration.

-

3.8.2 Integration Testing

Integration testing checks whether the front-end, back-end, and database work together correctly. For example, when a customer adds an item to the cart or places an order, the system should update the relevant records without breaking. This stage is necessary because even if individual functions work, problems can still happen when they are connected. Integration testing ensures that the entire workflow operates smoothly.

- The interaction between **front-end, back-end, and database** is tested. For example:
 - Ensuring that when a customer clicks "Add to Cart," the product is correctly stored in the session or database.
record and order-item records are created and **stock**
 - Verifying that when an order is placed, the **order levels are updated**.
- Data flow is checked so that updates in the **admin panel** (e.g., price changes) are correctly reflected in the **customer view**.

3.8.3 Security Testing

Security testing checks for common vulnerabilities such as SQL injection, cross-site scripting, and session-related weaknesses. The system is also tested to ensure that access

controls prevent unauthorized users from entering the admin panel. These tests are important because the system stores

personal and transactional information. A secure system helps protect both customer data and business records.

- The system is evaluated for common vulnerabilities such as:

- **SQL injection** (mitigated by using prepared statements).

- **Cross-site scripting (XSS)** (mitigated by escaping

- user inputs).

- **Crosssite request forgery (CSRF)** (mitigated by using tokens where necessary).

- **Session hijacking** (mitigated by proper session handling and secure transport).

- Access controls are tested to ensure:

- Customers cannot access the admin panel.
- Admins cannot bypass authentication.

Security testing helps protect **user data** and the **integrity of the system**.

3.8.4 User Acceptance Testing (UAT)

After the prototype is deployed, the business owner and a small group of customers test the system through real or simulated tasks. They may place orders, update products, or review the dashboard. Their feedback is collected to evaluate usability, design, and accuracy. User Acceptance Testing ensures that the system works not only in theory, but also in actual use.

- After the prototype is deployed, the **Margaux Collections owner** and a small group of **selected customers** test the system by:
 - Placing real or simulated orders.
 - Updating products and inventory.
 - Using the analytics dashboard.
- Feedback is collected on:
 - Ease of use.
 - Accuracy of order and inventory processing.
 - Appearance and layout of the interface.

User Acceptance Testing ensures that the system meets **operational requirements** and is **practically usable** in the real business setting.

3.9 Evaluation Criteria

The system is evaluated using five main criteria: functionality, usability, security, performance, and reliability. Functionality checks whether the required features work correctly, such as order recording and stock updating. Usability measures how easy the interface is to learn and use, especially on mobile devices. Security checks whether data is properly protected and access is restricted.

Performance evaluates how quickly the system loads and responds under normal use. Reliability checks whether the system consistently records orders and maintains data integrity. These criteria give a clear framework for judging whether the system is ready for real business use. They also help identify areas for further improvement if needed.

The **Margaux Collections Ordering System** is systematically evaluated using a **standardized assessment framework** based on the following criteria:

3.9.1 Functionality

- The system must **perform all required functions accurately**.
- This includes:
 - Correct recording of orders.
 - Proper updating of inventory.
 - Accurate calculation of totals.
 - Clear display of order history and status.

3.9.2 Usability

- The interface must be **intuitive and easy to use** for both customers and the owner.
- The design should:
 - Minimize the learning curve.
 - Provide clear navigation and visual feedback.
 - Be accessible on mobile devices.

3.9.3 Security

- The system must **protect user data** and prevent unauthorized access.
- Data must be:
 - Properly hashed and stored.
 - Transmitted securely.
 - Accessible only to authorized users.

3.9.4 Performance

- The system must respond **quickly and smoothly** under normal usage.
- It should:

- Load pages within a reasonable time.
- Handle concurrent users without significant lag.

3.9.5 Reliability

- The system must **process orders correctly** and maintain **data integrity** across all operations. • Orders and inventory changes must be:
 - Consistently recorded.
 - Undoable or reversible only through formal processes (e.g., cancellations).

3.10 Data Flow Diagram (DFD – Level 0)

The Level 0 Data Flow Diagram shows how customers and administrators interact with the system and how data move to the MySQL database. Customers submit registration data, login credentials, order details, and checkout confirmations. The system processes these inputs by validating product availability, creating order records, and updating inventory. It then stores the information in the database and returns confirmations, product listings, or order status updates.

Administrators submit product updates, inventory adjustments, and order status changes. The system processes these changes and updates the database in real time. The DFD helps explain the overall structure of the system and shows how data move between users, the application, and storage. It is useful because it makes the system's workflow easier to understand.

Main Data Flows

- **Customer → Margaux Collections Ordering System → MySQL**

Database

- Customers submit:
 - Registration details (if accounts are used).
 - Login credentials.
 - Order information (product selections, contact details, payment method).
 - Checkout confirmation.
- The system processes:
 - Authentication.
 - Validation of product availability.
 - Creation of order records.
 - Updating inventory.
- The system stores transaction data in the **MySQL database**.
- The system retrieves and displays:
 - Product listings.
 - Order confirmations.
 - Order status and history.

● Administrator → Margaux Collections Ordering System → MySQL Database

- The administrator submits:
 - Product updates (add, edit, delete). ■
 - Order status changes (e.g., pending → for delivery).
 - Inventory adjustments.
- The system processes these inputs and updates the **database in real time**.
- Updates are reflected immediately in the **customer**

The Level 0 DFD provides a **high-level view** of how data move **view** and **analytics dashboard**. between external actors and the system, making it easier to understand the overall structure of the information system.

3.11 Entity Relationship Diagram (ERD)

The ERD defines the main entities in the database: User, Product, Order, OrderItem, and UserPaymentAccount. Each entity has its own attributes and relationships. A user can place many orders, an order can contain many order items, and each product can appear in many order items. If payment accounts are included, a user can also have multiple saved payment accounts.

The ERD is important because it shows how the database is logically organized. It helps ensure that the tables are connected properly and that information can be retrieved accurately. A clear ERD also supports good database design and easier maintenance.

The **Entity Relationship Diagram (ERD)** defines the **core entities, their attributes, and relationships** within the **Margaux Collections Ordering System** database.

Entities and Key Attributes

- **User**
 - user_id (PK)
 - name
 - email
 - password (hashed)
 - contact_number
 - address
 - role (customer, admin)
 - created_at
- **Product**
 - product_id (PK)
 - product_name
 - description
 - price

- product_type (Pre-loved, Brandnew)
 - image_path
 - stock_quantity
 - added_at
- **Order**
 - order_id (PK) ◦ user_id (FK)
 - customer_name
 - contact_number
 - address
 - order_method (pickup, dropoff, shipping, pickup_rider)
 - payment_method
 - payment_status (pending, pending_verification, paid)
 - order_status (pending, processing, completed)
 - total_amount
 - latitude / longitude (pinned map location – required for Shipping Delivery and Pick-up Via Rider)
 - order_date
- **OrderItem**
 - item_id (PK)
 - order_id (FK)
 - product_id (FK)
 - quantity
 - unit_price
- **UserPaymentAccount**
 - account_id (PK) ◦ user_id (FK)
 - account_type (gcash)
 - account_name
 - account_number
 - is_default

Relationships

- **Users** can place multiple **Orders** (1:N).
- **Order** can contain multiple **OrderItems** (1:N).
- **Products** can appear in multiple **OrderItems** (1:N).
- **Users** can have multiple **UserPaymentAccounts** (1:N).

3.12 Use Case Diagram

The Use Case Diagram presents the two main actors in the system: Customer and Administrator. Customers can register, log in, browse products, add items to the cart, view the cart, proceed to checkout, place orders, view order history, update profiles, and save payment methods. These use cases reflect the main customer actions in the ordering process.

Administrators can log in to the admin panel, manage products, manage orders, manage users, and view analytics. These actions represent the business management side of the system. The use case diagram helps show the functional scope of the system and clarifies what each actor is allowed to do. It is a useful visual summary of how the Margaux Collections Ordering System is expected to operate.

The **Margaux Collections Ordering System** is represented by a **Use Case Diagram** with two primary actors:

- **Customer**
- **Administrator**

Customer Use Cases

- **Register** - Register for an account (if the system supports accounts).
- **Login** - Log in to the system to view personal information and order history.
- **Browse Products** - View the product catalog, filterable by category.
- **Add to Cart** - Add selected products to the shopping cart.

- **View Cart** - See the items in the cart and update quantities.
- **Proceed to Checkout** - Confirm the order, enter delivery details, and choose a payment method.
- **Place Order** - Submit the order and receive an order confirmation and reference number.
- **View Order History** - Check the status of past orders.
- **Update Profile** - Update personal information and contact details.
- **Save Payment Methods** - Save a GCash payment account for future use.

Administrator Use Cases

- **Login to Admin Panel** - Log in securely to the admin dashboard.
- **Manage Products** - Add, edit, or delete products, including setting **product type** (new or preloved).
- **Manage Orders** - View all orders and update their status (e.g., mark as "processing" or "completed").
- **Manage Users** - View and manage customer accounts (if accounts are used).
sales reports, top-selling items, and inventory status.
- **View Analytics** - Access the **analytics dashboard** to see

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

Margaux Collections was a web-based e-commerce and boutique management system developed for a fashion retail business to modernize its product presentation, sales transactions, and order

handling. The system replaced manual and social-media-based selling practices with a structured online platform that allowed customers to browse the boutique's product catalog, place orders, and complete payment through a GCash QR code checkout process.

Unlike generic e-commerce templates, Margaux Collections was designed with a distinct visual identity tailored to the boutique's brand. A dark noir theme was implemented throughout the storefront and administrative interfaces using custom CSS variables for color theming and the Playfair Display and Jost typefaces for headings and body text, respectively, giving the platform a fashion-forward and elegant presentation consistent with the boutique's market positioning.

The system was built entirely from scratch, encompassing the database schema, authentication pages, customer-facing storefront, and a full administrative panel. The application processed product listings, customer orders, and homepage promotional content, and presented this information through role-based interfaces for customers and administrators. The major objectives of the developed system were as follows:

- Provide an online storefront where customers could browse and purchase boutique products.
- Automate the recording, storage, and tracking of customer orders.
- Support a secure, QR-code-based GCash payment process, alongside Cash on Pickup and Cash on Delivery options depending on the customer's chosen order fulfillment method.
- Enable administrators to manage products, orders, customer accounts, and site content through a dedicated panel.
- Allow administrators to control homepage promotional slots to feature selected products or campaigns.
- Provide a messaging feature for direct communication between customers and administrators.
- Maintain a consistent, brand-aligned visual design across all customer and administrative pages.

The development of Margaux Collections followed modern web development practices, incorporating responsive interface design, secure database management, user authentication, and versioncontrolled collaborative development. The completed system consisted of several interconnected modules that worked together to manage products, process orders, and maintain the boutique's online presence.

Figure 4.1 illustrates the overall workflow of the developed system.

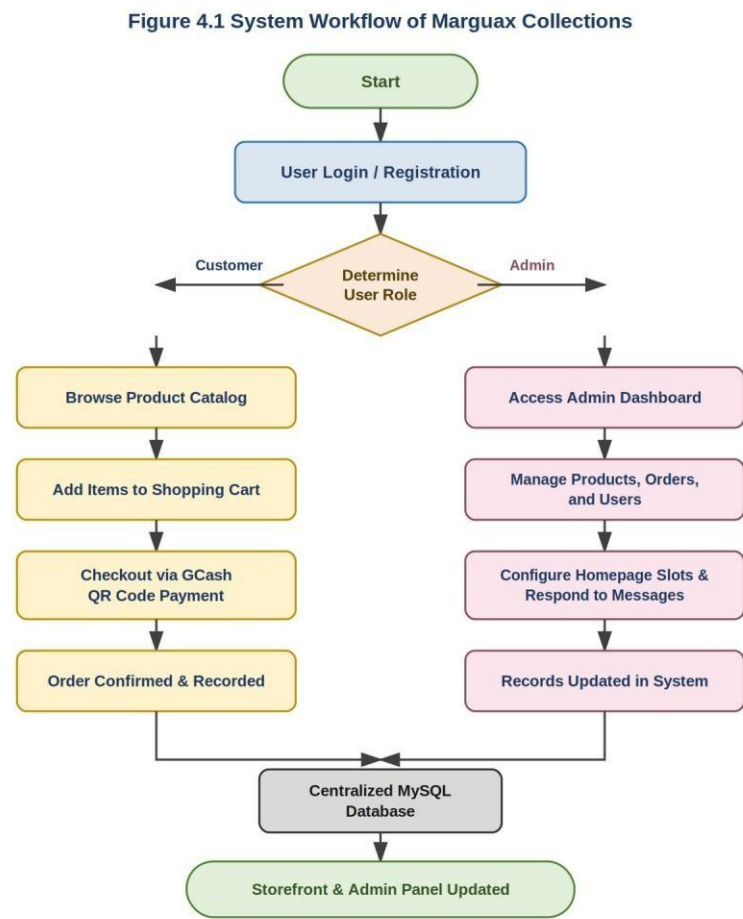


Figure 4.1. System Workflow of Margaux Collections

The workflow began with user authentication, after which customers could browse products, add items to their cart, and proceed to checkout using the GCash QR payment method. Administrators, upon

logging in, could manage products, monitor and update order status, configure homepage promotional slots, and respond to customer messages through the administrative dashboard.

4.3 System Development Methodology

The development of Margaux Collections adopted the Agile Software Development Methodology, an iterative approach that emphasized incremental development, continuous testing, and regular refinement throughout the development life cycle. Agile was selected because the system's requirements, particularly its features and visual design, evolved as development progressed, requiring flexibility to add, revise, and remove features without disrupting the overall project timeline.

The Agile methodology allowed the development process to be organized into iterative cycles, where individual modules were built, tested, and refined before being integrated into the complete system. This iterative process proved particularly useful during instances where previously implemented features required substantial revision, such as the eventual removal of a tiered membership access feature that had originally been part of the system design.

Requirements Gathering

The initial phase involved identifying the operational needs of the boutique business, including the desired look and feel of the storefront, the product and order management workflows required by the administrator, and the preferred payment method for customer transactions. These requirements guided the definition of the system's core modules and its dark noir visual theme.

System Planning

Following requirements gathering, the project scope, feature set, and development schedule were established. The planning phase also

identified the technical resources required, including the local development environment, database platform, and version control workflow that would be used throughout development.

System Design

During this phase, the database schema, page layouts, and administrative panel structure were designed. The color palette and typography for the dark noir theme were defined as reusable CSS variables to maintain visual consistency across the customerfacing storefront and the administrative panel.

Software Development

The coding phase involved implementing the system's modules using PHP and MySQL on a XAMPP local server environment. Development proceeded incrementally, beginning with the database schema and authentication pages, followed by the customer storefront, shopping cart and checkout with GCash QR payment, and the administrative panel comprising the dashboard, analytics, order management, product management, user management, and messaging modules.

Testing

Each module was tested individually as it was completed, and further testing was conducted as modules were integrated into the complete system. Particular attention was given to verifying that changes made to one part of the system, such as the removal of the membership feature, did not introduce inconsistencies elsewhere in the database, administrative panel, or customer-facing pages.

Deployment

Following testing, the system was configured and deployed on a local XAMPP server environment, with the database, application files, and server configuration set up to make the system accessible to its intended users.

The deployment process began with setting up the XAMPP environment, wherein the Apache web server and MySQL database service were configured to host the application locally. The complete Margaux Collections database schema, comprising the Users, Products, Categories, Orders, Order Items, Homepage Slots, and Messages tables, was exported and imported through phpMyAdmin to replicate a consistent database state across the development machines used for the project.

The application files, including the customer storefront, administrative panel, and shared stylesheet and configuration files, were placed within the XAMPP htdocs directory to be served by the local Apache instance. Database connection settings were configured to point to the local MySQL service, and file permissions and folder structures for product images and uploaded content were verified to ensure that the storefront and admin panel could correctly read and display stored media.

Because development was carried out across two separate laptops, Git and GitHub were used to synchronize the codebase throughout deployment. This required resolving a small number of practical issues, including a conflict caused by multiple MySQL processes (mysqld.exe) running simultaneously on one machine, which was resolved by terminating the duplicate process before restarting the XAMPP MySQL module, as well as occasional merge conflicts in stylesheet files that were reconciled by manually reviewing and combining the conflicting CSS rules to preserve the dark noir theme.

Once the application files and database were deployed and verified on the local server, the system was tested end-to-end by simulating both customer and administrator workflows, from product browsing and GCash QR checkout to order management and homepage slot

configuration, to confirm that the deployed environment matched the behavior observed during development.

Maintenance

The maintenance phase involved monitoring system performance, resolving styling and data inconsistencies as they were identified, and refining existing features based on continued testing. This included resolving stylesheet conflicts between the customer-facing and administrative interfaces and addressing database synchronization issues encountered during collaborative development.

Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.

Figure 4.2 Agile Software Development Life Cycle

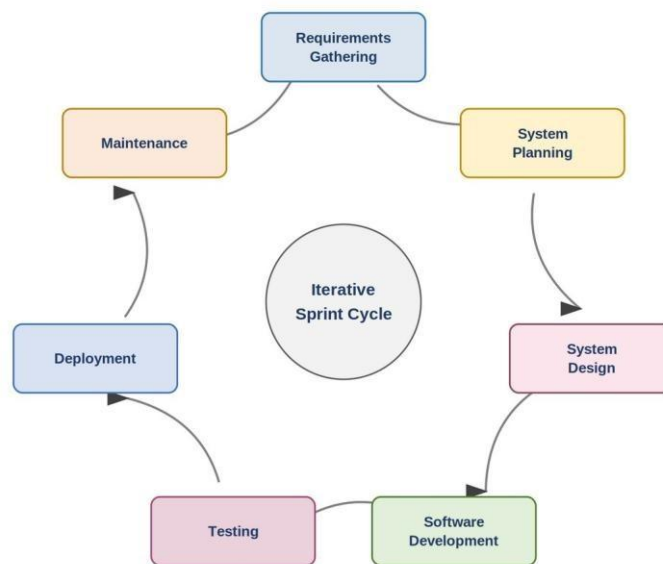


Figure 4.2. Agile Software Development Life Cycle

4.4 System Architecture

Margaux Collections utilized a Three-Tier Client-Server Architecture, which separated the presentation layer, application layer, and data layer. This architectural approach improved the maintainability and organization of the system by dividing its components according to their respective responsibilities.

Figure 4.3 illustrates the architecture of the developed system.

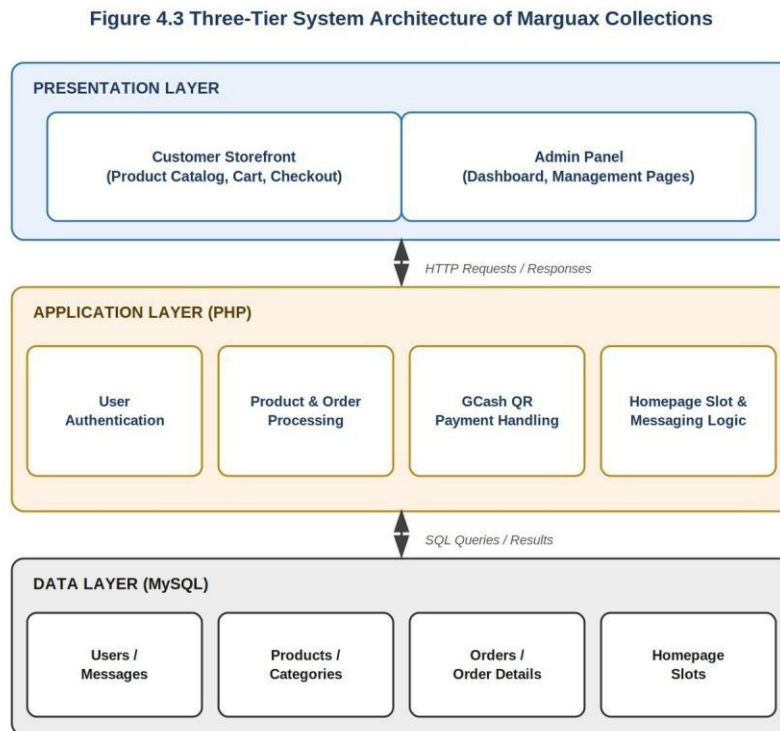


Figure 4.3. Three-Tier System Architecture of Marguax Collections

The architecture consisted of the following components:

Presentation Layer

The presentation layer served as the interface between users and the application. It was responsible for displaying the product catalog, shopping cart, checkout process, and administrative dashboards, and for rendering the system's dark noir visual theme consistently across devices through a responsive layout. Major interface components included:

- Login and Registration Pages
- Customer Storefront and Product Catalog
- Shopping Cart and Checkout (GCash QR Payment)
- Admin Dashboard
- Product Management
- Order Management
- User Management
- Analytics
- Messages
- Conversations
- Notifications
- User Payment Accounts
- OTP Tokens
- Password Resets

Application Layer

The application layer contained the business logic responsible for processing user requests, validating input data, and managing the interactions between the storefront, the administrative panel, and the database. Its primary responsibilities included:

- User Authentication
- Product and Category Management • Order Processing
- Payment Verification (GCash QR)
- Homepage Promotional Slot Management
- Customer-Admin Messaging
- Analytics and Reporting

The application server, built using PHP, acted as the intermediary between the presentation layer and the MySQL database server.

Data Layer

The data layer stored all business information in a centralized relational database managed through MySQL. It was responsible for maintaining data integrity and supporting efficient retrieval of

the information required by the storefront and administrative panel. Stored information included:

- User Accounts
- Products and Categories
- Orders and Order Items
- Homepage Promotional Slots
- Customer-Admin Messages

The architecture ensured that communication between the application server and the database remained organized and consistent, restricting unauthorized access to administrative functions and customer data.

4.5 Hardware and Software Requirements

The implementation of Margaux Collections required appropriate hardware and software resources to ensure efficient performance during product browsing, order processing, and administrative operations.

4.5.1 Hardware Requirements

Table 4.1 presents the recommended hardware specifications.

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	128 GB SSD	256 GB SSD or higher

Monitor	1366 x 768 Resolution	Full HD Display
Internet Connection	10 Mbps	25 Mbps or higher
Mobile Device	Android 8	Android 11 or higher

4.5.2 Software Requirements

Table 4.2 presents the software requirements.

Software	Purpose
Windows 10 / 11	Operating System
Visual Studio Code	Source Code Editor
MySQL (via XAMPP)	Database Management System
PHP	Backend Development
HTML, CSS, JavaScript	Frontend Development
Google Chrome	Web Browser
Git and GitHub	Version Control and Collaboration

XAMPP	Local Development Server
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The selection of these software tools was based on their stability, accessibility, and suitability for developing and locally hosting a PHP and MySQL web application.

4.6 Database Design

The database served as the foundation of Margaux Collections, organizing and storing product listings, customer accounts, order records, and site content. A relational database model was implemented using MySQL, with tables structured to reduce data redundancy and maintain consistency among related records.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.

Figure 4.4. Entity Relationship Diagram of Margaux Collections

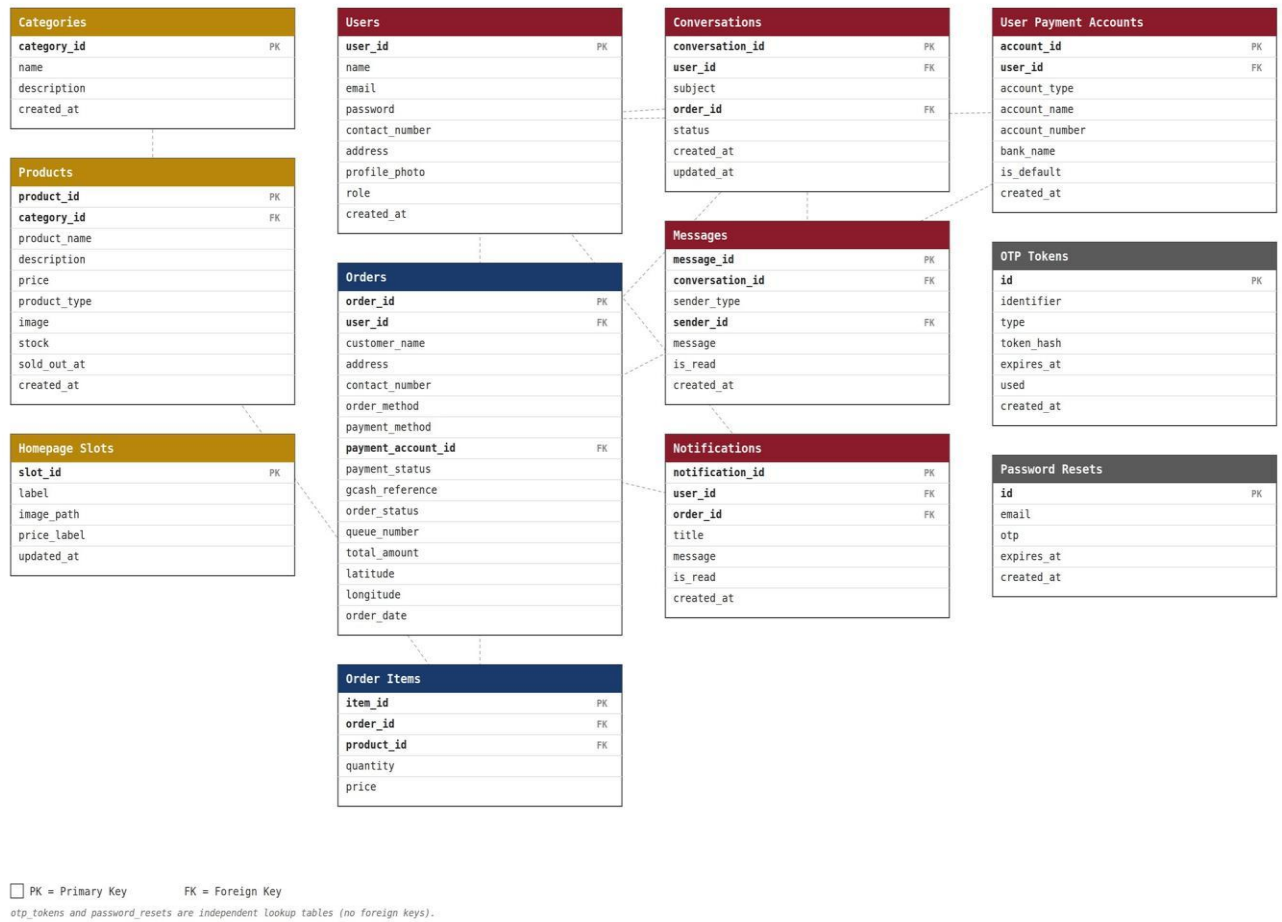


Figure 4.4. Entity Relationship Diagram of Margaux Collections

The primary entities of the database included:

- Users
- Products
- Categories
- Orders
- Order Items
- Homepage Slots
- Messages
- Conversations
- Notifications
- User Payment Accounts
- OTP Tokens

- Password Resets

Users Table

This table stored account credentials and role information for both customers and administrators.

Field	Description
user_id	Primary Key
name	Complete Name
email	Login Email Address
password	Encrypted Password
role	User Access Level (Customer / Admin)

Products Table

This table stored the boutique's product listings.

Field	Description
product_id	Primary Key
product_name	Product Name
category_id	Foreign Key – Product Category

price	Selling Price
Field	Description
stock	Available Inventory
image	Product Image
description	Product Description
product_type	Product Type (Pre-loved / Brandnew)
sold_out_at	Timestamp When Product Went Out of Stock

Categories Table

This table stored the product classifications used to organize the boutique's catalog.

Field	Description
category_id	Primary Key
name	Category Name

Orders Table

This table recorded customer purchase transactions.

Field	Description
order_id	Primary Key
Field	Description
user_id	Foreign Key
order_date	Date of Purchase
order_method	Order Fulfillment Method (Meet up / Drop off / Shipping Delivery / Pick-up Via Rider)
payment_method	Selected Payment Method (Cash on Pickup / Cash on Delivery / GCash)
total_amount	Total Order Amount
latitude	Pinned Delivery Latitude (Shipping / Pick-up Via Rider only)

longitude	Pinned Delivery Longitude (Shipping / Pick-up Via Rider only)
payment_status	Payment Status (Pending / Pending Verification / Paid)
order_status	Order Fulfillment Status

Order Items Table

This table recorded the individual items included in each order.

Field	Description
item_id	Primary Key
order_id	Foreign Key
product_id	Foreign Key
quantity	Quantity Ordered
price	Unit Price at Time of Purchase

Homepage Slots Table

This table stored administrator-configured promotional content displayed on the storefront's homepage.

Field	Description
-------	-------------

slot_id	Primary Key
label	Slot Label (e.g. Featured, Dresses, Tops)
image_path	Uploaded Promotional Image Path
price_label	Optional Display Price Text (e.g. "Starting at ₱299")
updated_at	Date Last Updated

Messages Table

This table stored messages exchanged between customers and administrators.

Field	Description
message_id	Primary Key
conversation_id	Foreign Key – Linked Conversation
sender_type	Sender Role (Customer / Admin)
sender_id	Foreign Key – Sending User

message	Message Text
created_at	Date and Time Sent

In addition to these primary entities, the database included several supporting tables that handled authentication and account security. These comprised Conversations, which grouped customer-admin messages into threads referenced by the Messages table; Notifications, which stored order-related alerts sent to customers; OTP Tokens and Password Resets, which supported onetime-password verification for login, registration, and password recovery; and User Payment Accounts, which stored customers' saved GCash details for checkout. These tables were excluded from Figure 4.4 for clarity but followed the same referential integrity principles as the primary entities.

The relationships among these tables allowed the application to associate customer orders with specific products, trace order histories, and control the promotional content displayed on the storefront. Foreign key constraints were used to maintain referential integrity between related records, ensuring that order details, promotional slots, and messages remained consistently linked to their corresponding users and products.

4.7 System Implementation

The system was implemented incrementally, beginning with the database schema and core authentication pages, followed by the customer-facing storefront and its shopping cart and checkout functionality, and concluding with the administrative panel. The GCash QR checkout process was integrated to allow customers to complete payment by scanning a displayed QR code at the point of

order confirmation. Four order fulfillment methods were supported – Meet up, Drop off (at MJM Building), Shipping Delivery, and Pickup Via Rider – each restricted to specific payment options: Meet up orders were limited to Cash on Pickup, Drop off orders required GCash payment, Shipping Delivery orders allowed either GCash or Cash on Delivery, and Pick-up Via Rider orders allowed either Cash on Pickup or GCash. For Shipping Delivery and Pick-up Via Rider orders, customers were required to pin their exact delivery location on an interactive map (built with Leaflet and OpenStreetMap), which captured latitude and longitude coordinates alongside the typed address to help ensure accurate rider and courier navigation.

The administrative panel was implemented as a set of interconnected pages, including the dashboard, analytics, order management, product management, user management, and messaging pages, each maintaining the system's dark noir theme through shared CSS variables to ensure a consistent visual identity across the platform.

The system was developed and maintained using Git and GitHub for version control, allowing development work to be synchronized across multiple machines. This collaborative workflow required resolving occasional synchronization issues, including conflicting local database processes and merge conflicts in stylesheet files, to maintain a consistent and functioning codebase throughout development.

4.8 Testing and Evaluation

Testing was conducted throughout development to verify that each module of Margaux Collections functioned as intended. Unit testing was performed on individual features such as product management, order processing, and the GCash QR checkout process, while

integration testing verified that these modules interacted correctly once combined into the complete system.

Particular attention was given to visual consistency testing across the customer-facing and administrative interfaces, following the identification of stylesheet conflicts in which customer-facing styling rules unintentionally overrode administrative styling rules, resulting in illegible or washed-out text on certain administrative pages. These issues were resolved by revising the affected stylesheets to properly scope styling rules to their intended pages.

Database-level testing was also conducted to confirm that order records, product listings, and homepage promotional content were stored and retrieved correctly, and that referential integrity was maintained across related tables following the removal of the membership feature.

4.9 Results and Discussion

The completed Margaux Collections system provided a functional online storefront through which customers could browse the boutique's product catalog, place orders, and complete payment through the GCash QR checkout process. The administrative panel enabled the boutique's staff to manage products, monitor and update orders, configure homepage promotional content, and communicate directly with customers through the built-in messaging feature.

The distinct dark noir visual theme, implemented through reusable CSS variables and the Playfair Display and Jost typefaces, was successfully maintained across the storefront and administrative interfaces following the resolution of earlier stylesheet conflicts. This consistency contributed to a cohesive brand presentation aligned with the boutique's intended market positioning.

The removal of the tiered membership feature simplified the customer experience by making the full product catalog accessible to all customers without requiring a membership tier, which was determined to better suit the boutique's business model. Overall, the results indicated that the system met its intended objectives of automating order recording, supporting a secure QR-based payment process, and providing administrators with the tools needed to manage the boutique's online presence.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter provided a comprehensive conclusion to the development of Margaux Collections. It synthesized the project's results, assessed whether the research objectives were achieved, and offered recommendations for future enhancements and implementation strategies to further improve the system's utility for the boutique business.

5.2 Summary of Findings

The development and evaluation of Margaux Collections yielded results aligned with the project's objectives:

- The system successfully provided an online storefront that allowed customers to browse the boutique's product catalog and complete purchases through a GCash QR-based checkout process.
- The administrative panel effectively supported product, order, and user management, along with configurable homepage promotional content.
- The system's dark noir visual theme was implemented consistently across the storefront and administrative interfaces following the resolution of stylesheet conflicts encountered during development.
- The removal of the tiered membership feature simplified the customer purchasing experience and reduced the complexity of the underlying database and administrative controls.

- Collaborative development through Git and GitHub allowed the project to be maintained across multiple machines, with synchronization issues identified and resolved during the course of development.

5.3 Conclusion

Margaux Collections proved to be a functional and brand-consistent solution for transitioning the boutique's sales process to an online platform. By providing a structured storefront, a secure GCash-based checkout process, and a comprehensive administrative panel, the system addressed the boutique's need for an accessible and visually distinctive e-commerce presence. The project successfully demonstrated that a small-scale fashion retail business could be supported by a purpose-built web application without reliance on generic, template-based e-commerce solutions.

5.4 Recommendations

System Improvements

- Implement automated low-stock notifications to assist in inventory management.
- Enhance account security by introducing additional authentication measures, such as one-time password verification.
- Develop reporting features that allow order and sales data to be exported in formats such as PDF or CSV for recordkeeping purposes.

Future Research Directions

- Explore the integration of automated payment verification to reduce manual confirmation of GCash transactions.
- Investigate the feasibility of a mobile-native application to extend accessibility for customers with limited desktop access.
- Conduct a follow-up study to assess the system's long-term impact on the boutique's sales performance and customer retention.

Implementation Strategies

- Boutique staff should be trained on consistent use of the administrative panel to ensure accurate product and order records.
- Regular database backups should be performed to prevent data loss and support system reliability.

5.5 Impact of the Study

Margaux Collections contributed to the practical application of web development principles in supporting a small-scale fashion retail business. By providing a tailored, brand-consistent ecommerce platform, the project demonstrated how purpose-built systems could address the specific operational and visual identity needs of small businesses more effectively than generic solutions, while highlighting the iterative refinement, such as feature removal and stylesheet correction, that such projects commonly required during development.

5.6 Summary

In summary, Margaux Collections represented the successful development of a brand-aligned, web-based e-commerce and boutique management system. It served as a demonstration of how iterative, requirements-driven development could produce a functional and visually cohesive platform for a small fashion retail business, providing a foundation for future enhancements to further support the boutique's operations.